ACORD Standards for US Life & Annuity

A Practical Business & Technology Architecture Study Guide

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Chapter 1 — ACORD Foundation

1.1 What ACORD Is

[Official ACORD concept] ACORD (Association for Cooperative Operations Research and Development) is a non-profit, industry-owned standards-setting body for the global insurance and related financial-services industries, founded in 1970. It was originally formed by carriers and agents to standardize the many proprietary paper forms used for new business and claims submission in US property & casualty insurance, and later expanded into electronic data standards. ACORD is not an insurer and does not process claims or provide coverage — it publishes standards, forms, and supporting tools. It maintains offices in New York/New Jersey and London, and today engages more than 36,000 participating organizations across 100+ countries — carriers, reinsurers, agents/brokers, software vendors, and other financial-services organizations. In 2016, ACORD Solutions Group was formed as an industry-owned extension of ACORD to help members actually implement the standards (reducing the risk/cost/time of deployment), separate from ACORD's standards-setting role itself.

1.2 Why ACORD Exists

[Industry-standard concept, applied to ACORD] The insurance value chain involves many independent parties that must exchange data reliably — carriers, agents/brokers, MGAs/MGUs, TPAs/BPOs, reinsurers, custodians/broker-dealers (for VA/RILA), and regulators. Without a shared standard, every pairwise integration is custom-built and brittle — the classic "N×N integration problem," where N systems require roughly N×(N-1) point-to-point mappings. A common standard turns this into an "N-to-1-to-N" problem: each party maps once to/from the shared standard, and the standard mediates everyone else. This is the core economic reason ACORD (and standards bodies generally) exist, and it's the same logic behind a canonical/reference-architecture approach inside a single enterprise.

1.3 ACORD Standards — the Two Big Buckets

[Official ACORD concept] For an L&A architect's purposes, ACORD's output splits into two related but distinct things:

- **ACORD Data Standards** — line-of-business-specific electronic message standards (XML, and historically EDI flat-file). Major lines: P&C/Surety, **Life, Annuity & Health (LAH)**, Reinsurance & Large Commercial, plus regional variants (e.g., APAC General Insurance, ANZ Life). The current LAH data-standard release is **ACORD LA XML v2.49.00 (Final, Dec 2025)**, following v2.48.00 (Dec 2024) and v2.47.00 (Dec 2023) — the standard is revised on a roughly annual cadence, and trading partners agree on and certify against a specific version for a given integration.
- **ACORD Reference Architecture** — a cross-line, technology-neutral *enterprise* architecture framework (distinct from the line-specific data standards), covered in 1.4 below.

Full schema files, Help Files, the Object Reference Model, and the Testing & Certification Facility (TCF) require an ACORD account tied to a membership/participation program. *[Requires verification against licensed/current ACORD documentation for any field-level schema content — this guide will not invent field names, type codes, or message structures it cannot verify from public sources.]*

1.4 ACORD Reference Architecture — the Seven Facets

[Official ACORD concept, partially verified — see note] ACORD describes the Reference Architecture as "a series of seven interrelated industry models, or facets," usable individually or together, and open to subscription by both members and non-members (access terms vary). Publicly confirmed facets:

1. **Business Glossary** — 6,000+ standardized, non-technical definitions of insurance concepts (e.g., "Beneficiary"), including synonyms and line-of-business-specific usage. Its job is shared vocabulary, not technical structure.
2. **Information Model** — a formal, technology-neutral, conceptual representation of data elements and their relationships across *all* lines of business (Policy, Product, Party, Claim, etc.). This is the semantic hub: the single source of meaning that both data models and message standards map back to, enabling any-to-any integration instead of point-to-point mapping.
3. **Data Model** — the more structure/storage-oriented model derived from the Information Model; used as the basis for canonical data stores, operational data stores, and data warehouses.
4. **Capability Model** — a business-capability taxonomy describing *what* an insurer does (not how, and not org-structure-dependent). Ten high-level capabilities, decomposed across three further tiers. Useful for API domain boundaries, target-operating-model design, and application-portfolio

rationalization ("which apps support which capability").

5. **Process Model** — standardized business-process flows (e.g., claims handling, underwriting, subrogation) aligned to the Capability Model and to Data Standards messages; recent updates have extended it to cover where AI/automation sits within a process (e.g., claims triage).
6. **Component Model** — decomposes business functions into modular, technology-neutral components with defined interfaces (e.g., "claims processing," "payment handling" as separable components). Used heavily in legacy consolidation/modernization to decide what to keep, refactor, or retire.
7. *Not confirmed from public sources at the time of this writing.* ***[Requires verification against current ACORD Reference Architecture documentation at acord.org/standards-architecture/reference-architecture]*** *— do not assume a name for this facet without checking the live source, since the Reference Architecture is periodically restructured (the 2023 update alone touched three facets at once).*

1.5 Canonical Data Models (the general concept, and how ACORD implements it)

[Industry-standard concept] A canonical data model is a single, system-agnostic representation of a business object — "Party," "Policy," "Financial Activity" — that every internal system and every external message maps to and from, rather than every system pair mapping directly to every other system. The ACORD **Information Model** is designed to serve exactly this role for insurance: a canonical layer sitting between heterogeneous policy administration systems (OIPA, Ingenium, Duck Creek, Majesco, or a legacy AS/400 book of business) and everything outside them — distributors, custodians/broker-dealers, TPAs, regulators, and increasingly AI/analytics layers.

1.6 Message Standards for Life & Annuity

[Official ACORD concept, with verification flags]

- The Life & Annuity message pattern is built around a request/response envelope commonly referred to as **TXLife / TXLifeResponse** — the wrapper carries authentication and source/destination routing information around the actual business content.
- The underlying business object model beneath TXLife is generally referred to as **OLifE** (the Life & Annuity object model) — encapsulating entities like Holding, Policy, and Party in a defined object hierarchy. *[Requires verification of current terminology/structure against licensed ACORD LA XML v2.49.00 documentation — object naming and hierarchy can shift release to release, and this guide will not reconstruct field-level structure from memory.]*
- A separate **EDI (flat-file) track** also exists for L&A — historically used, for example, in the ACORD L&A/DTCC EDI Public Specification for application, asset-pricing, and commission data between carriers and distribution/clearing partners.

1.7 API Standards

[Industry-standard concept, with a caution] ACORD's Life & Annuity data standards are XML/EDI-first by history, predating REST. There is not a single, universally-cited "ACORD REST standard" for the LA line in the way there's a citable LA XML version number. What exists in practice is *industry convention*: using the ACORD Information Model as the semantic backbone behind custom or vendor-built REST/JSON APIs. ACORD has been publishing broader "Next-Generation Digital Standards" material aimed at more granular, technology-agnostic, faster digital data exchange — *[requires verification of current scope/maturity against acord.org before treating any specific endpoint set as an official ACORD standard]*.

1.8 XML, JSON, REST, Web Services — how they relate here

- **XML** — the historical and still-current wire format for ACORD LA data standards, schema-validated against ACORD-published XSDs.
- **JSON** — increasingly used as a lighter serialization of the same underlying canonical concepts behind modern REST APIs; not itself "the" ACORD LA standard in the historical sense.
- **REST APIs / Web services** — the transport/interaction pattern. Older ACORD web-services implementations leaned SOAP; newer integration layers lean REST/JSON, generally still expressing the same Information Model concepts underneath.

1.9 Schema Concepts: Versioning, Extensions, Code Lists, Validation

- **Versioning** — the LA XML standard revises roughly annually (v2.47 → v2.48 → v2.49); trading partners must agree on and certify against a specific version.
- **Extensions** — ACORD schemas generally provide defined extension points so carriers can add proprietary fields without breaking base-schema validation. *[Exact mechanism requires verification against licensed schema documentation.]*
- **Code lists / type codes** — controlled vocabularies for enumerated fields (relationship-to-insured, transaction type, etc.) — referenced in public schema-documentation mirrors as "tc" (type code) attributes.
- **Validation & certification** — ACORD's **Testing & Certification Facility (TCF)** validates member-submitted XML first for schema/DTD syntax and structure, then for business-rule and data-quality compliance, before issuing a certification usable in production — a two-stage gate, not a single syntax check.

1.10 How the Pieces Relate — the Architecture Flow

[Illustrative — built for this guide, showing how the Reference Architecture facets and Data Standards work together end to end]

```
BUSINESS CAPABILITY
  "What the business does" — e.g., New Business Issuance, Claims Handling
```

```
  (ACORD Capability Model)
          │
          ▼
BUSINESS PROCESS
  "How it's done, step by step" — e.g., Application Intake → Underwriting → Issue
  (ACORD Process Model, aligned to the Capability above)
          │
          ▼
INFORMATION MODEL
  The technology-neutral MEANING of every concept touched by that process —
  Party, Policy, Coverage, Rider, Financial Activity, Requirement, etc.
  (the semantic hub / canonical layer — single source of meaning)
          │
          ▼
DATA MODEL
  The structured, storable representation of that meaning — logical/physical
  shape used for canonical data stores, ODS, and data warehouses
          │
          ▼
MESSAGE / API
  The wire format carrying the data between systems — LA XML (TXLife) message,
  EDI file, or a REST/JSON API — all expressing the same Information Model concepts
          │
          ▼
INSURANCE APPLICATION
  The system that actually executes the transaction — your PAS (OIPA, Ingenium,
  Duck Creek, Majesco) or a legacy AS/400 book of business
```

Read top-down: a Capability ("New Business Issuance") is realized by one or more Processes ("Application Intake," "Underwriting," "Policy Issue"). Each Process step consumes or produces concepts defined in the Information Model ("Party," "Coverage," "Requirement"). Those concepts are persisted per the Data Model, and exchanged between systems using a Message or API built on that same Information Model — ultimately landing in, or being triggered by, an actual PAS instance or legacy book of record.

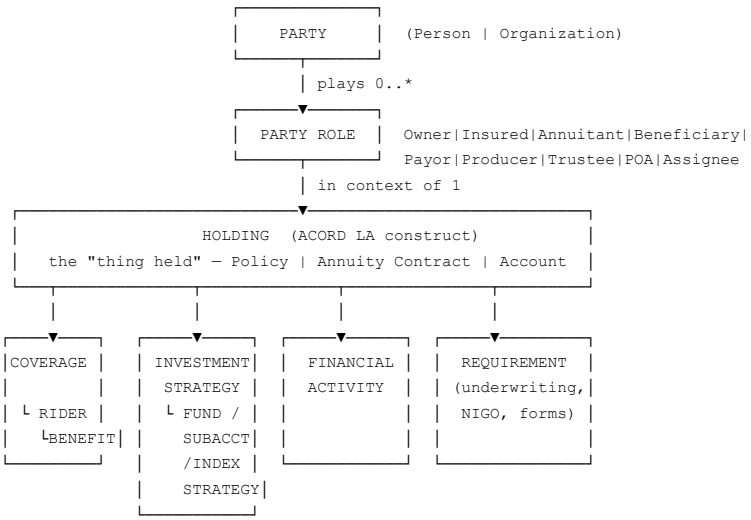
End of Chapter 1. Next: Chapter 2 — the Life & Annuity Domain Model (Party, Policy/Contract, Coverage, Rider, Financial Activity, and the rest of Part II's entity catalog, each with definition, purpose, relationships, cardinality, key attributes, typical PAS representation, and an example).

Sources referenced for Chapter 1: acord.org (About ACORD, Standards & Architecture, Reference Architecture, LAH Data Standards pages), ACORD news releases on Reference Architecture updates (2023–2025), and independent secondary summaries used only to corroborate publicly stated facts — not for any schema or field-level detail.

Chapter 2 — Life & Annuity Domain Model

*Scope note: the entity names below are the **conceptual** names used across the ACORD Information Model, ACORD LA XML, and general enterprise insurance modelling. Where a name is specifically an ACORD LA XML construct (e.g. Holding, OLifE) it is marked. Field-level attribute names in ACORD schemas are **not** reproduced — attributes listed are conceptual and marked [Illustrative] where they are this guide's naming.*

2.0 The Shape of the Model in One Picture



The single most important structural idea: PARTY is separate from ROLE, and ROLE is what connects a party to a contract. A person is not "an owner" — a person **is a party** who **plays the owner role** on a particular contract. Chapter 6 develops this fully; everything else in the model depends on getting it right.

2.1 Party Cluster

PARTY

- **Definition.** Any individual or legal entity of interest to the carrier — the abstract supertype of Person and Organization. **[Official ACORD concept]**
- **Purpose.** One record per real-world entity across the enterprise, independent of how many policies they touch or in what capacity.
- **Relationships.** PARTY 1..1 → 0..* PARTY ROLE. PARTY 1..1 → 0..* ADDRESS. PARTY 1..1 → 0..* COMMUNICATION. PARTY ↔ PARTY relationships (spouse, parent, employer, trustee-of).
- **Cardinality note.** A Party may hold zero roles (a prospect), or dozens (an agency that is producer on 40,000 contracts).
- **Key attributes (conceptual).** Party ID, party type (Person/Organization), legal name, tax ID (SSN/EIN), tax-ID type, date established (DOB/incorporation date), status, OFAC/sanctions screening result, source system ID(s).
- **Typical PAS representation.** OIPA: `Client` entity with `ClientGUID`. Ingenium: a client/name master keyed by client number. Legacy AS/400: usually an `OWNER_ID` / `CLIENT_NO` in a name-and-address master file, frequently duplicated per line of business.
- **Example.** Party P-0001 — Person, "Margaret L. Chen," SSN ●●●-●●-4417, DOB 1961-03-08 — who is Owner and Annuitant on contract A-77120, Beneficiary on her spouse's contract A-77121, and a prospect for a third.

PERSON

- **Definition.** A Party subtype representing a natural human being. **[Official ACORD concept]**
- **Purpose.** Carries mortality- and identity-relevant attributes that only apply to humans.
- **Relationships.** Subtype of PARTY. Referenced by the Insured and Annuitant roles specifically — **an Organization can never be an Insured or Annuitant** (there's no measurable life), though it can be Owner, Beneficiary, or Payor.
- **Key attributes.** First/middle/last/suffix, DOB, date of death, gender (for pricing where permitted), marital status, citizenship/residency, tobacco/risk class, occupation.
- **PAS representation.** OIPA `Client` with person-type role codes; legacy: `ANNUITANT_ID` pointing to a person master.
- **Example.** Person record behind Party P-0001 above, with risk class "Standard Non-Tobacco," used at issue for a life policy but irrelevant on a MYGA.

ORGANIZATION

- **Definition.** A Party subtype representing a legal entity — corporation, trust, estate, charity, plan sponsor, agency, broker-dealer. **[Official ACORD concept]**
- **Purpose.** Supports non-natural owners, which are common and consequential: a trust-owned annuity, an estate as default beneficiary, an employer as group-contract holder.
- **Relationships.** Subtype of PARTY. May be Owner, Beneficiary, Payor, Producer/Agency, Assignee, Plan Sponsor. **May not** be Insured or Annuitant.
- **Cardinality trap.** A trust-owned non-qualified deferred annuity has a *non-natural owner* — which under IRC §72(u) generally strips tax deferral unless the trust holds it as agent for a natural person. Your model must be able to represent "owner is an Organization" and let a rule fire on it; if `OWNER_ID` can only point at a person, you cannot even express the risk.
- **Key attributes.** Legal name, EIN, entity type (corp/LLC/trust/estate/qualified plan), state of formation, trust date, natural-person-agent indicator [Illustrative].
- **Example.** "The Chen Family Revocable Trust dtd 2019-06-11," EIN 87-●●●●●●●, Owner of contract A-77122; Margaret Chen is Annuitant. Tax-deferral rule requires evaluating §72(u).

PRODUCER / AGENT

- **Definition.** The Party (person or agency) credited with selling and servicing the contract; "Agent" is the individual writing producer, "Producer" often the broader term including agency/firm. **[Official ACORD concept — modelled as a Party Role, not a separate party type]**
- **Purpose.** Drives commission calculation, licensing/appointment compliance, suitability attestation, book-of-business servicing, and hierarchy-based overrides.
- **Relationships.** PARTY (Person or Org) → PRODUCER ROLE → 0..* HOLDING. Producers form **hierarchies** (writing agent → agency → IMO/FMO → broker-dealer), so PRODUCER 0..* ↔ 0..* PRODUCER via a hierarchy relationship with effective dating.

- **Cardinality.** A contract commonly has **multiple** producers with split percentages summing to 100% — model producer-on-contract as an associative entity carrying a split, not as a single foreign key.
- **Key attributes.** Producer ID/NPN, license states & lines, appointment status per carrier/state, FINRA registration (required for VA/RILA/VUL), hierarchy position, commission schedule ID, split %.
- **PAS representation.** OIPA: agent/producer roles with hierarchy tables; legacy: `AGENT_NO` plus a separate commission-hierarchy file.
- **Validation that matters.** At new business: is the producer licensed *in the owner's state*, appointed *with this carrier*, and — for a RILA or VA — securities-registered and affiliated with a selling broker-dealer? Failure here is the single most common NIGO cause after missing signatures.

CARRIER

- **Definition.** The insurance company legally issuing the contract and bearing the obligation. **[Official ACORD concept]**
- **Purpose.** Identifies the obligor; drives statutory reporting, state filing, claims-paying-ability disclosure, and — critically for TPAs — which client's book a policy belongs to.
- **Relationships.** CARRIER 1..1 → 0..* HOLDING. In a TPA/BPO context, one administration platform hosts **many** carriers, so carrier ID is a partitioning key across nearly every entity and every API path.
- **Key attributes.** Carrier ID, NAIC company code, state of domicile, licensed states, admitted/non-admitted status, rating.

DISTRIBUTOR

- **Definition.** The organization through which the contract was distributed — broker-dealer, bank, wirehouse, IMO/FMO, RIA firm. **[Industry-standard concept; represented via Party + role/hierarchy in ACORD terms]**
- **Purpose.** Distinguishes *who sold it* (producer) from *the firm that owns the relationship and supervises the sale* — different entities, different compliance obligations, different compensation flows.
- **Relationships.** DISTRIBUTOR 1..1 → 0..* PRODUCER; DISTRIBUTOR 1..1 → 0..* HOLDING.
- **Why separate.** Reg BI / suitability supervision belongs to the BD, not the individual. Trail compensation may be paid to the firm. Data feeds (DTCC) are exchanged at firm level.

2.2 Contract Cluster

HOLDING

- **Definition. [Official ACORD LA XML construct]** The generic container for "the thing a party holds" — a life policy, an annuity contract, or an investment account. It is the top-level business object that Policy and other specific structures hang beneath.
- **Purpose.** Lets one message structure carry heterogeneous held items (a life policy and an annuity in a single party's portfolio) without separate top-level schemas for each.
- **Relationships.** HOLDING 1..1 → 0..1 POLICY (for insurance holdings); HOLDING 1..1 → 0..* PARTY ROLE.
- *[Requires verification against licensed ACORD LA XML v2.49.00 documentation for the exact current subtype list and structure.]*
- **Modelling advice.** Even if you're not implementing ACORD XML literally, the Holding abstraction is worth stealing: it gives you a clean place to hang "portfolio view" queries (`GET /parties/{id}/holdings`) that span life and annuity without union-ing two unrelated tables.

POLICY / CONTRACT

- **Definition.** The legal agreement between carrier and owner. Convention: **"policy"** for life insurance, **"contract"** for annuities — the same modelled entity, different business vocabulary. **[Official ACORD concept]**
- **Purpose.** The spine of the entire model. Everything financial, every role, every coverage, every transaction attaches to it.
- **Relationships.**
  - POLICY 1..1 → 1..* COVERAGE
  - POLICY 1..1 → 0..* RIDER (or riders hang off Coverage — see 2.3)
  - POLICY 1..1 → 1..* PARTY ROLE (at minimum an Owner; life needs an Insured; deferred annuity needs an Annuitant)
  - POLICY 1..1 → 0..* FINANCIAL ACTIVITY
  - POLICY 1..1 → 0..* INVESTMENT STRATEGY / allocation
  - POLICY 1..1 → 0..1 TAX QUALIFICATION
- **Key attributes.** Policy/contract number, carrier ID, plan/product code, product type, issue date, effective date, issue state (**jurisdiction of the owner**

at issue — drives which filed form applies), status, status-effective date, maturity date, free-look expiry, surrender-charge schedule ID, MVA indicator, tax-qualification code, currency.

- **PAS representation.** OIPA: `Policy` with `PolicyGUID`, plan code linking to product configuration. Legacy AS/400: a policy master record keyed by `POLICY_NUMBER`, with `PLAN_CODE` encoding a bundle of product behaviours in a 3–6 character string.
- **The classic legacy failure.** In older systems `PLAN_CODE` simultaneously encodes product type, tax qualification, surrender schedule, and sometimes state variation. Chapter 5 explains why this must be decomposed.

LIFE POLICY

- **Definition.** A Policy subtype whose primary obligation is a death benefit contingent on an Insured's life. **[Official ACORD concept]**
- **Distinguishing structure.** Requires an **Insured** role; carries a face amount / death benefit option; may carry cash value (WL/UL/IUL/VUL) or not (Term); premium may be fixed (WL/Term) or flexible (UL family).
- **Key attributes.** Face amount, death benefit option (Level / Increasing / ROP), premium mode & amount, target premium, MEC status and 7-pay test result, no-lapse-guarantee status & shadow account value, cost of insurance basis.
- **Example.** IUL, face $500,000, DB Option B (increasing), monthly deduction of COI + expense + rider charges from account value, index credit applied annually on segment anniversary.

ANNUITY CONTRACT

- **Definition.** A Policy subtype whose primary obligation is accumulation and/or a stream of payments measured against an Annuitant's life. **[Official ACORD concept]**
- **Distinguishing structure.** Requires an **Annuitant**; has no face amount; has an accumulation value and a surrender value; has a maturity/annuitization date; distinguishes an **accumulation phase** from a **payout phase** — a state transition that changes which attributes are even meaningful.
- **Key attributes.** Accumulation/account value, cash surrender value, guaranteed minimum value (nonforfeiture), surrender charge schedule & remaining period, MVA parameters, free-withdrawal percentage, GLWB benefit base & rollup rate, income start date, annuitization option, payout frequency/amount, tax qualification, RMD indicator.
- **The phase problem — a modelling essential.** Pre-annuitization, `AccountValue` and `SurrenderValue` are the meaningful measures and the Owner controls the contract. Post-annuitization, account value may cease to exist entirely, surrender is typically impossible, and the Annuitant's survival drives payments. Models that treat these as one flat record accumulate nulls and contradictory rules. Prefer an explicit `contractPhase` state with phase-scoped attribute groups.

COVERAGE

- **Definition.** A distinct unit of insurance protection within a policy, with its own amount, term, and often its own insured life. **[Official ACORD concept]**
- **Purpose.** Enables multi-benefit policies without duplicating whole policies.
- **Relationships.** POLICY 1..1 → 1..* COVERAGE; COVERAGE 1..1 → 0..* RIDER; COVERAGE 1..1 → 1..* INSURED (usually 1, but 2 for survivorship).
- **Cardinality.** Base coverage 1; additional coverages 0..* (spouse term rider as a coverage, child term, additional-insured).
- **Key attributes.** Coverage ID, coverage type (base/supplemental), face or benefit amount, effective & expiry date, rate class, insured party role reference, status.
- **Annuity nuance.** Deferred annuities in accumulation are often modelled with a single nominal "coverage" or none at all — the value-bearing structures are Investment Strategy and the benefit-base riders. Don't force a life-shaped coverage model onto an accumulation annuity.

RIDER

- **Definition.** An optional contractual amendment adding, restricting, or modifying benefits, usually for an explicit charge. **[Official ACORD concept]**
- **Purpose.** Riders carry most of the product differentiation in the modern market — GLWBs on annuities, chronic-illness accelerators on life.
- **Relationships.** Attaches to POLICY or COVERAGE (0..*; generates BENEFIT (1..*)*) and FINANCIAL ACTIVITY (rider charges, benefit payments).
- **Key attributes.** Rider ID/code, rider type, effective date, termination date, charge basis (% of benefit base vs. % of account value vs. per-$1,000), charge rate, benefit base, rollup rate & rollup period, step-up basis & frequency, exercise status, elected-at-issue vs. added-later indicator.
- **Cardinality trap that bites in production.** A GLWB's **benefit base is not the account value.** It is a separate, rider-owned value with its own accumulation rules (rollup), its own reset events (step-up), and its own decrement rules on withdrawal (often pro-rata, which is why an excess withdrawal can permanently damage guaranteed income). If your data model stores one "value" per contract, you cannot administer a GLWB correctly. Model rider values as first-class, rider-scoped attributes.
- **Example.** GLWB rider, elected at issue, benefit base $100,000 rolling up 7% simple for 10 years or until first withdrawal, annual step-up to account value if higher, charge 1.25% of benefit base deducted quarterly from account value, lifetime withdrawal 5.0% at age 65 single life.

BENEFIT

- **Definition.** A specific promised payment or entitlement arising from a coverage or rider. **[Official ACORD concept]**
- **Purpose.** Separates *the promise* from *the vehicle that grants it* — one rider may produce several benefits (e.g. a linked-benefit LTC rider producing an acceleration benefit and an extension-of-benefits benefit).
- **Key attributes.** Benefit type, benefit amount or formula, trigger condition, waiting/elimination period, benefit period, maximum, taxability treatment.
- **Example.** Chronic-illness acceleration benefit: trigger = 2-of-6 ADL certification, monthly maximum = lesser of 2% of face or HIPAA per-diem cap, reduces death benefit dollar-for-dollar (or by a discounted factor, depending on form).

2.3 Role Cluster (summary here; developed fully in Chapter 6)

Role	Party type allowed	Cardinality per contract	Core rights/obligations	Notes
Owner	Person or Org	1..* (joint owners; TOD/JTWROS variants)	Controls the contract: surrender, allocation, beneficiary change, assignment	Non-natural owner triggers §72(u) analysis on NQ deferred annuities
Annuitant	**Person only**	1..* (joint annuitants)	Measuring life for payout & often death trigger	Cannot be an entity; changing the annuitant may be a taxable/contract-altering event
Insured	**Person only**	1..* (2 for survivorship)	Life on which death benefit is contingent	Underwriting attaches here
Beneficiary	Person or Org	0..* primary + 0..* contingent	Receives death proceeds	Percentages must total 100% *per class*; per stirpes/per capita must be modelled
Payor	Person or Org	0..1 typically	Remits premium	Often ≠ Owner (parent paying child's policy; employer on group)
Producer/Agent	Person or Org	1..* with splits	Sale & servicing; commission	Licensing/appointment/FINRA validation
Trustee	Person or Org	0..*	Acts for a trust owner	Authority evidenced by trust certification
Power of Attorney	Person	0..*	Acts for an incapacitated owner	Scope-limited; carrier must validate the POA document and often has form requirements
Assignee	Person or Org	0..*	Holds collateral or absolute assignment rights	Collateral assignment restricts owner's transaction rights — must be enforced as a rule
Plan Sponsor / Employer	Org	0..1	Group contract holder	Group life, 401(k)-funded annuities

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2.4 Financial & Investment Cluster

PREMIUM

- **Definition.** Consideration paid into a policy or contract. **[Official ACORD concept]**
- **Purpose.** Drives value, tax basis, MEC testing (life), and free-look/suitability tracking.
- **Relationships.** POLICY 1..1 → 0..* PREMIUM; each PREMIUM 1..1 → 1..1 FINANCIAL ACTIVITY; PREMIUM 1..1 → 0..* allocation splits across investment strategies.
- **Key attributes.** Premium amount, received date, effective/valuation date, premium type (initial/additional/scheduled/unscheduled/1035-in/rollover-in/transfer-in), source of funds, mode, tax-basis contribution (yes/no), tax year (for IRA contributions), allocation instruction.
- **Critical distinction.** A **1035 exchange in** and a **rollover in** are both money arriving, but their tax character is entirely different: a 1035 exchange **carries over basis**; a rollover into a qualified contract generally has **zero basis**; an after-tax contribution creates basis. Model `premiumType` and `basisAmount` separately — never infer basis from amount.

PAYMENT

- **Definition.** The movement of money in either direction (in: premium remittance; out: withdrawal, claim, annuity payment). **[Industry-standard**

concept]

- **Key attributes.** Payment method (check/ACH/wire/internal transfer), payee party, bank account reference (tokenized), amount, gross vs. net, status (pending/issued/cleared/stopped/returned), issue date, clearing date.
- **Modelling advice.** Keep PAYMENT distinct from FINANCIAL ACTIVITY. One withdrawal (a single financial activity) may produce **three** payments: net to the owner, federal withholding to the IRS, and state withholding to a state. Collapsing these loses reconcilability.

FINANCIAL ACTIVITY

- **Definition.** Any event that changes contract value or produces a money movement. **[Official ACORD concept — the central financial abstraction]**
- **Purpose.** A single uniform record type for premium, withdrawal, surrender, fee, interest credit, index credit, rider charge, transfer, loan, distribution, and adjustment. It is what makes `GET /policies/{id}/transactions` a coherent endpoint rather than a union of a dozen tables.
- **Relationships.** POLICY 1..1 → 0..* FINANCIAL ACTIVITY; FINANCIAL ACTIVITY 1..1 → 0..* PAYMENT; 1..1 → 0..* ACCOUNTING ENTRY; 0..1 → 1 TAX REPORTING RECORD; may reference a parent activity (reversal, adjustment).
- **Key attributes.** Activity ID, activity type, effective date, processed date, gross amount, net amount, taxable amount, non-taxable/basis amount, withholding amounts (fed/state), fee/charge amounts, source & destination fund allocations, reversal indicator & reversed-activity reference, originating transaction ID, actor/channel.
- **Non-negotiable design rule.** Financial activities are **append-only**. A correction is a new reversing activity plus a new correcting activity — never an update in place. Tax reporting, statement reproduction, audit, and dispute resolution all depend on being able to reconstruct the value history exactly as it stood on any past date.

TRANSACTION

- **Definition.** The business-level unit of work the customer or system requested (e.g. "a partial withdrawal"). **[Industry-standard concept; distinguished from Financial Activity deliberately]**
- **Why separate from Financial Activity.** One requested transaction may produce **many** financial activities: a $10,000 partial withdrawal from an FIA might generate (1) a pro-rata redemption from three index strategies, (2) a surrender charge, (3) an MVA adjustment, (4) federal withholding, (5) state withholding, (6) a net payment. Chapter 7 develops this hierarchy in full.

FUND / INVESTMENT / SUBACCOUNT

- **FUND [Official ACORD concept]** — the underlying investment vehicle (a mutual fund portfolio, a fixed account, a declared-rate bucket).
- **SUBACCOUNT** — the separate-account unit within a variable contract that invests in a fund; the owner holds *units* of a subaccount, not shares of the fund. Unit value (AUV) is struck daily; the owner's value = units × AUV.
- **INDEX STRATEGY** [Industry-standard; the FIA/RILA analogue] — a crediting bucket defined by index, term, crediting method, and limiter (cap / participation rate / spread / buffer / floor).
- **Key attributes.** Fund/strategy ID, name, type (separate-account subaccount / fixed / index strategy), index reference, crediting method, term length, cap rate, par rate, spread, buffer/floor %, segment start & maturity date, rate-guarantee period, current unit value.
- **Cardinality.** POLICY 1..1 → 0..* ALLOCATION → 1 FUND/STRATEGY. Allocations carry a percentage and an effective date; **allocation history must be retained** (needed to recompute credits and to defend a customer dispute).
- **Segment modelling (FIA/RILA).** Money entering an index strategy forms a **segment** with its own start date, start index value, term, and rate set. Multiple premiums create multiple, staggered segments with different maturity dates and different caps. This is the single most under-modelled structure in legacy annuity systems — a contract-level "index rate" field cannot represent it.

ACCOUNT VALUE / CASH VALUE / SURRENDER VALUE

These are three different numbers and conflating them causes real financial error.

Value	Definition	Formula shape	Where it's used
Account Value (AV) / Accumulation Value	Gross value of all buckets before any charge for early exit	Σ(allocations) + credits − charges − withdrawals	Statements, GLWB step-up basis, death benefit (often), rider charge basis
Cash Value	Value available on a life policy per contract terms	AV − outstanding loan & accrued interest (UL family)	Loans, non-forfeiture
Surrender Value (CSV)	What the owner actually receives on full surrender today	AV − surrender charge ± MVA − outstanding loan	Surrender processing, disclosure, 1035 out
Guaranteed Minimum Value	Statutory nonforfeiture floor	e.g. 87.5% of premium accumulated at the nonforfeiture rate	Floor test — CSV cannot fall below it
		Often AV with **no** surrender charge;	

Death Benefit Value	Payable on death	sometimes ROP or enhanced	Death claim
Benefit Base (rider)	Notional value for guaranteed income only	Premium + rollup, stepped up, decremented on withdrawal	GLWB income, rider charge — **never** payable as cash

The lesson: an annuity has at least six simultaneously valid "values." Any API returning a single `value` field is under-specified. See the `GET /policies/{id}/values` design in Chapter 9.

LOAN

- **Definition.** Borrowing against a life policy's cash value (annuities generally cannot be loaned; qualified plan annuities sometimes can). **[Official ACORD concept]**
- **Key attributes.** Loan balance, accrued interest, interest rate & charging basis (fixed/variable), loan type (standard/wash/preferred), collateral allocation, repayment history, capitalization schedule.
- **Why it matters structurally.** An unrepaid loan on a lapsing policy triggers **taxable income to the extent of gain** even though the owner received no cash at lapse — a notorious source of customer complaints and 1099-R disputes. The model must link LOAN → LAPSE → TAX REPORTING.

WITHDRAWAL / DISTRIBUTION

- **WITHDRAWAL** — an owner-initiated removal of value from a contract (partial surrender, systematic withdrawal, RMD, free-withdrawal-amount usage).
- **DISTRIBUTION** — the tax-reportable outflow view of the same event, plus death and annuitization payouts. Every distribution has a taxable/non-taxable split, a distribution code, and withholding.
- **Modelling advice.** These are two *views* of one financial activity, not two records. Model FINANCIAL ACTIVITY once, with a tax-reporting projection. Chapter 7 formalizes this.
- **Ordering rules that must be encoded.** Non-qualified deferred annuity withdrawals are **LIFO** (gain first, fully taxable) under IRC §72(e) — the opposite of the FIFO intuition most people bring. Pre-8/14/1982 contributions are grandfathered to FIFO. Qualified distributions are generally fully taxable unless basis exists. Roth distributions follow ordering rules (contributions → conversions → earnings) and the 5-year rules. **None of this can be inferred from the amount — it depends on tax qualification, contract dates, and basis tracking.**

TAX QUALIFICATION

- **Definition.** The tax treatment regime under which a contract is held. **[Official ACORD concept]** Fully developed in Chapter 5.
- **Key attributes.** Qualification type (NQ / Traditional IRA / Roth IRA / SEP / SIMPLE / 401(k) / 403(b) / 457 / Inherited IRA), plan/employer reference, cost basis, basis-as-of date, contribution-year tracking, Roth first-contribution year (5-year clock), RMD applicability, inherited-contract decedent info & applicable distribution regime (pre/post-SECURE Act), 72(q)/72(t) status.

2.5 Supporting Cluster

ADDRESS

- **Definition.** A physical or mailing location associated with a Party. **[Official ACORD concept]**
- **Cardinality.** PARTY 1..1 → 0..* ADDRESS, each with an address type (residence/mailing/business/temporary) and effective dating.
- **Why more than one matters.** State of *residence* drives premium tax, state withholding, and unclaimed-property escheatment; the *mailing* address drives correspondence. They are frequently different (snowbirds, PO boxes, third-party address). One address field = compliance error.
- **Key attributes.** Address type, lines, city, state/province, postal code, country, effective/expiry date, validated flag & standardization source, undeliverable/return-mail flag.
- **Special case.** A returned-mail flag combined with no contact is the trigger for the escheatment lifecycle — this must be a data-modelled state, not a comment field.

COMMUNICATION

- **Definition.** A non-address channel for reaching a Party — phone, email, and the record of correspondence sent. **[Official ACORD concept]**
- **Key attributes.** Communication type, value (tokenized where sensitive), preference & consent status (TCPA/e-delivery consent with timestamp), preferred-language, do-not-contact flag, delivery status.

REQUIREMENT

- **Definition.** An outstanding item that must be satisfied before a transaction can complete. **[Official ACORD concept]**
- **Purpose.** This is the entity that makes **NIGO (Not In Good Order)** processing tractable — the single largest operational cost driver in L&A new

business and post-issue service.

- **Relationships.** POLICY or TRANSACTION 1..1 → 0..* REQUIREMENT; each requirement 1..1 → 0..1 responsible PARTY.
- **Key attributes.** Requirement type (medical exam, APS, signature, spousal consent, trust certification, W-9, Medallion signature guarantee, suitability form), status (ordered/received/reviewed/waived/satisfied), ordered date, received date, due date, responsible party, waiver authority & reason.
- **Example.** A Roth conversion request arrives without a completed withholding election → requirement `WITHHOLDING_ELECTION` opened, status Outstanding, transaction held in Pending state, correspondence triggered, aging clock started.

UNDERWRITING INFORMATION

- **Definition.** Evidence and evaluation used to classify risk and decide issuance. **[Official ACORD concept]**
- **Relationships.** COVERAGE/INSURED 1..1 → 0..* UNDERWRITING INFO; drives rate class, ratings, exclusions, decline.
- **Key attributes.** Application answers, medical/lab results, APS, MIB/Rx/MVR results, financial justification, accelerated-UW model score & decision path, underwriter ID, decision, rate class, table rating, flat extra, exclusion rider.
- **Annuity nuance.** Deferred annuities are generally **not medically underwritten** — but they *are* suitability- and source-of-funds-reviewed, and **medically underwritten SPIAs** (impaired-risk / substandard annuities) do exist and pay enhanced income on shortened life expectancy. Don't model underwriting as life-only.
- **Privacy note.** Underwriting data is the most sensitive class in the model — it warrants separate access control, encryption, and retention policy from ordinary policy data.

End of Chapter 2.

Chapter 3 — Product Modeling

3.0 The Universal Decomposition

Every product in Chapter 3 decomposes down the same spine. Learning this once means you never have to memorize products individually — you derive them.

```
POLICY / CONTRACT          the legal agreement + tax wrapper + jurisdiction
      │
      ├── COVERAGE          the protection unit (life-dominant)
      │      └── RIDER      optional modification, usually charged
      │             └── BENEFIT   the specific promise produced
      │
      ├── INVESTMENT STRATEGY     how value grows (annuity/UL-dominant)
      │      └── FUND / SUBACCOUNT / INDEX STRATEGY / DECLARED RATE
      │             └── SEGMENT   a dated tranche with its own rate set
      │
      └── FINANCIAL ACTIVITY      every value change and money movement
```

The key insight for a business architect: the *same* structure serves all twelve products; what varies is **which branch carries the weight**.

- **Term Life** — Coverage branch heavy, Investment branch empty.
- **MYGA** — Investment branch is a single declared-rate bucket, Coverage branch nominal.
- **VUL** — *both* branches fully loaded (real coverage + real subaccounts).
- **SPIA** — Coverage and Investment both nearly empty; the contract *is* a Financial Activity schedule.

A product model that can't express "this branch is empty for this product" without nulls everywhere is the wrong model. Use optionality, not presence-of-null.

3.1 Product-by-Product Structural Mapping

Product	Coverage	Rider(s) typical	Investment Strategy	Financial Activity pattern	Distinguishing modelling need
Term Life	1 base, level face, fixed term	Waiver of premium, ADB, child term, conversion privilege	None	Level premium in; death benefit out; no value	Conversion privilege = a dated *option*, must be modelled with expiry
Whole Life	1 base, level face	Paid-up additions, waiver, term blend, LTC accel	Guaranteed cash value table + **dividend** (par)	Premium in, dividend credit, loan, surrender	Dividends & PUA create a *second growing coverage* — not just a credit

Universal Life	1 base, DB Option A/B/C	Waiver, ADB, chronic illness, no-lapse guarantee	Declared interest rate on account value	Flexible premium in; **monthly deduction** out (COI + expense + rider)	Monthly deduction is a recurring *financial activity*, not a static charge
Indexed UL	as UL	as UL + index-lock features	**Index strategies + segments** + fixed bucket	as UL, plus segment-maturity index credit	Segment structure (§3.2) + participating loan interaction
Variable UL	as UL	as UL	**Separate-account subaccounts (units)**	as UL, plus daily unit revaluation, transfers	Unit accounting; SEC/FINRA registration; prospectus delivery
Fixed Annuity (deferred)	nominal / none	Nursing home waiver, terminal illness waiver	Single declared rate, periodically reset	Premium in; interest credit; withdrawal; surrender ± SC/MVA	Rate-reset schedule and renewal-rate history
MYGA	none	rarely any	Single rate **guaranteed for a stated term** (3/5/7/10)	as above, plus **end-of-term window** decision	Term-end election window (renew / annuitize / 1035 / surrender) with a hard clock
FIA	none	**GLWB**, enhanced death benefit, LTC doubler	**Multiple index strategies + fixed bucket, segmented**	Premium in; segment-maturity index credit; rider charge; withdrawal	Segments; cap/par/spread per segment; **benefit base ≠ account value**
RILA	none	GLWB (less common), enhanced DB	Index strategies with **buffer or floor**; value **can decline**	as FIA, but with **negative** index credits possible	Interim/fair value between segment dates; SEC registration; prospectus
Variable Annuity	none	GLWB/GMIB/GMAB/GMDB	**Separate-account subaccounts** + fixed account	Premium in; daily unit revaluation; M&E and rider charges; transfers	Unit accounting + rider benefit bases + share class fee schedule
SPIA	none	Period certain, cash refund, COLA	None (reserve held by carrier)	**Single premium in; scheduled payments out**	The contract *is* a payment schedule; exclusion ratio per payment
DIA / QLAC	none	Death benefit before income start, COLA	None	Premium(s) in; **deferral period**; payments from a future date	Income start date as a first-class, changeable-within-limits attribute; QLAC caps

3.2 Worked Conceptual Example — Fixed Indexed Annuity with GLWB

[Illustrative — this guide's construction, not an ACORD artifact]

```
CONTRACT  A-77120
├── product: FIA, plan code FIA-10-GLWB
├── issueDate 2024-03-15 | issueState TX | taxQualification: NON_QUALIFIED
├── surrenderChargeSchedule: 10yr [9,9,8,7,6,5,4,3,2,1] % | MVA: true
│
├── PARTY ROLES
│     ├── Owner      → Party P-0001 (Person, Margaret Chen)
│     ├── Annuitant  → Party P-0001  ← same party, different role
│     ├── Beneficiary(primary, 100%)  → Party P-0002 (Person, spouse)
│     └── Producer   → Party P-9001 (agency, split 100%)
│
├── INVESTMENT STRATEGIES  (allocations must sum to 100%)
│     ├── S&P 500 Annual PTP w/ Cap        60%
│     │     └── SEGMENT SEG-001: start 2024-03-15, term 1yr,
│     │           startIndexValue 5117.09, cap 9.25%,
│     │           maturity 2025-03-15, principal $60,000
│     ├── Volatility-Controlled Index PTP w/ Par Rate   30%
│     │     └── SEGMENT SEG-002: term 2yr, parRate 115%, spread 0%
│     └── Fixed Declared Account            10%
│           └── rate 3.10%, guaranteed 1yr
│
├── RIDER  GLWB-2024
│     ├── benefitBase       $100,000  ← NOT the account value
│     ├── rollup            7.0% simple, 10yr or until first withdrawal
│     ├── stepUp            annual, to AV if AV > benefitBase
│     ├── charge            1.25% of benefitBase, deducted quarterly from AV
│     ├── withdrawalRate    5.0% single life at age 65
│     └── excessWithdrawalTreatment: pro-rata reduction of benefitBase
│
└── FINANCIAL ACTIVITIES  (append-only)
      ├── FA-001  PREMIUM        2024-03-15  +$100,000  basis $100,000
      ├── FA-002  RIDER_CHARGE   2024-06-15  -$312.50   (1.25%/4 × 100,000)
```

```
      ├── FA-003  RIDER_CHARGE   2024-09-15  -$312.50
      ├── FA-004  INDEX_CREDIT   2025-03-15  +$5,550    SEG-001 matured,
      │                                       index +11.2%, capped at 9.25%
      └── FA-005  BENEFIT_BASE_STEPUP 2025-03-15  benefitBase → $107,000
                                      (rollup applied; AV lower, no step-up)
```

What this example is teaching:

1. One party, two roles (Owner + Annuitant) — the Party/Role separation is not academic.
2. Segments, not a contract-level rate — SEG-001 and SEG-002 mature on different dates with different limiters.
3. Benefit base and account value diverge from day one and follow different rules.
4. The rider charge is deducted from the *account value* but computed on the *benefit base* — a cross-entity formula that a single-value model literally cannot express.
5. Every state change is a dated, append-only financial activity.

3.3 Worked Conceptual Example — SPIA

[Illustrative]

```
CONTRACT  S-44031
├── product: SPIA | taxQualification: NON_QUALIFIED
├── PARTY ROLES: Owner=P-0100, Annuitant=P-0100, Beneficiary=P-0101
├── PAYOUT
│     ├── option: Life with 20-Year Period Certain
│     ├── premium: $250,000  (basis $250,000)
│     ├── monthlyPayment: $1,432.00
│     ├── firstPaymentDate: 2026-04-01
│     ├── certainPeriodEnd: 2046-03-01
│     ├── exclusionRatio: 0.6721   ← derived from expected return
│     └── expectedReturn / basis recovery tracking
└── FINANCIAL ACTIVITIES
      ├── FA-001 PREMIUM   -  $250,000 in
      ├── FA-002 ANNUITY_PAYMENT 2026-04-01 gross $1,432.00
      │           taxable $469.51 | excluded (basis) $962.49
      └── … monthly, until basis fully recovered, after which
            payments become **100% taxable** even though nothing changed
            contractually
```

The lesson: the exclusion ratio expires when basis is exhausted (life-contingent payments continuing beyond life expectancy become fully taxable). This is a *derived, dated state change with no triggering transaction* — the system must detect it, not wait for an event. Most tax-reporting defects in SPIA administration live here.

3.4 Where Products Actually Differ — a Checklist for Requirements Work

When you're handed a new product to model, these eight questions determine ~90% of the build:

1. **Is there a measured life?** (Insured / Annuitant required, or not at all — SPIA vs. MYGA)
2. **Can the value go down from market movement?** (RILA/VA yes; FIA/MYGA no) → drives SEC registration, prospectus delivery, and whether your value engine needs negative crediting.
3. **Is crediting declared, indexed, or unit-based?** → drives the entire Investment Strategy branch.
4. **Are there segments?** → if yes, you need dated tranches, not a contract-level rate.
5. **Is there a guaranteed-income rider with its own benefit base?** → if yes, you need parallel value accounting.
6. **Is there a surrender charge and/or MVA?** → drives the surrender-value formula and the free-withdrawal carve-out.
7. **What's the phase model?** (accumulation-only, payout-only, or both with a transition)
8. **Which tax qualifications is it available under?** → NQ-only, IRA-eligible, or plan-eligible; drives RMD, withholding defaults, and contribution-limit validation.

Chapter 4 — Annuity Transactions

Template for each: ***Business event · Required information · Affected entities · Validations · Message/API · PAS impact · Downstream · Data flow.*** *All API paths and message names are* ***[Illustrative]*** *unless explicitly attributed to ACORD.*

4.0 The Universal Transaction Lifecycle

Every one of the transactions below follows the same eight-stage skeleton. Learn the skeleton; the transactions become variations.

```
1 INTAKE      request arrives (portal / call / form / API / batch file)
2 IDENTIFY    resolve contract + resolve requesting party + verify authority
3 VALIDATE    contract state · party authority · product rules · regulatory rules
4 REQUIRE     open Requirements for anything missing → NIGO if unsatisfied
5 CALCULATE   quote the effect (amount, charges, tax, withholding) — often a
              reversible "pending" state the customer can confirm or cancel
6 COMMIT      write FINANCIAL ACTIVITY(s) → update contract values → change state
7 SETTLE      payments out/in · accounting entries · tax records
8 NOTIFY      correspondence · downstream feeds · events published
```

Where NIGO lives: stages 3–4. **Where money moves:** stages 6–7. **Where reversals originate:** almost always a stage-3 rule that should have fired and didn't.

4.1 NEW BUSINESS

- **Business event.** An application for a new annuity contract is submitted and, once in good order and funded, a contract is issued.
- **Required information.** Applicant/owner identity & tax ID; annuitant; beneficiary designations; product & plan code; initial premium amount and **source of funds** (new money / 1035 / rollover / transfer); tax qualification; allocation instructions; rider elections; producer & split; suitability/Reg BI documentation; replacement disclosure if replacing existing coverage; state of solicitation; signatures & dates.
- **Affected entities.** PARTY (create/match), PARTY ROLE, HOLDING, POLICY/CONTRACT, COVERAGE (if any), RIDER, INVESTMENT STRATEGY + ALLOCATION, PREMIUM, FINANCIAL ACTIVITY, REQUIREMENT, PRODUCER, TAX QUALIFICATION.
- **Validations.** Product available in owner's state on the signed date; issue-age within product limits (age at nearest/last birthday per contract definition); premium within min/max; allocations sum to 100% and only to strategies open on that date; rider combination legal and age-eligible; producer licensed + appointed + (for RILA/VA) securities-registered; suitability completed; replacement forms present; **OFAC/AML screening on the owner and payor**; tax ID present & W-9 certified; for IRA — contribution year and limit; for a rollover — 60-day rule and one-per-year IRA rollover limit.
- **Message/API.** ACORD LA XML new-business application message family (TXLife request carrying Party + Holding + Policy) **[Official ACORD concept — exact transaction-type codes require verification against licensed LA XML v2.49.00 documentation]**. Illustrative REST: `POST /applications` → `POST /applications/{id}/submit` → `POST /policies` on issue.
- **PAS impact.** Creates the contract record; establishes surrender-charge schedule and free-look expiry; opens the first index segments (dated to the funding date, **not** the application date); establishes benefit base for elected riders; starts commission accrual.
- **Downstream.** Commission system; suitability/compliance archive; welcome-kit correspondence; state premium-tax accrual; reinsurance cession (life); DTCC feed to the distributor; data warehouse; producer book-of-business.
- **Data flow.** `Distributor order-entry → ACORD LA XML application → carrier/TPA intake → validation & requirement engine → (NIGO loop with distributor) → funding received → PAS issue → segment creation → welcome kit + DTCC position feed.`

4.2 PREMIUM (INITIAL)

- **Business event.** The money funding a newly issued contract is received and applied.
- **Required information.** Contract ID, amount, receipt date, funds source, payment method, allocation instruction, tax-basis characterization.
- **Affected entities.** PREMIUM, FINANCIAL ACTIVITY, ALLOCATION, SEGMENT (creation), TAX QUALIFICATION (basis), ACCOUNTING ENTRY.
- **Validations.** Amount matches application (or within tolerance); received within the application's good-through window; source of funds matches declared (a 1035 arriving as personal check is a red flag); AML thresholds; **effective date rules** — many carriers date the segment to the next available segment start date, not the receipt date.
- **Message/API.** `POST /policies/{id}/premiums` **[Illustrative]**.
- **PAS impact.** Establishes account value; strikes segment start index values; starts free-look clock (usually from delivery, not receipt — a frequent defect).
- **Downstream.** Cash reconciliation; commission trigger; premium tax; statement; 5498 if an IRA contribution.
- **Data flow.** `Lockbox/ACH/wire → cash app → premium activity → allocation → segment creation → GL posting.`

4.3 ADDITIONAL PREMIUM

- **Business event.** Subsequent money added to an existing flexible-premium contract.
- **Required information.** Contract ID, amount, date, allocation (new instruction or standing), tax year (if IRA), basis characterization.
- **Affected entities.** PREMIUM, FINANCIAL ACTIVITY, **new SEGMENT**, ALLOCATION, TAX QUALIFICATION (basis increment).
- **Validations.** Product accepts additional premium at all (**many MYGAs and most FIAs do not, or only within a window**); within annual/aggregate

maximum; owner age still within limits; not past the additional-premium cutoff; IRA contribution limits and earned-income requirement; for a MEC-sensitive life contract, 7-pay test.

- **PAS impact.** Creates a **new segment with the current rate set** — this is why one contract can hold six segments with six different caps. Some products apply a separate surrender-charge schedule per deposit ("rolling" surrender charges), which multiplies the schedule structure.
- **Downstream.** 5498 reporting; trail commission; statement; segment-maturity calendar.
- **Trap.** Additional premium into an FIA typically **does not** increase a GLWB benefit base at the same rate as the initial premium, and sometimes not at all. Rider-specific.

4.4 WITHDRAWAL (PARTIAL SURRENDER)

- **Business event.** Owner requests a one-time removal of part of the contract's value.
- **Required information.** Contract ID, requesting party & authority, gross-vs-net election, amount or percentage, source (pro-rata across strategies or specified), withholding elections (federal W-4R and state), payment instructions.
- **Affected entities.** FINANCIAL ACTIVITY (parent + children), ALLOCATION/SEGMENT (redemption), ACCOUNT VALUE, SURRENDER VALUE, **RIDER benefit base (decrement!)**, TAX QUALIFICATION (basis), PAYMENT ×3, TAX REPORTING (1099-R), REQUIREMENT.
- **Validations.** Contract in force and in accumulation phase; not in free-look; requester is Owner (or authorized POA/trustee — **verify authority, not just identity**); **collateral assignment does not block it**; amount ≤ available surrender value; minimum-remaining-value rule; free-withdrawal amount available this contract year; surrender charge & MVA computed on the excess; **GLWB excess-withdrawal test** — is this within the allowed annual amount, or does it trigger a pro-rata benefit-base reduction?; age <59½ → 10% additional tax warning (§72(q)/§72(t)) — disclose, don't withhold it; withholding election present or default applied (federal default currently 10% for non-periodic NQ distributions unless elected otherwise — *verify current-year rate*).
- **Message/API.** `POST /withdrawals` **[Illustrative]**, with `Idempotency-Key`, returning a quote first (`POST /withdrawals/quote`) then a commit.
- **PAS impact.** Reduces account value; applies SC/MVA; may break segments mid-term (**interim value calculation** — a major RILA complexity); decrements benefit base; consumes free-withdrawal allowance.
- **Downstream.** Payment issuance; fed/state withholding remittance; 1099-R accrual; confirmation letter; **GLWB impact disclosure**; data warehouse; distributor feed.
- **Data flow.** `Request → authority check → quote (SC/MVA/tax/rider impact) → owner confirms → FA-parent + FA-children (redemption, SC, MVA, withholding) → 3 payments → GL → 1099-R accrual → confirm letter.`
- **The one that generates complaints.** A $12,000 withdrawal on a contract with a $10,000 free amount and a 5% GLWB allowance: the owner sees a surrender charge *and* a permanent income reduction they didn't expect. Your quote step must show both, in writing, before commit. This is as much a UX/architecture requirement as a compliance one.

4.5 SYSTEMATIC WITHDRAWAL (SWP)

- **Business event.** A recurring scheduled withdrawal at a set frequency/amount.
- **Required information.** All withdrawal fields **plus** frequency, start date, end date/perpetual, amount type (fixed $, % of AV, interest-only, GLWB-income), recalculation rule, standing withholding election.
- **Affected entities.** A **SCHEDULE / STANDING INSTRUCTION** entity (distinct from the activities it spawns) + one FINANCIAL ACTIVITY per occurrence.
- **Validations.** Per-occurrence re-validation — a SWP set up when the contract had value must **stop** when value is exhausted or the contract lapses; annual free-amount tracking across occurrences; if funding a GLWB income stream, amounts must track the allowed rate as it changes (age band, step-up).
- **PAS impact.** Batch-driven; each cycle produces a full withdrawal activity.
- **Trap.** SWPs and RMDs are frequently the *same money* but modelled as different instructions — leading to double distribution. Model an explicit relationship: "this SWP satisfies the RMD obligation."

4.6 FULL SURRENDER

- **Business event.** Owner terminates the contract and takes the full surrender value.
- **Required information.** Contract ID, authority, signature (often Medallion-guaranteed above a threshold), withholding elections, payment instructions, spousal consent (qualified plan contracts).
- **Affected entities.** POLICY (status → Surrendered), all SEGMENTS (liquidated), all RIDERS (terminated), FINANCIAL ACTIVITY, PAYMENT, TAX REPORTING, LOAN (offset if any).
- **Validations.** Authority; assignment/lien check; outstanding loan offset; **free-look window — if inside it, this is a free-look cancellation, not a surrender, with completely different tax treatment**; guaranteed-minimum-value floor test; final rider charge proration.
- **PAS impact.** Terminal state. Riders and benefit bases die with it — an owner who surrenders a contract with a $180,000 benefit base and a $120,000 account value receives **$120,000 less charges**. The disclosure obligation here is significant.

- **Downstream.** 1099-R (full gain reportable); commission chargeback if within the chargeback period; contract-termination correspondence; escheatment de-registration; conservation/retention workflow *before* processing.
- **Data flow.** `Request → conservation attempt → authority + requirements → surrender quote → commit → liquidate segments → SC/MVA → withholding → payment → status change → 1099-R.`

4.7 PARTIAL SURRENDER

Functionally 4.4; the term is used where the contract language distinguishes a "partial surrender" (reduces the contract permanently, may reduce the death benefit proportionally) from a "withdrawal" (against a free amount). **Model them as one activity type with a subtype**, and let product configuration decide the death-benefit treatment (dollar-for-dollar vs. pro-rata reduction) — that single configuration choice changes claim amounts years later.

4.8 RMD (REQUIRED MINIMUM DISTRIBUTION)

- **Business event.** A distribution required by the Code from a qualified contract once the owner reaches the applicable age, or from an inherited contract under the applicable regime.
- **Required information.** Prior-year 12/31 fair market value (**including the actuarial value of certain riders — a genuinely hard calculation for FIA/VA with living benefits**), owner DOB, applicable table (Uniform Lifetime / Joint & Last Survivor if spouse >10 yrs younger and sole beneficiary / Single Life for inherited), aggregation election, distribution instruction, withholding.
- **Affected entities.** TAX QUALIFICATION, an **RMD OBLIGATION** entity (calculated annually), FINANCIAL ACTIVITY, TAX REPORTING, SCHEDULE (if satisfied via SWP).
- **Validations.** Is the contract RMD-subject? (Traditional IRA/SEP/SIMPLE/qualified plans yes; **Roth IRA no during owner's lifetime**; inherited Roth yes under the beneficiary regime); applicable age per current law (SECURE 2.0 moved this — *verify the current-year age and the excise-tax rate rather than hard-coding*); has the obligation already been satisfied by another distribution this year?; is the owner aggregating across IRAs elsewhere (carrier can't know — must be an owner election with an attestation).
- **PAS impact.** Annual calculation cycle, typically Q4-heavy; produces a distribution or an unsatisfied-obligation flag.
- **Downstream.** 1099-R; RMD notice correspondence (required annually); 5498 FMV reporting; **unsatisfied-RMD reporting and follow-up**.
- **Architecture note.** RMD is the clearest case for **deterministic rules, never an LLM** (see Chapter 14). The calculation is legally prescribed; a probabilistic system is categorically the wrong tool.

4.9 ROTH IRA CONVERSION

- **Business event.** Assets in a Traditional/SEP/SIMPLE IRA are converted to Roth status, creating a taxable event in the conversion year.
- **Required information.** Source contract, destination (same-carrier Roth contract, new Roth contract, or external Roth), amount (full or partial), **fair market value of the converted amount** (for an annuity, potentially including rider actuarial value), withholding election (converting owners usually elect 0% and pay from outside funds — the system must allow 0%), basis in the traditional IRA (Form 8606 interaction), conversion date.
- **Affected entities.** Source CONTRACT (distribution), destination CONTRACT (contribution), **TAX QUALIFICATION on both**, FINANCIAL ACTIVITY ×2 (out and in) linked by a conversion reference, TAX REPORTING (**1099-R on the source with the conversion distribution code, 5498 on the destination**), Roth **5-year clock** initiation, REQUIREMENT.
- **Validations.** Source is convertible (SIMPLE IRAs have a 2-year rule from first contribution); **recharacterization of conversions is no longer permitted** — this is irreversible, so the confirmation step is a real control, not a formality; destination Roth exists or is created; owner is the same person on both (**you cannot convert into someone else's Roth**); FMV determination methodology documented; withholding election captured explicitly (defaulting to 10% on a conversion actively harms the customer by reducing the converted amount and creating a taxable+penalized distribution of the withheld portion).
- **Message/API.** `POST /roth-conversions` **[Illustrative]** — body: sourcePolicyId, destinationPolicyId|newContractRequest, conversionType (FULL|PARTIAL), amount, fmvBasis, withholding{federalPct, statePct}, effectiveDate, requestedBy.
- **PAS impact.** If in-place (same carrier), the tax-qualification attribute changes and a paired distribution/contribution is recorded. **The contract number may or may not change — a significant design decision.** Changing it preserves clean tax lineage but breaks customer familiarity and every downstream reference; keeping it requires the qualification attribute to be *dated and versioned*, not overwritten.
- **Downstream.** 1099-R (source), 5498 (destination), conversion confirmation, Roth 5-year-clock record, cost-basis tracking, state tax reporting, data warehouse.
- **Full end-to-end trace: see Chapter 15.**

4.10 IRA ROLLOVER

- **Business event.** Qualified assets move from one qualified vehicle to another without becoming taxable — either **direct (trustee-to-trustee)** or **indirect (60-day)**.
- **Required information.** Source institution & account, destination contract, amount, rollover type (direct/indirect), distributing plan type, whether it includes after-tax basis, date distributed (for the 60-day clock), owner attestation for indirect rollovers.

- **Affected entities.** PREMIUM (type = ROLLOVER_IN), TAX QUALIFICATION (type, basis = generally $0 unless after-tax), FINANCIAL ACTIVITY, TAX REPORTING (5498 code differs by rollover type), REQUIREMENT (transfer paperwork, acceptance letter).
- **Validations. Direct vs. indirect matters enormously** — a direct trustee-to-trustee transfer between IRAs is not even a reportable rollover in the same sense and is not subject to the one-per-12-month limit; an indirect 60-day rollover **is** limited to one per 12 months across all the owner's IRAs (the carrier cannot verify this — attestation required) and must land within 60 days; plan-to-IRA rollovers of pre-tax money to a **Roth** IRA are a conversion, not a rollover (different reporting); RMDs are **not** rollover-eligible.
- **Message/API.** `POST /policies/{id}/rollovers` **[Illustrative]**.
- **PAS impact.** Establishes qualification, basis, and — for indirect rollovers — a 60-day compliance record.
- **Downstream.** 5498; acceptance letter to the ceding institution; ACATS/DTCC where applicable.

4.11 1035 EXCHANGE

- **Business event.** A tax-free exchange of one annuity (or life policy) for another under IRC §1035, **carrying basis over**.
- **Required information.** Ceding carrier & contract, absolute assignment form, surrender/transfer authorization, **cost basis reported by the ceding carrier**, amount, partial-vs-full, replacement disclosure, new contract application.
- **Affected entities.** Both contracts; PREMIUM (type = EXCHANGE_1035_IN with carried basis); TAX QUALIFICATION (basis); FINANCIAL ACTIVITY on both sides; REQUIREMENT (assignment, basis statement); PRODUCER (replacement compliance).
- **Validations.** Exchange direction is permitted (**annuity → life is NOT a valid 1035; life → annuity IS**); NQ↔NQ and qualified↔qualified only — you cannot 1035 across tax qualification; ceding basis received (**if basis is not provided, the receiving carrier must not guess — it becomes a customer tax problem and a carrier liability**); state replacement regulations and, for partial exchanges, the 180-day rule (a withdrawal from either contract within 180 days can retroactively taint the exchange); suitability on replacement is heavily scrutinized.
- **Message/API.** `POST /1035-exchanges` **[Illustrative]** — often actually executed over DTCC/industry transfer rails rather than a bilateral API.
- **PAS impact.** New contract funded with carried-over basis; new surrender-charge schedule begins (**the customer restarts the surrender clock — a core suitability concern**).
- **Downstream.** No 1099-R taxable amount but **reportable with the exchange distribution code**; replacement register; ceding-carrier correspondence; conservation attempt on the ceding side.
- **The basis problem.** Basis carryover is the #1 data-quality failure in exchanges. Architect it as a **required, sourced, auditable field with the ceding carrier's statement retained** — not an optional numeric.

4.12 TRANSFER

- **Business event.** Movement of value between investment strategies **within** a contract (distinct from an exchange between contracts).
- **Required information.** Contract, source strategy/strategies, destination, amount or %, effective date.
- **Affected entities.** ALLOCATION, SEGMENT, FINANCIAL ACTIVITY (transfer-out + transfer-in pair, net zero to account value).
- **Validations.** Transfers permitted at this time (**FIA/RILA index strategies typically only allow transfers at segment maturity — this is the single most common customer misunderstanding**); VA transfer limits and market-timing/excessive-trading policies; minimum balances; destination strategy open.
- **PAS impact.** VA: units sold and bought at the next-computed AUV. FIA/RILA: old segment closes, new segment opens with the **current** rate set.
- **Downstream.** Confirmation; no tax consequence (inside a tax wrapper); fund-company trade files for VA.

4.13 ANNUITIZATION

- **Business event.** Conversion of accumulated value into a payment stream — the phase transition.
- **Required information.** Contract, election of payout option (life only / life w/ period certain / joint & survivor / period certain / cash refund / installment refund), payment frequency, start date, joint annuitant if applicable, payout rate basis (**current vs. guaranteed annuity purchase rates — take the more favorable**), withholding election.
- **Affected entities.** POLICY (phase → Payout), PAYOUT SCHEDULE (new entity), all SEGMENTS (liquidated), RIDERS (mostly terminate — but a GLWB may instead continue as a withdrawal stream rather than true annuitization), TAX QUALIFICATION (exclusion ratio for NQ), FINANCIAL ACTIVITY, PAYMENT (recurring).
- **Validations.** Contract eligible (minimum deferral period met; not past maturity); annuitant alive and age-eligible; option available for this qualification (a period-certain longer than life expectancy fails qualified-plan rules); **irreversibility disclosure** — annuitization is generally permanent and eliminates liquidity; surrender charges typically waived on annuitization (product-specific); exclusion-ratio computation for NQ.
- **Message/API.** `POST /policies/{id}/annuitizations` **[Illustrative]**.
- **PAS impact.** The most significant state change in the model. Account value ceases to be the governing measure; a schedule takes over. Systems that model this as "just another transaction" struggle for years.

- **Downstream.** Recurring payment setup; 1099-R with periodic-payment coding; reserve reclassification (actuarial); statement format change; **survivor verification / proof-of-life process begins**.

4.14 DEATH CLAIM

- **Business event.** The death of an Owner or Annuitant triggers contractual obligations.
- **Required information.** Certified death certificate; decedent identity & role; date of death; claimant identity, relationship, and tax ID; beneficiary election (lump sum / stretch / 5-year / 10-year / spousal continuation / annuitized); withholding; payment instructions.
- **Affected entities.** PARTY (date of death), POLICY (status), all RIDERS (enhanced death benefit evaluation), FINANCIAL ACTIVITY, PAYMENT (per beneficiary), TAX QUALIFICATION (**inherited regime — a new qualification state, not a deletion**), REQUIREMENT (claim forms per beneficiary), BENEFICIARY roles.
- **Validations. Which death matters?** Owner-driven vs. annuitant-driven death benefit differs by contract and is a frequent defect source — on an owner-driven contract with a different annuitant, the *owner's* death triggers §72(s) distribution rules; beneficiary designations current and summing to 100%; deceased beneficiary → contingent or per-stirpes resolution; **surrender charges usually waived on death** (product-specific); enhanced/ratchet death benefit calculation as of date of death vs. date of claim (contract language governs — and the difference can be tens of thousands of dollars in a volatile market); spousal continuation option available and elected; **SECURE Act beneficiary regime** — eligible designated beneficiary vs. designated beneficiary vs. non-designated, driving lifetime stretch vs. 10-year vs. 5-year; slayer statutes; minor beneficiaries requiring a guardian/UTMA custodian.
- **Message/API.** `POST /claims` + `POST /claims/{id}/beneficiary-elections` **[Illustrative]**.
- **PAS impact.** Value frozen or continued per contract; per-beneficiary sub-accounting (**one contract may split into N beneficiary interests — model this as N derived holdings, not as N payments off one record**).
- **Downstream.** 1099-R to each beneficiary; escheatment clock if a beneficiary can't be located; **death master file matching obligations**; commission chargeback rules; correspondence to each claimant.
- **This is the highest-emotion, highest-litigation transaction in the book.** Requirement tracking and clear per-beneficiary status are worth more architectural investment than almost anything else.

4.15 BENEFICIARY CHANGE

- **Business event.** The owner changes who receives death proceeds.
- **Required information.** Contract, requester & authority, complete new designation set (primary and contingent, with percentages and relationships), signature, date, spousal consent where required.
- **Affected entities.** PARTY (create/match new beneficiaries), PARTY ROLE (**terminate old roles with an end date; create new — never overwrite**), REQUIREMENT.
- **Validations.** Requester is the Owner; contract not annuitized (usually locked or restricted); **irrevocable beneficiary present → requires that beneficiary's consent**; percentages total 100% per class; spousal consent for qualified plan contracts (ERISA) and community-property states; minors flagged; **a beneficiary designation naming an estate has major tax consequences (loses designated-beneficiary status) — disclose**.
- **PAS impact.** Role versioning. **The dated history is the entire point** — a death claim two years later must be adjudicated against the designation in force at date of death, which means you must be able to reconstruct it. Overwriting beneficiaries is a litigation-generating design defect.
- **Downstream.** Confirmation letter to owner (and to a removed irrevocable beneficiary); audit log.

4.16 OWNERSHIP CHANGE

- **Business event.** Transfer of contract ownership to another party.
- **Required information.** Contract, current & new owner, ownership-change form, new owner's tax ID & W-9, reason, effective date, consent of assignee/irrevocable beneficiary.
- **Affected entities.** PARTY ROLE (owner terminate/create), TAX QUALIFICATION (**potentially a taxable event!**), PARTY, REQUIREMENT.
- **Validations. Tax consequences must be evaluated, not assumed** — gratuitous transfer of a NQ deferred annuity to a non-spouse generally triggers recognition of gain under §72(e)(4)(C); transfer to a spouse or incident to divorce generally does not; **IRAs cannot change ownership** except by divorce decree or death (an "ownership change" request on an IRA is almost always a misfiled request — reject and redirect); transfer to a non-natural owner triggers §72(u); assignment restrictions; new owner AML/OFAC screening.
- **PAS impact.** Role change plus a potential taxable-event financial activity with **no cash movement** — a case where a 1099-R is issued and no money left the contract. Systems that couple tax reporting to payment issuance fail here.
- **Downstream.** 1099-R if taxable; new-owner welcome correspondence; producer servicing rights.

4.17 ADDRESS CHANGE

- **Business event.** A party's address is updated.

- **Required information.** Party, address type, new address, effective date, source of request, authority.
- **Affected entities.** ADDRESS (**end-date old, create new — retain history**), PARTY, potentially TAX QUALIFICATION (state withholding).
- **Validations.** Requester authority (an address change is a classic **account-takeover vector** — this is a fraud control, not an administrative task); address standardization/validation; **state change → recalculate state withholding defaults and premium-tax jurisdiction**; foreign address → W-8BEN requirement and possible 30% NRA withholding; APO/FPO and PO-box handling; **a change immediately followed by a withdrawal request should trigger enhanced verification**.
- **Downstream.** Correspondence redirect; state-withholding recalculation for scheduled payments; return-mail clearance; escheatment clock reset.

4.18 RIDER ADDITION

- **Business event.** A rider is added to an in-force contract.
- **Required information.** Contract, rider code, effective date, election form, evidence of insurability (if required), new charge acknowledgment.
- **Affected entities.** RIDER, BENEFIT, FINANCIAL ACTIVITY (charge stream begins), POLICY, REQUIREMENT.
- **Validations.** Rider available for this product/plan/issue-state on this date; **most GLWB riders can only be elected at issue** — post-issue addition is the exception, not the rule; age eligibility at addition; anti-selection windows; benefit base at addition (usually current account value, **not** original premium); charge disclosure & signed acknowledgment; state approval of the rider form in the owner's state.
- **PAS impact.** New charge stream; new benefit-base accounting; possible re-rating.
- **Downstream.** Amended contract pages/endorsement issuance; statement change; commission (some riders pay).

4.19 RIDER REMOVAL / TERMINATION

- **Business event.** A rider is dropped, expires, or terminates by its own terms.
- **Required information.** Contract, rider, reason (owner election / automatic termination / benefit exhaustion / annuitization), effective date.
- **Affected entities.** RIDER (terminate with reason), FINANCIAL ACTIVITY (final prorated charge), BENEFIT BASE (**extinguished — value is lost**).
- **Validations. Is removal permitted at all?** Many GLWBs are irrevocable once elected; some allow removal only after a waiting period or on a contract anniversary; automatic terminations (owner change, annuitization, full surrender, benefit-base exhaustion) must fire without a request; **removal is usually irreversible** — a strong confirmation control is required.
- **Downstream.** Charge cessation; disclosure of forfeited benefit base; endorsement; statement change.
- **Trap.** Automatic rider termination on ownership change catches people out badly — a routine ownership change silently destroys a $200,000 benefit base. Make that consequence explicit in the ownership-change quote step (§4.16), not discoverable afterward.

4.20 ALLOCATION CHANGE

- **Business event.** The owner changes how *future* money is directed (as distinct from a transfer of existing value).
- **Required information.** Contract, new allocation set with percentages, effective date, scope (future premium only vs. also rebalance existing).
- **Affected entities.** ALLOCATION (versioned), STANDING INSTRUCTION.
- **Validations.** Sums to 100%; only open strategies; minimum-percentage increments (often 5% or 10%); **timing** — for FIA/RILA, a change to existing money is only effective at segment maturity, while a change to future premium is effective immediately. **These are two different transactions that customers describe with the same words.** Disambiguate at intake.
- **Downstream.** Confirmation; segment-maturity processing picks up the new instruction.

4.21 REBALANCING

- **Business event.** Existing value is redistributed back to target percentages, one-off or on an automatic schedule.
- **Required information.** Contract, target allocation, frequency (quarterly/semi-annual/annual), start date, tolerance band.
- **Affected entities.** STANDING INSTRUCTION, ALLOCATION, FINANCIAL ACTIVITY (transfer pairs), SEGMENT.
- **Validations.** Available for this product (**common on VA, restricted or unavailable on FIA/RILA because of segment mechanics**); interaction with living-benefit riders (**many GLWB riders require an approved model portfolio and rebalancing is mandatory, not optional — deviating can void the guarantee**); transfer limits.
- **PAS impact.** Batch-driven transfer pairs, net zero.
- **Downstream.** Confirmation; fund trade files; rider-compliance monitoring.

4.22 INDEX STRATEGY ALLOCATION (SEGMENT MATURITY / RENEWAL)

- **Business event.** A segment reaches term end; value plus credit is determined and reallocated per standing or new instruction into a new segment at the **new** rate set.

- **Required information.** Contract, maturing segment(s), ending index value, applicable cap/par/spread/buffer, standing renewal instruction or a new election received in the window.
- **Affected entities.** SEGMENT (close + open), FINANCIAL ACTIVITY (INDEX_CREDIT — **which can be negative on a RILA**), ACCOUNT VALUE, ALLOCATION, RIDER (step-up evaluation).
- **Validations.** Correct index values on the correct dates (**including market-holiday and index-discontinuation handling — every FIA/RILA contract has substitution language, and it has been invoked in practice**); crediting method applied correctly per strategy (annual PTP, monthly sum with a monthly cap, monthly average, performance trigger, high-water mark); cap/par/spread as of segment start, **not** current; floor of 0% on FIA (never negative), buffer/floor arithmetic on RILA; renewal window election deadline; renewal rates communicated to the owner in advance where required.
- **PAS impact.** The highest-volume calculation event in FIA/RILA administration and the highest-risk for defect. **This is where your reference-architecture entity design earns or loses its keep.**
- **Downstream.** Credit confirmation/renewal-rate notice; statement; benefit-base step-up evaluation; data warehouse.

4.23 RENEWAL (RATE RENEWAL — MYGA/FIXED)

- **Business event.** A guaranteed-rate period ends and a new rate applies for the next period.
- **Required information.** Contract, expiring rate & period, new declared rate, renewal window dates, owner election.
- **Affected entities.** INVESTMENT STRATEGY (rate versioned), FINANCIAL ACTIVITY (interest credits at the new rate), POLICY (possibly a new surrender-charge period).
- **Validations.** New rate ≥ contractual guaranteed minimum; **advance notice period met** (state-regulated); renewal window election options presented (renew / surrender penalty-free / 1035 / annuitize); **whether a new surrender-charge period attaches on renewal is a major consumer-protection issue and must match the filed form**.
- **Downstream.** Renewal notice (a hard regulatory deadline); election processing; retention workflow.

4.24 MATURITY

- **Business event.** The contract reaches its stated maturity/annuity-commencement date.
- **Required information.** Contract, maturity date, owner election (annuitize / extend / surrender / default option).
- **Affected entities.** POLICY (status/phase), PAYOUT SCHEDULE (if annuitized), FINANCIAL ACTIVITY.
- **Validations.** Advance notice sent (typically 30–90 days); default option applied if no election (**usually automatic annuitization under a specified option — and applying a default that the owner didn't understand is a recurring complaint source**); maturity extension permitted and within limits; **qualified contracts cannot extend past applicable RMD rules**.
- **Downstream.** Maturity notice; election processing; annuitization or continuation.

4.25 LAPSE

- **Business event.** The contract terminates for insufficient value/premium (primarily a UL/life concern; annuities in accumulation generally cannot lapse, but a UL-chassis contract or a contract with charges exceeding value can).
- **Required information.** Contract, value, charges due, grace period start, notices sent, no-lapse-guarantee status.
- **Affected entities.** POLICY (status → Lapsed), LOAN (offset), FINANCIAL ACTIVITY, TAX REPORTING, RIDER (terminated).
- **Validations.** Grace period observed (state-mandated, typically 31 days); lapse notice delivered per state rules; **secondary/no-lapse guarantee test before lapsing** — the account value may be zero while the guarantee is fully intact, and lapsing such a policy is a serious error; **third-party/designated-notice recipient** requirements (many states require notice to a designated person, particularly for older insureds); **loan-related phantom income** — a lapse with an outstanding loan produces taxable gain with no cash to the owner.
- **Downstream.** 1099-R if gain; lapse notice; reinstatement offer window; conservation.

4.26 REINSTATEMENT

- **Business event.** A lapsed contract is restored.
- **Required information.** Contract, reinstatement application, evidence of insurability (life), back premium + interest, reinstatement window compliance.
- **Affected entities.** POLICY (status restored), COVERAGE/RIDER (restored or not — **riders often do not come back**), FINANCIAL ACTIVITY, UNDERWRITING, LOAN.
- **Validations.** Within the reinstatement window (commonly 3–5 years, contract-specific); evidence of insurability satisfactory; all back charges paid with interest; **contestability and suicide clauses may restart** — a material consumer disclosure; original issue-age rates vs. attained-age.
- **Downstream.** Reinstatement confirmation; restored statement; commission treatment.

4.27 FREE-LOOK CANCELLATION

- **Business event.** The owner cancels within the statutory free-look/right-to-examine period.
- **Required information.** Contract, delivery date, cancellation request date, refund basis.
- **Affected entities.** POLICY (status → Free-Look Cancelled — **not "Surrendered"**), FINANCIAL ACTIVITY, PAYMENT, PREMIUM (reversal), TAX REPORTING (**generally none, or return-of-premium coding**).
- **Validations. Within the window** — length varies by state and by whether the contract is a replacement (often extended to 30 days for replacements or for senior buyers); refund basis per state — **return of premium** in most states vs. **return of account value** in some (a critical distinction on a RILA where the value may have fallen); delivery-date evidence is the linchpin and is frequently the weakest data in the whole file.
- **PAS impact.** Effectively unwinds issuance. Should reverse, not delete — the contract existed.
- **Downstream.** Full commission chargeback; premium-tax reversal; 1035 unwind coordination if funded by exchange (**genuinely messy — the ceding contract may already be terminated**); regulatory reporting.

4.28 TAX WITHHOLDING

- **Business event.** Federal and/or state tax is withheld from a distribution and remitted.
- **Required information.** Distribution amount & taxable portion, payee, W-4R / state election on file, residency state, foreign status (W-8BEN), election effective date.
- **Affected entities.** FINANCIAL ACTIVITY (child activities), PAYMENT (to IRS and to state), TAX REPORTING, PARTY (elections & residency).
- **Validations.** Correct **default** applied when no election exists (defaults differ by distribution type — periodic vs. non-periodic vs. eligible rollover distribution, the last of which carries **mandatory 20%** federal withholding when paid to the participant); **mandatory state withholding states** (several states require withholding whenever federal is withheld, or set their own mandatory rules — must be table-driven and maintained, never hard-coded); no valid TIN → backup withholding; non-resident alien → treaty rate and W-8BEN validity period; election cannot reduce mandatory withholding below the floor.
- **Downstream.** IRS/state remittance on the applicable deposit schedule; 945 reporting; 1099-R boxes; year-end reconciliation.
- **Architecture note.** Withholding rules change annually and vary by 50+ jurisdictions. This is **configuration and reference data with effective dating**, not code. Any design that embeds rates in program logic will be wrong every January.

4.29 TAX REPORTING

- **Business event.** Year-end (and event-driven) reporting of contract activity to the IRS, states, and contract owners.
- **Required information.** All reportable activities for the year, payee TINs, distribution codes, taxable amounts, basis, FMV as of 12/31, corrections.
- **Affected entities.** TAX REPORTING RECORD (derived from FINANCIAL ACTIVITY), PARTY, TAX QUALIFICATION.
- **Key forms. 1099-R** (distributions — the distribution code in Box 7 is the entire tax story and the most defect-prone field in L&A); **5498** (IRA contributions, rollovers, conversions, FMV, RMD indicator); **1099-INT/1099-MISC** (interest on delayed claims); **W-2** in limited employer contexts; **1042-S** for non-resident aliens; state equivalents.
- **Validations.** TIN matching against IRS records (mismatch → B-notice → backup withholding); distribution-code correctness by transaction type, age, and qualification; **corrected-return handling with a full audit trail**; filing deadlines; aggregation rules per payee per contract.
- **Downstream.** IRS transmittal; state filings; owner statements; corrected forms; B-notice workflow.
- **Design rule.** Tax reporting is a **projection over the append-only financial activity ledger**, computed and stored as its own dated record — never a live query over mutable balances. If you can't reproduce last year's 1099-R exactly, the design is wrong.

4.30 Cross-Transaction Design Principles

1. **Quote before commit.** Every value-affecting transaction should have a reversible calculation step showing charges, tax, and rider impact before anything is written. This is a compliance control and the single best defect-reducer.
2. **Authority ≠ identity.** Verify *who is entitled to act*, not merely who is calling. POAs, trustees, irrevocable beneficiaries, and assignees all constrain owner rights.
3. **Append-only financial ledger.** No in-place updates. Reversal + correction, always.
4. **Version roles and allocations with effective dates.** Beneficiaries and allocations must be reconstructable as of any past date.
5. **Effective date ≠ processing date ≠ receipt date.** Store all three. Most backdating and reconciliation defects trace to collapsing them.
6. **Tax reporting is decoupled from payment.** Taxable events with no cash movement exist (ownership change, lapse with loan). Payments with no tax exist (return of basis).
7. **Requirements are first-class.** NIGO is a state of the transaction, not an email thread.
8. **Idempotency on every write API.** Retries at the distributor edge are guaranteed to happen; duplicate withdrawals are unrecoverable customer harm.

Chapter 5 — Tax Qualification

5.1 The Central Thesis: Four Orthogonal Dimensions

The most consequential modelling error in Life & Annuity systems is **conflating product type, contract type, tax qualification, and transaction type into a single code**. They are four independent dimensions.

Dimension	Answers	Owned by	Changes over time?	Example values
Product type	*What was designed and filed?*	Product management / actuarial	No — fixed at issue	FIA, RILA, MYGA, VA, SPIA, IUL
Contract type	*What legal form was issued?*	Legal / compliance	No — fixed at issue	Individual deferred annuity, group annuity certificate, individual life policy
Tax qualification	*Under what tax regime is it held?*	Tax / operations	**Yes** — conversion, inheritance, divorce	NQ, Traditional IRA, Roth IRA, SEP, SIMPLE, 401(k), 403(b), 457, Inherited IRA
Transaction type	*What is happening right now?*	Operations	N/A — per event	Withdrawal, conversion, rollover, annuitization

Why they must not be conflated — the four-way proof:

- The **same product** (one FIA) is sold as NQ, Traditional IRA, and Roth IRA. → product ≠ qualification.
- The **same qualification** (Traditional IRA) holds a MYGA, an FIA, and a VA. → qualification ≠ product.
- The **same transaction** (a $10,000 withdrawal) is LIFO-taxable on an NQ annuity, fully taxable on a Traditional IRA, and potentially tax-free on a Roth. → transaction ≠ qualification.
- **Qualification changes; product does not.** A Roth conversion changes the qualification of a contract whose product is unchanged. An inheritance changes it again.

The legacy anti-pattern. A `PLAN_CODE` of `FIA10Q` encoding "FIA, 10-year surrender, Qualified" seems efficient until: (a) a Roth conversion makes the code wrong but the product right; (b) you need to report AUM by product across qualifications; (c) a new qualification type requires a new plan code for every product — combinatorial explosion; (d) nobody can tell whether "Q" means Traditional IRA or a 403(b), because the encoding lost the distinction the day it was created. **Decompose it. Every time.**

5.2 The Qualification Taxonomy

Qualification	Contribution character	Growth	Distribution taxation	RMD (owner alive)	Key modelling attributes
Non-Qualified (NQ)	After-tax; creates **basis**	Deferred	**LIFO** — gain first, fully taxable; basis returns tax-free	**No**	costBasis, pre-8/14/1982 grandfather flag, exclusion ratio (if annuitized), §72(u) non-natural-owner flag
Traditional IRA	Pre-tax (usually); after-tax possible → basis	Deferred	Ordinary income; basis pro-rated (Form 8606 pro-rata rule)	**Yes**	contributionYear, deductible vs. non-deductible basis, aggregation attestation
Roth IRA	After-tax	Tax-free if qualified	Tax-free if qualified (5 years + trigger); else ordering rules	**No**	firstContributionYear (5-yr clock), per-conversion 5-yr clocks, contribution/conversion/earnings layers
SEP IRA	Employer, pre-tax	Deferred	Ordinary income	**Yes**	employer/plan reference; taxed as a Traditional IRA on distribution
SIMPLE IRA	Employee deferral + employer	Deferred	Ordinary income	**Yes**	**2-year rule** from first contribution (early distributions/rollovers penalized at 25%, and no rollover to non-SIMPLE within 2 years)
401(k)	Pre-tax and/or designated Roth	Deferred	Per source	**Yes** (with still-working exception for non-owners)	plan reference, source-level accounting (pre-tax / Roth / after-tax / match), loan provisions, spousal-consent (ERISA)
403(b)	Pre-tax and/or Roth	Deferred	Per source	**Yes**	Historically annuity-centric (TSA); pre-1987 balance grandfathering; 15-year service catch-up

457(b)	Pre-tax and/or Roth	Deferred	Ordinary income	**Yes**	**Governmental vs. non-governmental matters enormously** — non-governmental 457(b) assets remain employer assets subject to creditors and cannot roll to an IRA
Inherited IRA (Traditional or Roth)	N/A	Deferred/tax-free	Per underlying type	**Yes, per beneficiary regime**	decedent identity & DOD, beneficiary classification (EDB/DB/non-DB), applicable regime (life expectancy / 10-year / 5-year), pre-vs-post SECURE, decedent's RMD-begin status

[Tax rules cited are conceptual and change with legislation. Applicable ages, limits, and rates must be verified against current-year IRS guidance — treat them as effective-dated reference data, never as code constants.]

5.3 Events That Change Qualification (and What They Must Record)

Event	From → To	Taxable?	Must record
Roth conversion	Traditional/SEP/SIMPLE → Roth	**Yes**, on the converted FMV	Conversion date, converted FMV, per-conversion 5-yr clock, 1099-R + 5498 pair, **irreversibility acknowledgment**
Rollover (direct)	Plan → IRA, or IRA → IRA	No	Source plan type, direct/indirect, after-tax basis carried, 5498 code
Rollover (indirect, 60-day)	Any → same type	No if within 60 days	Distribution date, receipt date, **one-per-12-month attestation**, 60-day compliance record
1035 exchange	NQ → NQ (or Qual → same Qual)	No	**Carried basis with ceding-carrier documentation**, 180-day partial-exchange window
Death / inheritance	Any → Inherited-{type}	On distribution	Decedent, DOD, beneficiary class, applicable regime, decedent's RMD status
Divorce (QDRO / §408(d)(6))	Split between spouses	No, if properly executed	Court order reference, split methodology, both parties' basis allocation
Ownership change (NQ)	NQ → NQ, new owner	**Often yes** (§72(e)(4)(C) gratuitous transfer)	New owner, relationship, gain recognized, 1099-R **with no cash movement**

The single design implication: `taxQualification` must be a **dated, versioned attribute with a change history and a reason code** — not a static column. Every qualification change is itself an auditable event with tax reporting attached.

5.4 Basis: The Attribute Everyone Underestimates

Basis (the after-tax amount already taxed, returnable tax-free) is:

- **Created** by NQ premium, non-deductible IRA contributions, after-tax plan contributions.
- **Carried** by 1035 exchanges — and **lost forever if the ceding carrier doesn't report it**.
- **Recovered** by LIFO ordering (NQ withdrawals — gain first, so basis comes out *last*), by the exclusion ratio (NQ annuitization), and pro-rata (IRA under §72 / Form 8606).
- **Not** inferable from premium totals, because withdrawals, exchanges, and conversions all modify it.

Model it as: `basisAmount`, `basisAsOfDate`, `basisSource` (premium / 1035-carried / after-tax-contribution / conversion), and a **basis movement ledger** parallel to the financial activity ledger. Carriers that store only a single mutable `COST_BASIS` field cannot defend a customer dispute or reconstruct a prior-year 1099-R.

5.5 The Modelling Pattern

```
CONTRACT  (product type: FIA — immutable)
   │
   └── TAX_QUALIFICATION  (versioned, dated)
         ├── v1  2018-04-02 → 2026-08-15   TRADITIONAL_IRA
         │       basis $0 | RMD subject: yes | plan ref: —
         │
         └── v2  2026-08-15 → open          ROTH_IRA
                 basis $0 | RMD subject: no (owner alive)
                 firstContributionYear 2026 | conversionClock 2026
                 changeReason: ROTH_CONVERSION
                 changeActivity: FA-0442 (linked)
```

The product never changed. The contract never changed. **Only the tax regime did** — and the model records exactly when, why, and which financial

activity effected it.

Chapter 6 — Parties and Roles

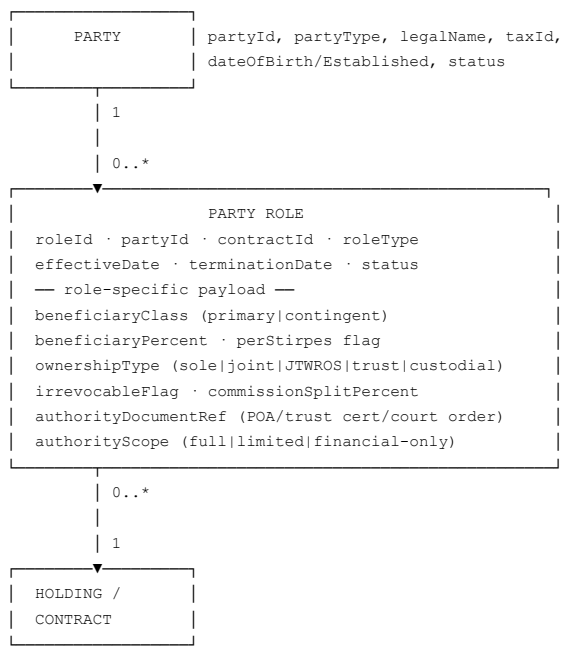
6.1 Why PARTY Is Separated From ROLE

The naive model puts `owner_name`, `annuitant_name`, `beneficiary_name` on the policy record. It fails immediately and permanently:

1. **The same person appears in many capacities.** Margaret is Owner and Annuitant on one contract, Beneficiary on her spouse's, and Trustee of a trust that owns a third. Four rows in a naive model, four places to update her address, four chances to be wrong.
2. **Roles are time-bounded; parties are not.** Beneficiaries change; owners change; the person does not. A death claim must be adjudicated against the designation **in force at date of death** — impossible if roles are overwritten.
3. **Roles carry role-specific data.** A beneficiary has a percentage and a class; an owner has none. A producer has a commission split. Bolting all of these onto the party record creates a sparse mess.
4. **Cardinality differs by role.** One owner or several. One annuitant or two. Zero to dozens of beneficiaries. Fixed columns cannot express this.
5. **Authority derives from role, not identity.** "Can this caller request a withdrawal?" is answered by *role on this contract*, not by *who they are*.

The rule: `PARTY` (who they are, once, enterprise-wide) → `PARTY ROLE` (what capacity, on which contract, for what period, with what role-specific data) → `HOLDING/CONTRACT`.

6.2 The Model



6.3 One Person, Many Roles — Worked Example

[Illustrative]

Party **P-0001 — Margaret L. Chen** (one party record, one address, one tax ID):

Contract	Role	Effective	Terminated	Role payload
A-77120 (her FIA)	**Owner**	2024-03-15	—	sole ownership
A-77120	**Annuitant**	2024-03-15	—	measuring life
A-77120	**Payor**	2024-03-15	—	ACH from her account
A-77121 (spouse's VA)	**Beneficiary**	2019-01-10	2023-06-02	primary, 100%
A-77121	**Beneficiary**	2023-06-02	—	primary, 50% (changed)
A-77122 (trust-owned)	**Trustee**	2019-06-11	—	authority: trust cert TC-4417

A-77122	**Annuitant**	2019-06-11	—	measuring life; Owner is the *trust*
L-30011 (mother's policy)	**Power of Attorney**	2025-02-20	—	scope: financial only; doc POA-8891

Read what this makes possible that a flat model cannot:

- Her address change updates **one** record and correctly propagates to all eight relationships.
- The 2023 beneficiary change is **preserved with dates** — a claim on A-77121 for a death in May 2023 pays 100%; a death in July 2023 pays 50%. Both are reconstructable.
- On A-77122 the **Owner is an Organization** and the Annuitant is a Person — §72(u) analysis is expressible.
- Her POA authority on L-30011 is **scoped and document-backed**, so "can she surrender it?" is answerable by rule.

6.4 Role Compatibility and Constraint Rules

Constraint	Rule	Why
Owner = Annuitant	**Allowed and typical**	The default retail case
Owner ≠ Annuitant	Allowed	Triggers §72(s) owner-driven death rules; a frequent defect source
Annuitant is an Organization	**Prohibited**	No measurable life
Insured is an Organization	**Prohibited**	Same
Owner is an Organization	Allowed	§72(u) tax-deferral analysis required on NQ deferred annuities
Owner = Beneficiary	Discouraged/often prohibited	Circular — the estate ends up as default payee
Annuitant = Beneficiary	Allowed but analyze	On an annuitant-driven contract this is self-referential
Irrevocable Beneficiary present	Owner's rights are **constrained**	Beneficiary consent needed for change, surrender, assignment
Collateral Assignee present	Owner's rights are **constrained**	Assignee's interest must be satisfied first
IRA owner change	**Prohibited** except divorce/death	IRAs are individual by statute
Joint owners on an IRA	**Prohibited**	Same reason
Minor as owner	Requires custodial structure	UTMA/UGMA custodian is the acting party

Architectural point: these belong in a **rules engine driven by role metadata**, not scattered through application code. Chapter 10 places that engine; Chapter 14 explains why it must stay deterministic.

6.5 Authority — the Distinction That Prevents Fraud

Three separate questions, often collapsed into one and shouldn't be:

1. **Authentication** — is the caller who they claim to be? (identity proofing, MFA)
2. **Role** — what capacity does that party hold on this contract? (Owner? Beneficiary? POA?)
3. **Authority** — does that role, *as scoped and documented*, permit this specific action right now?

A beneficiary authenticates perfectly, holds a real role, and has **no authority** to request a withdrawal while the owner lives. A POA holder may have financial authority but **not** beneficiary-change authority (many POA forms exclude it, and many carriers require specific language). An owner with an irrevocable beneficiary has authority to request but **not to complete** a beneficiary change alone.

Model authority as a derived decision: `authority = f(roleType, roleScope, documentValidity, contractState, requestedAction)` — evaluated per request, logged with its inputs.

Chapter 7 — Financial Activity Model

7.1 The Five-Level Hierarchy — the Chapter's Core Contribution

These five terms are used interchangeably in most organizations, and that imprecision is the source of an enormous amount of reconciliation pain. They are **five distinct levels of the same event**.

```
1. BUSINESS TRANSACTION      "I want to take out $10,000."
   ↓  what the customer asked for — one per request, has a status,
      can be pending/NIGO/cancelled, is what the customer sees
```

```
2. POLICY TRANSACTION        the contract-level effect of granting it
   ↓  changes contract state and values; the unit of PAS processing

3. FINANCIAL ACTIVITY        the value-changing components (1..* per policy txn)
   ↓  redemption from SEG-001, redemption from SEG-002, surrender charge,
      MVA adjustment, federal withholding, state withholding

4. ACCOUNTING TRANSACTION    the double-entry GL postings (1..* per activity)
   ↓  Dr Annuity Reserve / Cr Cash; Dr Cash / Cr Surrender Charge Income;
      Dr Cash / Cr Withholding Payable

5. API / MESSAGE EVENT       the integration-visible representations
      POST /withdrawals (request) · WithdrawalCompleted (event) ·
      1099-R accrual record · statement line · DTCC position update
```

7.2 Worked Decomposition — One $10,000 Withdrawal

[Illustrative]

Level 1 — Business Transaction `TXN-88201`

> Owner P-0001 requests $10,000 net from contract A-77120, pro-rata, 10% federal withholding, no state, ACH to account on file.

Level 2 — Policy Transaction `PTX-88201`

> Contract A-77120, effective 2026-08-22. Gross required to net $10,000 after charges and withholding: **$11,847.15**. Contract phase: accumulation. Post-transaction AV: $98,152.85. GLWB benefit base decremented pro-rata.

Level 3 — Financial Activities (six children under one parent)

ID	Type	Source/Target	Amount	Notes
FA-1001	WITHDRAWAL (parent)	Contract	−11,847.15	gross
FA-1002	SEGMENT_REDEMPTION	SEG-001 (60%)	−7,108.29	interim value applied (mid-term)
FA-1003	SEGMENT_REDEMPTION	SEG-002 (30%)	−3,554.15	interim value applied
FA-1004	SEGMENT_REDEMPTION	Fixed (10%)	−1,184.71	no interim adjustment
FA-1005	SURRENDER_CHARGE	Contract	−735.44	7% on the $10,505 excess over free amount
FA-1006	FEDERAL_WITHHOLDING	Contract	−1,111.71	10% of taxable $11,117.15
FA-1007	BENEFIT_BASE_ADJUSTMENT	GLWB rider	base 107,000 → 94,432	pro-rata; **not a cash movement**

Level 4 — Accounting Transactions

Dr	Cr	Amount
Annuity Contract Reserve	Cash	11,847.15
Cash	Surrender Charge Income	735.44
Cash	Federal Withholding Payable	1,111.71
(net cash out)		10,000.00

Level 5 — API / Message Events `POST /withdrawals` (request, idempotency-keyed) → `POST /withdrawals/{id}/confirm` → published events `withdrawal.completed`, `policy.value.changed`, `rider.benefitBase.changed`, `tax.reportable.created` → 1099-R accrual → statement line → confirmation letter → DTCC position update.

What this decomposition buys you: the customer sees one transaction; operations reconciles seven activities; finance posts four entries; tax reports one distribution; the distributor receives one position update. **Each audience gets the right granularity from the same underlying ledger** — which is only possible if the levels are modelled separately.

7.3 Activity Type Catalogue

Category	Activity types	Direction	Cash?	Taxable?
Money in	PREMIUM_INITIAL, PREMIUM_ADDITIONAL, ROLLOVER_IN, EXCHANGE_1035_IN, TRANSFER_IN, LOAN_REPAYMENT	In	Yes	No

Money out	WITHDRAWAL, SYSTEMATIC_WITHDRAWAL, RMD, FULL_SURRENDER, PARTIAL_SURRENDER, LOAN_DISBURSEMENT, DEATH_BENEFIT, ANNUITY_PAYMENT, EXCHANGE_1035_OUT, ROLLOVER_OUT	Out	Yes	Depends
Value credits	INTEREST_CREDIT, INDEX_CREDIT (**can be negative on RILA**), DIVIDEND_CREDIT, PERSISTENCY_BONUS, PREMIUM_BONUS	Neither	No	No (inside wrapper)
Value charges	SURRENDER_CHARGE, MVA_ADJUSTMENT (**±**), RIDER_CHARGE, M&E_CHARGE, ADMIN_FEE, COI_DEDUCTION, LOAN_INTEREST	Neither	No	No
Internal moves	TRANSFER (paired out/in), REBALANCE (paired), SEGMENT_MATURITY, SEGMENT_CREATION, ALLOCATION_APPLIED	Neither	No	No
Tax	FEDERAL_WITHHOLDING, STATE_WITHHOLDING, BACKUP_WITHHOLDING	Out	Yes	N/A
Notional (no cash, no AV)	BENEFIT_BASE_ROLLUP, BENEFIT_BASE_STEPUP, BENEFIT_BASE_ADJUSTMENT, BASIS_ADJUSTMENT, TAXABLE_EVENT_NO_CASH	Neither	**No**	Sometimes!
Corrections	REVERSAL, ADJUSTMENT, RESTATEMENT	Varies	Varies	Varies

The two categories people forget: *notional* activities (a GLWB rollup changes a contractual value with no cash and no account-value movement — but it must be a dated ledger entry or you cannot audit the benefit base) and *taxable-with-no-cash* events (gratuitous ownership transfer, lapse with loan). A model that assumes "financial activity ⟹ money moved" produces wrong 1099-Rs.

7.4 Special Cases Worth Explicit Design

- **Interest crediting** — daily accrual vs. anniversary posting; store the *method*, and post as a dated activity so any historical value is reproducible.
- **Index crediting** — computed **at segment maturity** from the segment's own start value and its own rate set; the activity must reference the segment ID, not just the contract.
- **Rider charges** — computed on one value (benefit base), deducted from another (account value), at a frequency independent of both. Three separate configured facts.
- **MVA** — can be positive or negative; must be modelled signed, with the formula's rate inputs snapshotted at calculation time for later defence.
- **1035 exchange** — one economic event, two contracts, two carriers, two activity records that must reference each other, and a basis transfer that has no cash counterpart.
- **Roth conversion** — a distribution and a contribution occurring simultaneously, taxable, and (usually) with no net cash leaving the enterprise. Chapter 15.
- **Reversal** — never mutate. Post a mirror activity with `reversalOf: <activityId>` and let derived values recompute. The reversed activity remains visible in history because a customer statement showing it was mailed.

Chapter 8 — Message Architecture

8.1 The ACORD L&A Message Pattern

[Official ACORD concept — structural description only]

ACORD Life & Annuity XML follows a **request/response envelope** pattern. Publicly documented terminology:

- **`TXLife`** — the request envelope. Carries a wrapper containing user authentication and routing metadata, plus one or more transaction blocks.
- **`TXLifeRequest`** — an individual transaction within the envelope, identified by a transaction type code, carrying the business payload.
- **`TXLifeResponse`** — the response envelope, carrying result/status information and optionally the requested data.
- **`OLifE`** — the business object model beneath the transaction: the hierarchy containing Party, Holding, Policy, Relation, and related objects.

[Requires verification against licensed ACORD LA XML v2.49.00 documentation for: the exact current element hierarchy, the transaction-type code list, mandatory/optional cardinality, and the wrapper's authentication structure. This guide deliberately does not reconstruct these from memory or from third-party schema mirrors.]

8.2 The Conceptual Message Anatomy

Every well-formed insurance message — ACORD or otherwise — has the same six parts:

```
┌─ ENVELOPE ──────────────────────────────────────────┐
│ 1 HEADER      who sent it, to whom, when, which version, │
│               correlation ID, authentication             │
├─ TRANSACTION ──────────────────────────────────────────┤
```

```
| 2 INTENT      transaction type + mode (submit/query/     |
|               quote/cancel) + effective date              |
├─ PAYLOAD ──────────────────────────────────────────────────┤
| 3 PARTY       the parties involved and their roles        |
| 4 HOLDING     the contract(s) in scope                    |
| 5 ACTIVITY    the financial/business content of the event |
├─ RESULT (response only) ──────────────────────────────────┤
| 6 STATUS      success/failure, result codes, errors,      |
|               warnings, and any returned data             |
└──────────────────────────────────────────────────────────┘
```

The relationship between Party and Holding is by reference, not containment. A party appears once in the message and is referenced by ID from each role on each holding. This is the message-level expression of the Chapter 6 Party/Role separation — and it's why the same person appearing as owner of one contract and beneficiary of another doesn't get duplicated in the payload.

8.3 Illustrative XML — Withdrawal Request

> ⚠️ **Illustrative ACORD-aligned example — not an official ACORD schema.** Element names below are this guide's construction demonstrating the *structural pattern*. Do not use for implementation. Actual element names, type codes, and cardinality must be taken from licensed ACORD LA XML documentation.

```
<InsuranceMessage>
  <!-- 1. HEADER -->
  <MessageHeader>
    <MessageId>MSG-2026-08-22-0001</MessageId>
    <CorrelationId>TXN-88201</CorrelationId>
    <SentDateTime>2026-08-22T14:22:09Z</SentDateTime>
    <StandardVersion>ILLUSTRATIVE-1.0</StandardVersion>
    <Sender><PartyId>BD-4471</PartyId><Name>Example Broker-Dealer</Name></Sender>
    <Receiver><PartyId>CAR-0099</PartyId><Name>Example Carrier</Name></Receiver>
    <Authentication><Method>OAuth2-Bearer</Method></Authentication>
  </MessageHeader>

  <!-- 2. INTENT -->
  <Transaction>
    <TransactionType>PARTIAL_WITHDRAWAL</TransactionType>
    <TransactionMode>SUBMIT</TransactionMode>   <!-- QUOTE | SUBMIT | CANCEL -->
    <EffectiveDate>2026-08-22</EffectiveDate>
    <IdempotencyKey>a3f9c2e1-7b44-4d12-9c88-0e2b41f7dd10</IdempotencyKey>

    <!-- 3. PARTY (declared once, referenced by role) -->
    <Parties>
      <Party id="P-0001">
        <PartyType>Person</PartyType>
        <Person>
          <FirstName>Margaret</FirstName><LastName>Chen</LastName>
          <BirthDate>1961-03-08</BirthDate>
          <TaxId type="SSN" masked="true">***-**-4417</TaxId>
        </Person>
        <Address type="Residence">
          <Line1>1420 Elm Street</Line1><City>Austin</City>
          <StateProvince>TX</StateProvince><PostalCode>78704</PostalCode>
        </Address>
      </Party>
      <Party id="P-9001">
        <PartyType>Organization</PartyType>
        <Organization><LegalName>Example Agency LLC</LegalName></Organization>
      </Party>
    </Parties>

    <!-- 4. HOLDING -->
    <Holding id="H-1">
      <HoldingType>AnnuityContract</HoldingType>
      <Policy>
        <PolicyNumber>A-77120</PolicyNumber>
        <CarrierId>CAR-0099</CarrierId>
        <ProductType>FIXED_INDEXED_ANNUITY</ProductType>
        <PlanCode>FIA-10-GLWB</PlanCode>
        <IssueDate>2024-03-15</IssueDate>
        <IssueState>TX</IssueState>
        <TaxQualification>NON_QUALIFIED</TaxQualification>
        <ContractPhase>ACCUMULATION</ContractPhase>
      </Policy>
      <!-- ROLES: reference parties, never duplicate them -->
```

```
      <Relation partyRef="P-0001" roleType="Owner"     effectiveDate="2024-03-15"/>
      <Relation partyRef="P-0001" roleType="Annuitant" effectiveDate="2024-03-15"/>
      <Relation partyRef="P-9001" roleType="Producer"  splitPercent="100"/>
    </Holding>

    <!-- 5. ACTIVITY -->
    <FinancialActivity>
      <ActivityType>PARTIAL_WITHDRAWAL</ActivityType>
      <AmountType>NET</AmountType>          <!-- GROSS | NET -->
      <RequestedAmount currency="USD">10000.00</RequestedAmount>
      <SourceMethod>PRO_RATA</SourceMethod> <!-- PRO_RATA | SPECIFIED -->
      <WithholdingElection>
        <Federal type="PERCENT">10.0</Federal>
        <State type="PERCENT">0.0</State>
        <ElectionFormOnFile>W-4R</ElectionFormOnFile>
      </WithholdingElection>
      <Disbursement>
        <Method>ACH</Method>
        <PayeeRef>P-0001</PayeeRef>
        <AccountToken>tok_9f2a44c1</AccountToken>
      </Disbursement>
    </FinancialActivity>
  </Transaction>
</InsuranceMessage>
```

8.4 Illustrative XML — Response with Partial Failure

> ⚠️ **Illustrative ACORD-aligned example — not an official ACORD schema.**

```
<InsuranceMessageResponse>
  <MessageHeader>
    <MessageId>MSG-RSP-2026-08-22-0001</MessageId>
    <InResponseTo>MSG-2026-08-22-0001</InResponseTo>
    <CorrelationId>TXN-88201</CorrelationId>
  </MessageHeader>

  <TransactionResult>
    <TransactionType>PARTIAL_WITHDRAWAL</TransactionType>
    <ResultStatus>PENDING_REQUIREMENTS</ResultStatus>
    <!-- SUCCESS | PENDING_REQUIREMENTS | REJECTED | PARTIAL -->

    <Errors>
      <Error severity="BLOCKING" code="AUTH-014">
        <Description>Requesting party is not authorized for this action.</Description>
        <Element>Transaction/RequestedBy</Element>
      </Error>
    </Errors>

    <Warnings>
      <Warning code="RIDER-201">
        <Description>Requested amount exceeds the GLWB annual allowance;
          benefit base will be reduced pro-rata and guaranteed income
          permanently lowered.</Description>
        <ProjectedBenefitBaseBefore>107000.00</ProjectedBenefitBaseBefore>
        <ProjectedBenefitBaseAfter>94432.00</ProjectedBenefitBaseAfter>
      </Warning>
      <Warning code="TAX-108">
        <Description>Owner is under age 59½; distribution may be subject
          to a 10% additional federal tax.</Description>
      </Warning>
    </Warnings>

    <Requirements>
      <Requirement id="REQ-5512" status="OUTSTANDING">
        <Type>MEDALLION_SIGNATURE_GUARANTEE</Type>
        <Reason>Amount exceeds unguaranteed threshold</Reason>
        <ResponsiblePartyRef>P-0001</ResponsiblePartyRef>
        <DueDate>2026-09-21</DueDate>
      </Requirement>
    </Requirements>
  </TransactionResult>
</InsuranceMessageResponse>
```

Design lessons embedded above:

1. **Errors block; warnings inform.** A GLWB benefit-base reduction is not an error — it's a consequence the customer must be told about. Conflating them either blocks legitimate business or hides material harm.
2. `PENDING_REQUIREMENTS` **is a first-class status,** not a failure. This is NIGO, modelled.
3. **The response carries the requirement objects themselves**, so the distributor can display and resolve them without a second call.
4. **Correlation IDs thread the whole conversation**, including the eventual asynchronous completion event.

8.5 Illustrative JSON — the Same Transaction

> ⚠️ **Illustrative ACORD-aligned example — not an official ACORD schema.**

```
{
  "messageHeader": {
    "messageId": "MSG-2026-08-22-0001",
    "correlationId": "TXN-88201",
    "sentDateTime": "2026-08-22T14:22:09Z",
    "sender":   { "partyId": "BD-4471" },
    "receiver": { "partyId": "CAR-0099" }
  },
  "transaction": {
    "type": "PARTIAL_WITHDRAWAL",
    "mode": "SUBMIT",
    "effectiveDate": "2026-08-22",
    "idempotencyKey": "a3f9c2e1-7b44-4d12-9c88-0e2b41f7dd10",
    "parties": [
      { "id": "P-0001", "type": "Person",
        "person": { "firstName": "Margaret", "lastName": "Chen",
                    "birthDate": "1961-03-08",
                    "taxId": { "type": "SSN", "masked": "***-**-4417" } } }
    ],
    "holdings": [
      { "id": "H-1", "type": "AnnuityContract",
        "policy": { "policyNumber": "A-77120", "productType": "FIXED_INDEXED_ANNUITY",
                    "taxQualification": "NON_QUALIFIED", "contractPhase": "ACCUMULATION" },
        "relations": [
          { "partyRef": "P-0001", "roleType": "Owner",     "effectiveDate": "2024-03-15" },
          { "partyRef": "P-0001", "roleType": "Annuitant", "effectiveDate": "2024-03-15" }
        ] }
    ],
    "financialActivity": {
      "activityType": "PARTIAL_WITHDRAWAL",
      "amountType": "NET",
      "requestedAmount": { "value": 10000.00, "currency": "USD" },
      "sourceMethod": "PRO_RATA",
      "withholding": { "federalPercent": 10.0, "statePercent": 0.0 },
      "disbursement": { "method": "ACH", "payeeRef": "P-0001", "accountToken": "tok_9f2a44c1" }
    }
  }
}
```

XML vs. JSON — the honest comparison for L&A:

	XML (ACORD LA)	JSON (modern API)
Schema validation	Mature — XSD, ACORD-published, TCF-certifiable	JSON Schema / OpenAPI; no industry certification equivalent
Industry adoption in L&A	**Dominant** — DTCC, carrier-distributor rails	Growing, mostly internal and greenfield
Verbosity	High	Low
Extension model	Defined ACORD extension points	Ad hoc unless disciplined
Tooling in legacy estates	Extensive (existing transforms, mappings)	Requires new investment

The pragmatic answer: use JSON at the API edge for developer experience, keep the canonical model in the middle, and retain XML where the industry rails require it. The canonical model is what makes both possible — see Chapter 11.

8.6 Error-Handling Taxonomy

Class	Meaning	Retry?	Example

Syntax/schema	Malformed message	No — fix and resend	Missing mandatory element
Semantic	Well-formed but meaningless	No	Withdrawal amount is negative
Authorization	Caller lacks authority	No	Beneficiary requesting a withdrawal
Business rule	Valid request, disallowed by product/contract	No	Transfer requested mid-segment on an FIA
Requirement	Valid, but incomplete → NIGO	**Yes, after satisfying**	Missing Medallion guarantee
State conflict	Contract state prevents it	Sometimes	Contract already surrendered
Transient/system	Downstream unavailable	**Yes, with idempotency key**	PAS timeout

Rule: the message must let the caller distinguish "never retry this" from "retry after fixing X" from "retry as-is." A single generic failure code forces the distributor into blind retries — which, on a withdrawal, means duplicate money movement.

Chapter 9 — APIs

9.1 From Message Standards to Resources

Traditional insurance messaging is **transaction-oriented**: one envelope, one transaction-type code, one big payload. REST is **resource-oriented**: nouns with a small verb set. The translation is not mechanical, and the common mistake is to expose `POST /txlife` with a type code in the body — which is RPC in a REST costume and inherits all the message model's coupling with none of REST's benefits.

The mapping that works:

Message concept	REST equivalent
Holding / Policy	`/policies/{id}` — a resource
Party	`/parties/{id}` — a resource
Party role on a contract	`/policies/{id}/participants` — a sub-collection
Financial activity	`/policies/{id}/transactions` — a sub-collection
Query transaction	`GET` on the relevant resource
Submit transaction	`POST` to an **action resource** (`/withdrawals` , `/roth-conversions`)
Transaction status	`GET /withdrawals/{id}` — the request itself is a resource with a lifecycle
Requirement	`/withdrawals/{id}/requirements`

The key design decision: model each *business transaction* as a **resource with its own lifecycle**, not as a verb on the policy. A withdrawal request is a thing that exists, has a status (`quoted` → `pending_requirements` → `confirmed` → `settled`), can be cancelled, and can be polled. `POST /policies/{id}/withdraw` returning 200-or-error cannot represent NIGO; `POST /withdrawals` returning a resource can.

9.2 Read Endpoints

`GET /policies/{id}` — contract summary.

```
{
  "policyNumber": "A-77120",
  "carrierId": "CAR-0099",
  "productType": "FIXED_INDEXED_ANNUITY",
  "planCode": "FIA-10-GLWB",
  "status": "IN_FORCE",
  "contractPhase": "ACCUMULATION",
  "issueDate": "2024-03-15",
  "issueState": "TX",
  "taxQualification": {
    "type": "NON_QUALIFIED", "effectiveDate": "2024-03-15",
    "costBasis": 100000.00, "rmdSubject": false
  },
  "surrenderChargeSchedule": { "scheduleId": "SC-10YR-A", "currentYear": 3,
                               "currentRate": 0.08, "expiryDate": "2034-03-15" },
  "freeLookExpiry": "2024-04-14",
  "_links": { "values": "...", "participants": "...", "transactions": "...",
              "allocations": "...", "riders": "..." }
```

```
}
```

`GET /policies/{id}/values` — **the endpoint that must not return a single number.** Per Chapter 2.4, an annuity has at least six simultaneously valid values.

```
{
  "asOfDate": "2026-08-22",
  "accountValue": 110000.00,
  "cashSurrenderValue": 101200.00,
  "surrenderCharge": 8800.00,
  "marketValueAdjustment": -1150.00,
  "guaranteedMinimumValue": 92500.00,
  "deathBenefitValue": 110000.00,
  "freeWithdrawalRemaining": 1342.85,
  "riderValues": [
    { "riderId": "GLWB-2024", "benefitBase": 107000.00,
      "annualIncomeAmount": 5350.00, "incomeRatePercent": 5.0,
      "excessWithdrawalThreshold": 5350.00 }
  ],
  "strategies": [
    { "strategyId": "SP500-APTP-CAP", "allocationPercent": 60,
      "segments": [ { "segmentId": "SEG-001", "value": 66000.00,
                      "startDate": "2025-03-15", "maturityDate": "2026-03-15",
                      "capRate": 0.0925, "interimValueApplied": true } ] }
  ]
}
```

`GET /policies/{id}/participants` — roles, dated, never flattened.

```
{
  "participants": [
    { "partyId": "P-0001", "roleType": "OWNER", "effectiveDate": "2024-03-15",
      "terminationDate": null, "ownershipType": "SOLE" },
    { "partyId": "P-0001", "roleType": "ANNUITANT", "effectiveDate": "2024-03-15" },
    { "partyId": "P-0002", "roleType": "BENEFICIARY", "beneficiaryClass": "PRIMARY",
      "percent": 100, "perStirpes": false, "irrevocable": false,
      "effectiveDate": "2024-03-15" }
  ],
  "constraints": { "irrevocableBeneficiaryPresent": false,
                   "collateralAssignmentPresent": false }
}
```

Note `asOf` support: `GET /policies/{id}/participants?asOf=2023-05-01` must return the designation **in force on that date** — this is what makes death-claim adjudication defensible.

`GET /policies/{id}/transactions` — the financial activity ledger, paginated, filterable by type and date range, showing parent/child structure and reversals.

9.3 Write Endpoints — the Transaction Resources

`POST /withdrawals` — two-phase by design.

```
// Request
{
  "policyNumber": "A-77120",
  "requestedBy": { "partyId": "P-0001", "roleType": "OWNER" },
  "amountType": "NET",
  "amount": 10000.00,
  "sourceMethod": "PRO_RATA",
  "withholding": { "federalPercent": 10.0, "statePercent": 0.0 },
  "disbursement": { "method": "ACH", "accountToken": "tok_9f2a44c1" },
  "mode": "QUOTE"
}

// 201 Created — a resource, with the full consequence disclosed
{
  "withdrawalId": "WD-88201",
  "status": "QUOTED",
  "quoteExpiresAt": "2026-08-22T18:00:00Z",
  "breakdown": {
    "grossAmount": 11847.15, "surrenderCharge": 735.44,
    "marketValueAdjustment": 0.00, "federalWithholding": 1111.71,
    "stateWithholding": 0.00, "netToOwner": 10000.00,
```

```
    "taxableAmount": 11117.15, "basisReturned": 0.00
  },
  "impacts": [
    { "type": "RIDER_BENEFIT_BASE_REDUCTION", "severity": "MATERIAL",
      "riderId": "GLWB-2024", "before": 107000.00, "after": 94432.00,
      "annualIncomeBefore": 5350.00, "annualIncomeAfter": 4721.60,
      "description": "Exceeds annual allowance; permanent income reduction." },
    { "type": "EARLY_DISTRIBUTION_TAX", "severity": "INFORMATIONAL",
      "description": "Owner under 59½; 10% additional federal tax may apply." }
  ],
  "requirements": [],
  "_links": { "confirm": "/withdrawals/WD-88201/confirm",
              "cancel":  "/withdrawals/WD-88201" }
}
```

Then `POST /withdrawals/WD-88201/confirm` with the `Idempotency-Key` header commits it.

Why two-phase is non-negotiable here: the customer is about to permanently reduce guaranteed lifetime income by $628/year to receive $10,000. A single-call API cannot show them that first. **This is a compliance control expressed as an API design.**

Other write endpoints, same pattern:

Endpoint	Notable design points
`POST /beneficiary-changes`	Full designation set (not a delta); validates 100% per class; requires irrevocable-beneficiary consent; **versions** rather than overwrites
`POST /roth-conversions`	Source + destination; FMV determination; **explicit** withholding election (no silent 10% default); irreversibility acknowledgment flag; returns paired distribution/contribution activity IDs
`POST /1035-exchanges`	Ceding carrier, contract, **ceding basis with document reference**; direction validation; replacement disclosure reference
`POST /rmd-elections`	Calculation year, aggregation attestation, satisfaction method (this contract / elsewhere / via SWP), withholding
`POST /surrenders`	Same quote/confirm; requires conservation-attempt flag; full rider-forfeiture disclosure
`POST /allocation-changes`	Explicit `scope: FUTURE_PREMIUM_ONLY` vs. `EXISTING_VALUE` — **never infer** (§4.20)
`POST /address-changes`	Elevated verification; triggers withholding recalculation; flagged if followed by a distribution request within N days

9.4 Cross-Cutting API Concerns

Idempotency. Every state-changing POST requires an `Idempotency-Key`. Store the key with the resulting resource for at least 24 hours; a repeated key returns the **original** result, not a new transaction. Without this, a distributor's retry on a timeout creates a duplicate withdrawal — unrecoverable customer harm and a reportable incident.

Authentication & authorization. OAuth2 client credentials for system-to-system; per-request the API must evaluate the Chapter 6.5 authority function. Distributor tokens should be **scoped to their own book of business** — a broker-dealer must not be able to `GET` a policy sold by another firm. In a TPA/BPO context, add a carrier-partition claim.

Validation layering. (1) Schema — is the payload well-formed? (2) Semantic — do the values make sense? (3) Authority — may this caller do this? (4) Business rule — does the product/contract allow it? (5) Regulatory — does the law allow it here, now? Return the *most specific* failure, and return **all** failures at once where possible rather than one per round-trip.

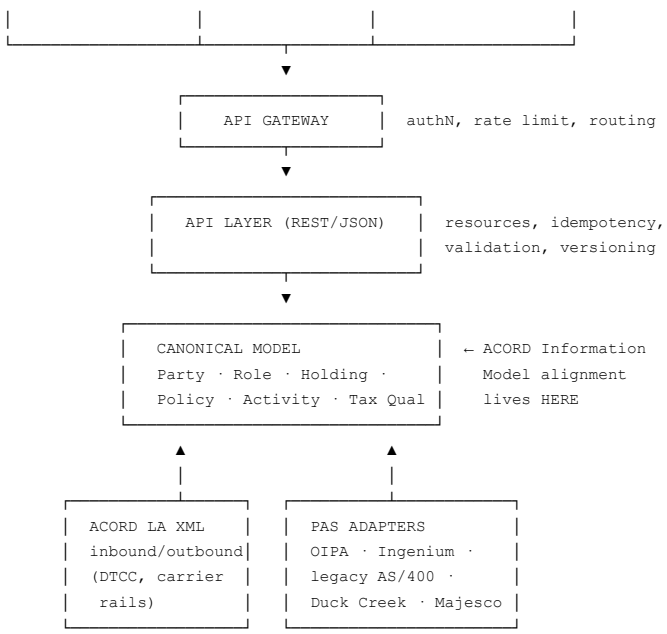
Versioning. URI-path versioning (`/v1/policies`) for major breaking changes; additive fields without a version bump; **never** repurpose an existing field's meaning. Publish a deprecation policy with a sunset date and honour it — distributors integrate once and revisit rarely.

Eventing. Synchronous APIs handle the request; asynchronous events handle the consequences. Publish `policy.issued`, `withdrawal.completed`, `policy.value.changed`, `rider.benefitBase.changed`, `beneficiary.changed`, `claim.opened`, `requirement.outstanding`, `tax.reportable.created`. Consumers: statements, correspondence, data warehouse, distributor feeds, AI/analytics layer. **Events should carry IDs and enough context to act, not full payloads** — consumers re-read the API for detail, which keeps the event contract stable.

Error format. Consistent envelope with a stable machine code, human message, offending field path, severity, and a retry hint (§8.6).

9.5 How the Canonical Model Sits Behind It All

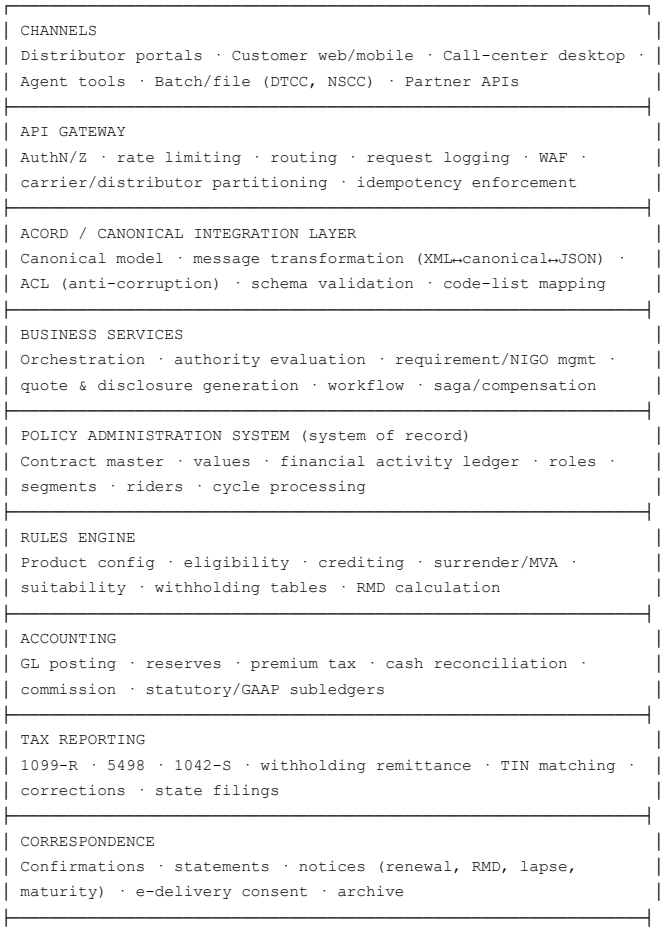
```
    Distributor          Customer          Internal           AI/Analytics
    order entry           portal            service              layer
```



The point: the canonical model is the **only** thing that knows what a "policy" means. The API layer translates it to JSON; the message layer translates it to ACORD XML; the adapters translate it to whatever each PAS calls it. Add a new channel — you write one adapter. Add a new PAS (an acquisition, a new carrier onboarded to the TPA platform) — you write one adapter. Neither touches the other. This is the entire economic argument for the architecture, and it's the argument a Business Architect has to be able to make to a CFO in two sentences.

Chapter 10 — PAS Architecture

10.1 The Layered Reference Architecture



```
| DATA & ANALYTICS                                                    |
| ODS · warehouse · lakehouse · BI · actuarial extracts ·             |
| regulatory reporting · AI/ML feature store                          |
└─────────────────────────────────────────────────────────────────────┘
```

10.2 Layer Responsibilities — and the Boundary Rules That Matter

Layer	Owns	Must NOT own
Channels	Presentation, capture, UX	Business rules — a portal that decides eligibility will diverge from the call center
API Gateway	Cross-cutting concerns	Business logic of any kind
Canonical/ACORD layer	**Meaning.** Translation between representations, code-list mapping, structural validation	Business decisions — it translates, it doesn't decide
Business services	Orchestration, sequencing, compensation, requirements, authority, disclosure	Value calculation (that's PAS/rules) and persistence of record
PAS	**The system of record.** Contract state, the append-only ledger, values, cycle processing	Channel-specific logic; presentation; integration protocol concerns
Rules engine	Product configuration & deterministic calculation	Being bypassed. If any layer hard-codes a rate, the engine is decorative
Accounting	Double-entry truth, reserves, reconciliation	Deciding what a transaction *is* — it posts what it's told
Tax reporting	Reportable projections over the ledger, forms, remittance	Being derived from live mutable balances (§4.29)
Correspondence	Generation, delivery, consent, archive	Recalculating values — it renders what it's given, or statements and the PAS will disagree
Data & analytics	Read-optimized copies, history, ML features	Being a write path. Analytics never writes back to the record

The three boundary rules that prevent most architectural decay:

1. **The PAS is the system of record; everything else is a projection.** If two layers can both claim to know a contract's value, you will eventually reconcile them by hand, forever.
2. **Rules are configuration, not code, and live in one place.** Withholding rates, cap rates, surrender schedules, RMD tables, state variations — all effective-dated reference data. Every January proves whether you did this.
3. **The canonical layer translates; the business layer decides.** Blur this and your integration layer accumulates business logic that nobody knows is there — the single most common way an ACL becomes a second, undocumented PAS.

10.3 Where TPA/BPO Architecture Differs

In a third-party administration context (multiple carriers, one platform), three things change materially:

- **Carrier is a partition key everywhere** — data, config, rules, correspondence templates, tax reporting entity, and API authorization scope. Product configuration is carrier-scoped; a "plan code" is only unique *within* a carrier.
- **Multi-tenancy vs. multi-instance is the defining architectural decision.** Shared-schema multi-tenancy gives operational leverage and cross-carrier analytics but requires rigorous partition enforcement and complicates carrier-specific regulatory requests. Per-carrier instances are safer but multiply upgrade and support cost. Most L&A TPAs run a hybrid — shared services, partitioned data.
- **The canonical model earns its keep faster than at a single carrier.** Onboarding carrier #6 with its own legacy extract format is a one-adapter problem instead of a re-platform. This is the clearest ROI story for the reference-architecture investment, and worth quantifying explicitly in any business case.

Chapter 11 — Legacy Modernization

11.1 The Situation

A carrier (or TPA) runs an annuity book on an IBM i / AS/400 policy administration system written in RPG, with DB2 for i files, green-screen operations, and 25 years of accumulated business rules — many undocumented, some encoded only in the behaviour of a program nobody has read since its author retired. The book is profitable, the system is reliable, and replacing it is a five-to-eight-year, nine-figure programme with a high failure rate.

The ACORD proposition is not "replace the PAS." It is "stop letting the PAS's internal model be the enterprise's model." Every new channel, partner, AI initiative, and analytics use case currently negotiates directly with a 1990s data structure. The canonical layer breaks that coupling — and once broken,

replacement becomes *optional and incremental* rather than *mandatory and monolithic*.

11.2 The Pattern Stack

1. Canonical model. The ACORD-aligned Information Model becomes the enterprise's shared vocabulary. Nothing outside the PAS speaks `PLAN_CODE`; everything speaks `productType` + `taxQualification` + `surrenderChargeSchedule`.

2. Anti-corruption layer (ACL). A deliberate translation boundary that prevents legacy concepts from leaking outward *and* modern concepts from leaking inward. The ACL is where `QUALIFICATION_CODE = 'Q'` becomes `taxQualification: TRADITIONAL_IRA` — and where the ambiguity of that legacy code gets confronted, documented, and resolved once rather than in fifty places.

3. API façade. REST/JSON resources (Chapter 9) fronting the legacy system. The façade exposes the canonical model; behind it, adapters call RPG programs, read DB2 files, or consume extracts. **Consumers never know.**

4. Adapters. Per-source translators. On IBM i, realistic options in rough order of preference:

- Expose existing RPG business logic as callable services (IBM i program call, stored procedures, or an integration gateway) — **reuses proven logic**, avoids re-implementing rules
- Read DB2 for i directly for query paths (safe, read-only, fast to build)
- Consume batch extracts where real-time isn't required (statements, analytics)
- CDC (change data capture) off journals for event generation — **this is how a batch system starts emitting events without being rewritten**

5. Message transformation. Canonical ↔ ACORD LA XML for industry rails; canonical ↔ JSON for APIs; canonical ↔ legacy record layouts for the PAS. Three transforms, one hub — versus N×N without it.

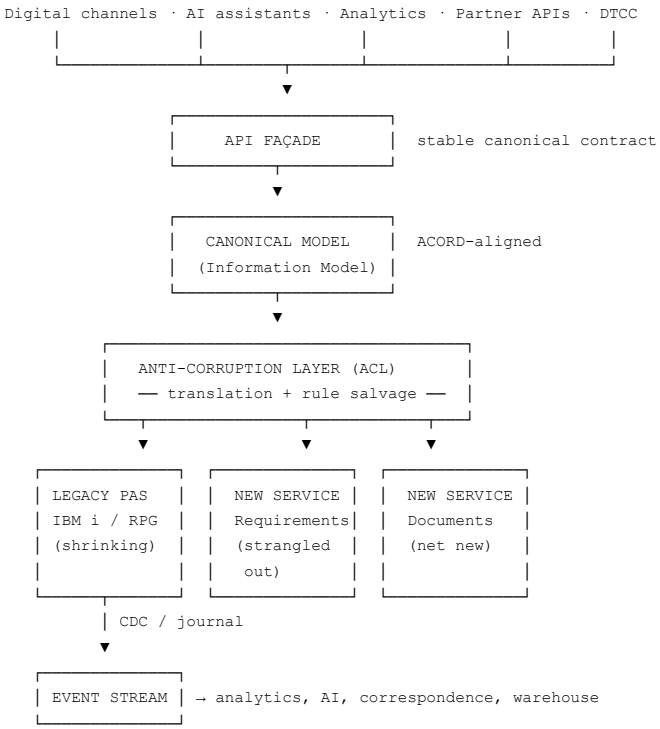
6. Event-driven architecture. Journal-based CDC or explicit event emission turns state changes into published events (`policy.value.changed`, `withdrawal.completed`). This is the highest-leverage single move in a legacy estate: it lets modern consumers react to a batch system in near-real-time without the batch system changing.

7. Data mapping & legacy field mapping. Chapter 12 — the concrete artifact.

8. Microservices. Selectively. New capabilities (a quote engine, a requirements/NIGO service, a document service) are built as services against the canonical model. **Do not** carve the core ledger into microservices — financial consistency is exactly what monoliths are good at.

9. Strangler pattern. New functionality is built outside; existing functionality is migrated capability-by-capability behind the stable façade; the legacy system's scope shrinks until it is small enough to retire — or until nobody cares that it's still there, which is a legitimate end state.

11.3 The Target State



11.4 Worked Example — Legacy Annuity PAS to Digital

Stage 0 (today). Digital portal reads a nightly extract. Values are a day stale. Adding a new distributor takes four months of custom file work. An AI

assistant is impossible because nothing is queryable in real time.

Stage 1 — Read façade (3–6 months). Build the ACL and canonical model for the read path only. Expose `GET /policies/{id}`, `/values`, `/participants`, `/transactions`. Adapter reads DB2 for i directly. **Nothing in the PAS changes.** Portal now shows real-time values; a second channel costs weeks, not months.

Stage 2 — Events (3–4 months). Journal-based CDC publishes value and status changes. Correspondence, warehouse, and distributor feeds move from nightly batch to event-driven. **Still nothing in the PAS changes.**

Stage 3 — Write path, low-risk transactions (6–9 months). Address change, beneficiary change, allocation change: `POST` through the façade, ACL translates, adapter calls existing RPG programs. Requirements/NIGO handled in a **new** service outside the PAS — the first strangled capability.

Stage 4 — Write path, financial transactions (9–18 months). Withdrawals and surrenders with the two-phase quote/confirm pattern (§9.3). The quote calculation is the delicate part: either call existing RPG calculation programs (preferred — proven logic) or re-implement in the rules engine with rigorous parallel-run reconciliation. **Never re-implement financial calculations without a parallel run against production history.**

Stage 5 — Selective replacement (ongoing). New products launch on a modern platform behind the same façade. The legacy book runs off naturally. The façade means consumers never know which contract lives where.

What made this work: at no point was there a big-bang cutover, and at every stage the business got something. That is the difference between a modernization programme that survives a budget cycle and one that doesn't.

11.5 Risks Worth Naming Up Front

Risk	Mitigation
The ACL becomes a second PAS	Hard rule: ACL translates, never decides. Review it for business logic quarterly
Undocumented rules lost in translation	Parallel-run everything financial; treat legacy behaviour as the spec until proven otherwise
Canonical model designed by committee, never finished	Model only what the current increment needs. A perfect unused model is worth zero
Legacy team disengagement	The RPG developers hold the business rules. They are the most valuable people in the programme, not obstacles to it
Strangler never finishes	Acceptable — if the legacy scope is stable and small. Name it as a decision rather than a failure
Calculation drift	Automated daily reconciliation between legacy and new calculation paths during any overlap

---

Chapter 12 — Data Mapping Exercise

12.1 The Hypothetical Legacy Record

[Illustrative — a realistic composite of legacy annuity master layouts, not any specific carrier's file]

```
ANNMAST  (DB2 for i physical file, 512-byte record)
─────────────────────────────────────────────────────
POLICY_NUMBER      CHAR(12)     A-77120
OWNER_ID           CHAR(9)      000440017
ANNUITANT_ID       CHAR(9)      000440017
PLAN_CODE          CHAR(6)      FIA10Q
ISSUE_DATE         DEC(7,0)     1240315        (CYYMMDD)
QUALIFICATION_CODE CHAR(1)      Q
ACCOUNT_VALUE      DEC(11,2)    00011000000    (implied decimal)
SURRENDER_VALUE    DEC(11,2)    00010120000
RIDER_CODE         CHAR(4)      GLWB
STATUS             CHAR(1)      A
STATE              CHAR(2)      TX
RMD_INDICATOR      CHAR(1)      Y
```

12.2 The Mapping Table

Legacy field	Business meaning	Canonical entity	Canonical attribute	Transformation	Validation	T
`POLICY_NUMBER`	Contract identifier, carrier-unique	POLICY	`policyNumber`	Trim; **prefix with carrierId** — not globally unique in a TPA estate	Non-blank; matches carrier format; unique within carrier	`C` `P`

`OWNER_ID`	Client number of the owner	PARTY + PARTY ROLE	`partyId` + `roleType: OWNER`	**Explode into a Party reference and a dated Role record.** Legacy holds one owner; canonical must support joint	Party exists; role effective-dated from issue; **detect joint-owner cases hidden in a secondary file**	/
`ANNUITANT_ID`	Client number of the annuitant	PARTY ROLE	`roleType: ANNUITANT`	Same explode. **If equal to OWNER_ID, create two roles for one party — do not deduplicate**	Party is a Person (never Org); alive at issue	/
`PLAN_CODE`	**Overloaded**: product + surrender schedule + qualification	POLICY (×4 attributes)	`productType` , `planCode` , `surrenderChargeScheduleId` , *(qualification cross-check)*	**Decompose via a maintained lookup table.** `FIA10Q` → productType `FIXED_INDEXED_ANNUITY` , schedule `SC-10YR` , implies qualified	Lookup must be exhaustive — **an unmapped plan code is a hard stop, never a default**; cross-check the implied qualification against `QUALIFICATION_CODE` and **flag disagreements for manual review**	p
`ISSUE_DATE`	Contract issue date	POLICY	`issueDate`	**CYYMMDD → ISO 8601.** C=1 means 20xx, C=0 means 19xx	Valid date; ≤ today; ≥ carrier's earliest issue	i
`QUALIFICATION_CODE`	Tax regime, 1 char	TAX_QUALIFICATION	`type` (+ version record)	**The hardest mapping.** `Q` →? Traditional IRA, SEP, or 403(b)? The legacy code lost the distinction. **Requires a data-archaeology exercise** against source documents and 5498 history	**Never guess.** Where ambiguous, map to a `QUALIFIED_UNSPECIFIED` holding value and drive a remediation project. `N` → `NON_QUALIFIED` , `R` → `ROTH_IRA`	t
`ACCOUNT_VALUE`	Gross accumulation value	POLICY VALUES	`accountValue`	Implied decimal → decimal(2). **As-of date is the last cycle date, not "now"** — carry it	Non-negative (FIA); reconcile to segment sum; **must equal Σ segments**	C
`SURRENDER_VALUE`	Value on full surrender	POLICY VALUES	`cashSurrenderValue`	Same conversion	≤ accountValue; ≥ guaranteedMinimumValue; **recompute rather than trust if the cycle date is stale**	/
`RIDER_CODE`	Single rider slot	RIDER (collection)	`riders[]`	**One field → an array.** Legacy allows one rider; canonical must allow many. Look up code → rider definition, charge basis, benefit-base rules	**`GLWB` present but no benefit base in this file** → benefit base lives elsewhere (or nowhere). This is the mapping that exposes the worst legacy gap	/
`STATUS`	Contract status, 1 char	POLICY	`status` + `statusEffectiveDate`	`A` → `IN_FORCE` , `S` → `SURRENDERED` , `D` → `DEATH_CLAIM` , `L` → `LAPSED` , `P` → `PENDING` , `F` → `FREE_LOOK_CANCELLED`	Exhaustive mapping; **status effective date is usually not in this record — source it from a history file or the model loses "since when"**	s
`STATE`	Jurisdiction (**which one?**)	POLICY / ADDRESS	`issueState` **and/or** owner `address.stateProvince`	**Ambiguous by design.** Determine whether this is issue state or current residence — they diverge when owners move, and they drive *different* rules	Valid 2-char code; **if it's issue state, current residence must be sourced separately for withholding**	i
	Subject to				**Cross-check against qualification type** — a `Y` on a Roth or an `N` on a Traditional IRA for an	

`RMD_INDICATOR`	RMD, Y/N	TAX_QUALIFICATION	`rmdSubject`	`Y` →true, `N` →false	owner past the applicable age is a data-quality defect, and both exist in real books	`t`

12.3 The Canonical Output

[Illustrative]

```
{
  "carrierId": "CAR-0099",
  "policyNumber": "A-77120",
  "productType": "FIXED_INDEXED_ANNUITY",
  "planCode": "FIA10Q",
  "issueDate": "2024-03-15",
  "issueState": "TX",
  "status": "IN_FORCE",
  "statusEffectiveDate": "2024-03-15",
  "contractPhase": "ACCUMULATION",
  "taxQualification": {
    "type": "QUALIFIED_UNSPECIFIED",
    "sourceCode": "Q",
    "remediationRequired": true,
    "remediationReason": "Legacy single-char code cannot distinguish Traditional IRA / SEP / 403(b)",
    "rmdSubject": true,
    "effectiveDate": "2024-03-15"
  },
  "surrenderChargeSchedule": { "scheduleId": "SC-10YR", "derivedFrom": "PLAN_CODE" },
  "participants": [
    { "partyId": "000440017", "roleType": "OWNER",     "effectiveDate": "2024-03-15" },
    { "partyId": "000440017", "roleType": "ANNUITANT", "effectiveDate": "2024-03-15" }
  ],
  "riders": [
    { "riderCode": "GLWB", "riderType": "GUARANTEED_LIFETIME_WITHDRAWAL_BENEFIT",
      "benefitBase": null,
      "dataGap": "Benefit base not present in ANNMAST — source required" }
  ],
  "values": {
    "asOfDate": "2026-08-21",
    "accountValue": 110000.00,
    "cashSurrenderValue": 101200.00,
    "reconciliationStatus": "PENDING_SEGMENT_SUM_CHECK"
  },
  "_provenance": { "sourceSystem": "ANNMAST", "extractDate": "2026-08-22",
                   "mappingVersion": "1.4" }
}
```

12.4 What This Exercise Actually Teaches

The mapping table is not clerical work. **It is a discovery instrument.** Working through twelve fields surfaced:

1. **Overloaded codes** (`PLAN_CODE`) that must be decomposed — and the decomposition lookup becomes a permanent, maintained artifact.
2. **Lost distinctions** (`QUALIFICATION_CODE = 'Q'`) that **cannot be recovered by transformation** and require a data-remediation project with a business owner and a budget.
3. **Cardinality gaps** (one `RIDER_CODE` , one `OWNER_ID`) where the legacy model is strictly less expressive than reality — meaning some contracts in production are *misrepresented today*.
4. **Missing data** (rider benefit base) — the most serious finding, because it means a contractual value the carrier owes may not be reliably stored.
5. **Ambiguous semantics** (`STATE`) that drive different regulatory outcomes depending on the answer.
6. **Absent temporality** (no status effective date, stale value as-of date) — the legacy model knows *what is* but not *when it became so*.

Make these findings the headline of your mapping deliverable, not an appendix. A mapping document that only says “field A → field B” has wasted the discovery. The gaps are the value; they’re what justify the programme, and they’re what a Business Architect is uniquely positioned to surface and translate into business risk.

Chapter 13 — Business Architect Perspective

13.1 What a Business Architect Needs From ACORD

Not schema memorization. **Six things:**

1. **A shared vocabulary you didn't have to invent or defend.** The Business Glossary means "beneficiary" means one thing across operations, IT, actuarial, and compliance. Arguing definitions is pure waste; ACORD ends the argument by citation.
2. **A capability map you can inherit.** The Capability Model gives a defensible baseline for "what does this organization do," which is the foundation of application-portfolio rationalization, target-operating-model work, and API domain boundaries.
3. **A canonical model as an architectural decision, not a build.** The Information Model lets you say "we will have one definition of Policy" and point at something real, rather than proposing a two-year internal modelling effort.
4. **Traceability from capability to data element.** Capability → Process → Information Model entity → Data Model attribute → Message/API field → PAS field. That chain is what lets you answer "if we change X, what breaks?" — the question a Business Architect exists to answer.
5. **A neutral reference in vendor conversations.** "Does your platform express Party and Role separately?" is a better question than "is it flexible?", and ACORD gives you the standing to ask it.
6. **A migration and integration blueprint.** Everything in Chapters 11–12.

13.2 The Capability → Data Element Trace

[Illustrative]

```
CAPABILITY          Policy Servicing
    └ SUB-CAP       Contract Value Management
        └ PROCESS   Process Partial Withdrawal
            ├ INFORMATION MODEL   Policy · Party · PartyRole ·
            │                     FinancialActivity · TaxQualification · Rider
            ├ DATA MODEL          policy.account_value · activity.gross_amount ·
            │                     activity.taxable_amount · rider.benefit_base
            ├ API                 POST /withdrawals · GET /policies/{id}/values
            ├ MESSAGE             ACORD LA XML withdrawal transaction
            ├ PAS FIELD           ANNMAST.ACCOUNT_VALUE · RIDERF.BENEFIT_BASE
            ├ RULE                Surrender charge · MVA · GLWB excess test ·
            │                     free-withdrawal allowance · withholding default
            └ CONTROL             Two-phase quote/confirm · authority check ·
                                  idempotency · material-impact disclosure
```

Use this trace three ways: impact analysis ("we're changing the GLWB excess rule — what's affected?"), gap analysis ("we have no control on material-impact disclosure"), and requirements traceability ("this requirement maps to these fields and these tests").

13.3 Role Distinctions — Who Needs to Know What

Role	Primary ACORD concern	Depth needed	Typical deliverable
Business Analyst	**Business Glossary**; transaction-level business rules; the specific fields in their scope	Working knowledge of the entities their process touches; can read a mapping table and spot a wrong definition	Requirements, process flows, test scenarios, NIGO rules, UAT
Business Architect	**Capability Model, Process Model, Information Model**; the traceability chain end-to-end	Broad across all facets; deep in none of the technical schemas. Must be able to explain the canonical-model business case to a CFO and the domain to an engineer	Capability maps, target operating model, canonical model governance, roadmap, business case, impact analysis
Solution Architect	**Component Model, Information Model, Message/API standards**	Deep on how a specific solution realizes the standard; makes the build/buy/integrate calls	Solution designs, integration patterns, NFRs, sequencing, ADRs
Data Architect	**Information Model, Data Model, Business Glossary**; code lists	Deepest on the data facets; owns canonical entity definitions, lineage, quality, MDM, retention	Logical/physical models, lineage, data-quality rules, migration mapping, master-data strategy
Integration Architect	**Message standards (LA XML), API standards, versioning, extensions, certification**	Deepest on the wire — schemas, transforms, error handling, idempotency, TCF certification	Interface specs, transformation maps, ACL design, event contracts, partner onboarding

The distinction that matters most in practice: the Business Architect is the only role accountable for the trace being *complete* — that a business capability actually connects, through defined processes and entities, to real fields in real systems. Analysts own slices; solution and data architects own layers; **the Business Architect owns the vertical.**

13.4 Where ACORD Knowledge Pays Off Commercially

For someone positioning toward independent L&A consulting, these are the engagements where the material in this guide is directly billable:

- **PAS modernization assessment** — current-state mapping, gap analysis, target architecture, phased roadmap (Chapters 11–12 are the deliverable skeleton).
- **Carrier onboarding for a TPA** — canonical mapping of a new carrier's book; the mapping table *is* the deliverable.
- **Data migration** — legacy-to-target field mapping with the discovery findings of §12.4 as the risk register.
- **Integration/partner enablement** — DTCC/distributor connectivity, API design, ACORD certification.
- **Product launch readiness** — the §3.4 eight-question checklist applied to a new product, driving configuration, rules, and reporting requirements.
- **Business-rule extraction from legacy** — Chapter 14's disciplined approach, with humans authoritative.
- **AI readiness assessment** — semantic-layer design, RAG grounding, the deterministic boundary.

Chapter 14 — AI + ACORD

14.1 Why a Standardized Information Model Is an AI Prerequisite

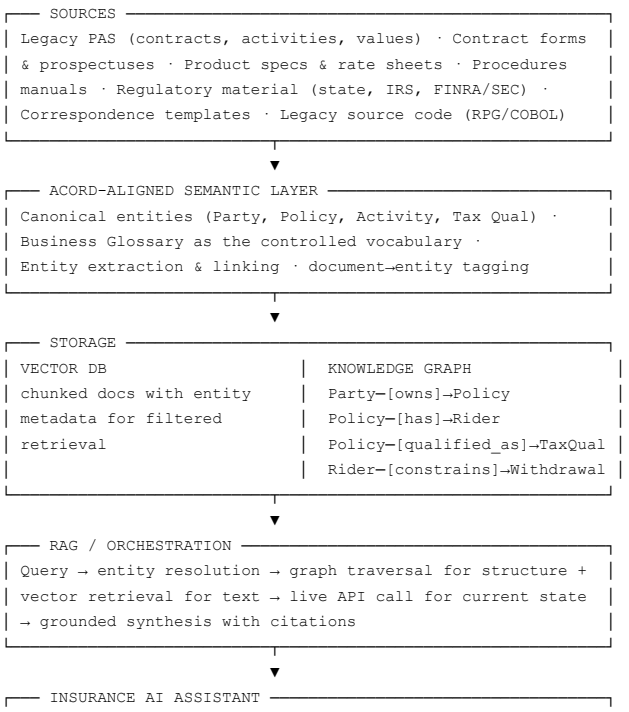
An LLM applied to a legacy insurance estate fails for a specific, diagnosable reason: **the data has no semantics.** `PLAN_CODE = 'FIA10Q'` means nothing to a model, and worse, means nothing *consistently* — because as Chapter 12 showed, it doesn't reliably mean one thing to the business either.

A canonical layer changes the input from opaque codes to typed, defined, related concepts. This is the difference between a system that can retrieve text about policies and one that can reason over policy structure.

The four capabilities a semantic layer unlocks:

Capability	Without canonical model	With canonical model
RAG	Retrieves document chunks; can't ground against contract state	Retrieves documents **and** structured contract facts; answers are grounded and citable
Knowledge graph	Not constructible — no defined entities or relationships	Entities and relationships are already defined; the Information Model **is** the ontology
Agents	Can only call opaque APIs; can't reason about consequences	Can reason over typed resources and read explicit constraints
Legacy comprehension	Reads RPG as text	Maps RPG variables to canonical concepts; extracted rules are expressible in business terms

14.2 Reference Architecture



```
| Service rep copilot · BA requirements assistant · legacy     |
| code explainer · document/NIGO triage · training             |
└──────────────────────────────────────────────────────────────┘
```

14.3 Where AI Helps — Concretely

- **Service-rep copilot.** "What happens if this customer takes $15,000?" → resolve contract, call the quote API (**deterministic calculation**), explain the result in plain language (**LLM**). The LLM narrates; it does not compute.
- **Requirements/BA assistant.** Draft user stories from a change request, mapped to canonical entities; find every place a rule is referenced across specs.
- **Legacy code comprehension.** Summarize what an RPG program does; **propose** a mapping of its variables to canonical concepts; identify candidate business rules. Output is a **hypothesis for human verification**, always.
- **Document triage.** Classify inbound service requests, extract fields, detect missing requirements → route as NIGO. High-volume, low-risk, reversible: the ideal AI use case.
- **Business-rule extraction.** Mine code, specs, and procedure manuals for rules; produce a candidate rule inventory with source citations. **The inventory is a draft for the rules engine, not a load into it.**
- **SLM/fine-tuning.** A small model trained on canonical schemas and glossary terms handles entity extraction and classification cheaply and predictably — often better than a large general model for structured tasks.

14.4 The Deterministic Boundary — the Most Important Section in This Chapter

Rule: anything that is legally prescribed, financially consequential, or must be identically reproducible must be computed deterministically. The LLM may explain the result. It must never produce it.

Must be deterministic	May be AI-assisted (with human review)
Surrender charge & MVA calculation	Explaining what a surrender charge is
Index credit at segment maturity	Summarizing how a crediting method works
GLWB benefit base, rollup, step-up, excess-withdrawal reduction	Describing the consequence to a customer
RMD amount	Reminding a customer that an RMD is due
Taxable amount, basis recovery, withholding	Explaining LIFO vs. exclusion ratio
1099-R distribution codes	Drafting a customer explanation of a 1099-R
Eligibility, suitability rules, authority	Flagging a case for review
Beneficiary determination & claim adjudication	Summarizing the file for the adjudicator
Anything reported to the IRS or a regulator	Drafting internal documentation about it

Four reasons, in order of severity:

1. **Legal correctness.** An RMD is defined by statute. A probabilistic approximation of a statutory formula is not a "mostly right answer" — it is a compliance failure.
2. **Reproducibility.** The same input must yield the same output in 2026 and in an audit in 2033. Non-determinism is disqualifying.
3. **Auditability.** A regulator asks *why* a number is what it is. "The model produced it" is not an answer. A rule engine's decision trace is.
4. **Consequence asymmetry.** A wrong explanation is corrected in a follow-up. A wrong withholding amount is an IRS penalty, a corrected 1099-R, and a harmed customer.

The architectural expression of this boundary:

```
Customer question
      ↓
 AI orchestrator     ← understands intent, resolves entities
      ↓
 ┌──────────────────────────────────┐
 | Calls DETERMINISTIC services:  |
 |   • quote API (rules engine)    |  ← the NUMBERS come from here
 |   • policy values API           |
 |   • RMD calculation service     |
 └──────────────────────────────────┘
      ↓
 AI synthesizes the EXPLANATION   ← the WORDS come from here
```

```
      ↓
Response, with the computed figures shown verbatim
and their source service cited
```

The LLM never recomputes, never rounds, never "adjusts" a returned figure. If it does, the design is wrong.

14.5 Governance Requirements

- **Provenance on every answer** — which contract, which API call, which document, which version.
- **Confidence and abstention** — the assistant must be able to say "I don't have that" rather than fabricating. Retrieval misses must surface as misses.
- **PII handling** — the semantic layer touches SSNs, health data, and financial detail. Retrieval must respect the same authorization as the API (§9.4); an AI assistant that bypasses entitlement checks is a data breach with a chat interface.
- **Human-in-the-loop by risk tier** — informational answers autonomous; customer-facing explanations reviewed; anything transactional gated.
- **Evaluation against known-correct cases** — maintain a regression suite of real historical transactions with verified outcomes.
- **Change control on the semantic layer** — it's now part of the control environment, not a side project.

Chapter 15 — Practical Case Study

"Customer Converts a Traditional IRA Deferred Annuity into a Roth IRA"

[Illustrative end-to-end trace — architecture and data elements are this guide's construction]

15.0 The Scenario

Margaret Chen (Party P-0001), age 65, holds contract **A-77120**, a fixed indexed annuity issued 2024-03-15 in Texas, currently qualified as a **Traditional IRA**, with an account value of $110,000 and a GLWB rider with a benefit base of $107,000. She wants to convert the full contract to a **Roth IRA** in tax year 2026, paying the resulting tax from outside funds.

Her financial professional initiates the request through a distributor portal.

STAGE 1 — Customer Request

What happens. Margaret and her advisor complete a Roth conversion request: full conversion, destination = new Roth IRA contract with the same carrier, federal withholding **0%**, state withholding **0%**, tax paid from a taxable brokerage account.

Entities touched. PARTY (P-0001), POLICY (A-77120), TAX_QUALIFICATION (current: TRADITIONAL_IRA).

Data elements captured. `requestingPartyId`, `sourcePolicyNumber`, `conversionType: FULL`, `destinationType: NEW_ROTH_CONTRACT`, `withholding.federalPercent: 0`, `withholding.statePercent: 0`, `taxPaymentSource: EXTERNAL`, `requestedEffectiveDate`, `irreversibilityAcknowledged: true`, `signatureDate`.

Critical control at this stage. Recharacterization of a conversion is no longer permitted. The acknowledgment is not boilerplate — it is the customer's confirmation that this cannot be undone. Capture it as a discrete, dated, auditable field, not as a checkbox buried in a PDF.

STAGE 2 — Channel

What happens. The distributor portal validates locally what it can (required fields, signature present) and submits.

Entities touched. PRODUCER (P-9001), DISTRIBUTOR (BD-4471).

Validations at the channel. Advisor is authorized on this contract; contract is visible within the distributor's entitlement scope; basic completeness.

What the channel must NOT do. Calculate the tax, decide eligibility, or default the withholding. Channels capture; they don't decide (§10.2).

STAGE 3 — API

Request.

```
POST /roth-conversions
Idempotency-Key: 7c1e-...-a29f
Authorization: Bearer <distributor-scoped token>

{
  "sourcePolicyNumber": "A-77120",
```

```
  "requestedBy": { "partyId": "P-0001", "roleType": "OWNER" },
  "conversionType": "FULL",
  "destination": { "type": "NEW_CONTRACT", "productCode": "FIA-10-GLWB",
                   "taxQualification": "ROTH_IRA" },
  "withholding": { "federalPercent": 0.0, "statePercent": 0.0,
                   "electionSource": "OWNER_ELECTION_W4R" },
  "taxPaymentSource": "EXTERNAL",
  "irreversibilityAcknowledged": true,
  "effectiveDate": "2026-08-22",
  "mode": "QUOTE"
}
```

Why `mode: QUOTE` first. The taxable amount is the single most important number in this transaction and the customer must see it before committing. Two-phase is mandatory here (§9.3).

API-layer responsibilities. AuthN/Z, idempotency key registration, schema validation, distributor scope check, correlation ID assignment.

STAGE 4 — Canonical Message / Model

The API payload is translated into the canonical model — the point at which distributor vocabulary becomes enterprise vocabulary.

```
CONVERSION_REQUEST
├── requestId: CONV-99341          correlationId: TXN-99341
├── sourceContract → POLICY A-77120
│     ├── productType: FIXED_INDEXED_ANNUITY   (unchanged by this transaction)
│     ├── taxQualification v1: TRADITIONAL_IRA (2024-03-15 → open)
│     ├── contractPhase: ACCUMULATION
│     └── riders: [GLWB-2024, benefitBase 107,000]
├── destinationContract → (to be created) taxQualification: ROTH_IRA
├── party → P-0001 (Owner, Annuitant)
└── conversionTerms → FULL, 0% federal, 0% state, external tax payment
```

Design note. If this were arriving over industry rails rather than a direct API, the same canonical structure would be populated from an ACORD LA XML message instead. **The canonical model is identical either way** — that's the entire point of the layer (§9.5).

STAGE 5 — Validation

Run in strict order; the most specific failure is returned.

#	Validation	Result	Notes
1	**Authority** — is P-0001 the Owner?	✅	Chapter 6.5 authority function; not merely authenticated
2	**Source convertible?**	✅	Traditional IRA is convertible. *(A SIMPLE IRA within 2 years of first contribution would fail here)*
3	**Same-person constraint**	✅	Destination Roth owner must be P-0001. **Cannot convert into another person's Roth**
4	**Contract state**	✅	In force, accumulation phase, not in free-look, no pending transaction
5	**Assignment / irrevocable beneficiary**	✅	None present
6	**FMV determination method**	⚠️	**The hard one — see below**
7	**Withholding election explicit?**	✅	0% elected by owner; **the system must permit 0% and must not silently default to 10%**
8	**Irreversibility acknowledged**	✅	Dated, captured
9	**RMD interaction**	⚠️	Owner is 65 — **below the current applicable RMD age**, so no RMD is outstanding. *If she were RMD-age, the RMD would have to be distributed first and is NOT convertible*
10	**Destination contract creatable**	✅	Product available in TX, age-eligible, Roth-eligible

Validation 6 — the FMV problem, in detail. For a conversion, the taxable amount is the **fair market value** of what's converted. For a plain MYGA that's

the account value. For an **annuity with a living-benefit rider**, Treasury guidance requires that the value include the actuarial value of additional contract features — meaning the FMV may exceed the account value. Here, the account value is $110,000 but the GLWB benefit base is $107,000 with guaranteed income attached.

What the architecture must do: call a **deterministic actuarial valuation service** with a documented, consistently-applied methodology, retain the inputs and result, and never let the number be estimated ad hoc per case. *[The specific valuation methodology must be verified against current IRS guidance and the carrier's actuarial policy — this guide does not prescribe a formula.]* For this trace, assume the valuation service returns **FMV = $112,400**.

Requirement raised. None blocking — the request is complete. *(Had the withholding election been missing, a `REQUIREMENT` of type `WITHHOLDING_ELECTION` would open, the transaction would go to `PENDING_REQUIREMENTS`, and correspondence would fire — the NIGO path of §4.0 stage 4.)*

STAGE 6 — Quote Returned, Customer Confirms

```
{
  "conversionId": "CONV-99341",
  "status": "QUOTED",
  "quoteExpiresAt": "2026-08-22T21:00:00Z",
  "valuation": {
    "accountValue": 110000.00,
    "fairMarketValue": 112400.00,
    "fmvMethodology": "ACCOUNT_VALUE_PLUS_RIDER_ACTUARIAL_VALUE",
    "fmvValuationRef": "VAL-2026-08-22-0117"
  },
  "taxImpact": {
    "taxableAmount": 112400.00,
    "basisInSourceIRA": 0.00,
    "federalWithholding": 0.00,
    "stateWithholding": 0.00,
    "netTransferredToRoth": 112400.00,
    "note": "Taxable in 2026. Tax to be paid from external funds."
  },
  "impacts": [
    { "type": "IRREVERSIBLE", "severity": "MATERIAL",
      "description": "Conversions cannot be recharacterized." },
    { "type": "ROTH_FIVE_YEAR_CLOCK", "severity": "MATERIAL",
      "description": "A 5-year clock applies to this conversion for penalty-free access to converted amounts." },
    { "type": "RIDER_CONTINUITY", "severity": "MATERIAL",
      "description": "GLWB rider benefit base of 107,000 continues on the new contract; rider terms unchanged." },
    { "type": "RMD_ELIMINATED", "severity": "INFORMATIONAL",
      "description": "Roth IRAs are not subject to RMDs during the owner's lifetime." }
  ]
}
```

Margaret confirms: `POST /roth-conversions/CONV-99341/confirm` with the same idempotency key.

STAGE 7 — PAS Execution

The structural decision. Two viable designs:

Design	Mechanics	Pros	Cons
A — New contract	Source terminated; new contract issued with Roth qualification	Clean tax lineage; unambiguous 1099-R/5498 pairing	New contract number; **surrender-charge schedule treatment must be carefully specified** (should continue, not restart)
B — In-place requalification	Same contract; `taxQualification` versioned to Roth	Customer familiarity; no downstream reference breakage	Requires dated, versioned qualification (§5.5); some reporting systems assume qualification is static

This trace uses Design A with an explicit rule that the surrender-charge schedule and rider benefit base **carry over unchanged** — because restarting a surrender clock on an internal conversion would be an unjustifiable customer harm and a suitability problem.

PAS operations, in order:

```
1. Snapshot source contract state @ 2026-08-22 (immutable record)
2. Liquidate segments on A-77120 at conversion FMV basis
3. Post FA-0442  DISTRIBUTION_ROTH_CONVERSION_OUT   A-77120  −112,400.00
                 taxableAmount 112,400.00 | withholding 0.00
```

```
                linkedConversionId CONV-99341
4. Terminate A-77120: status → CONVERTED (not "surrendered" — the
   distinction matters for reporting, retention metrics, and commission)
5. Create A-77330: productType FIA (unchanged), taxQualification ROTH_IRA
     ├── firstRothContributionYear: 2026
     ├── conversionClock: { year: 2026, amount: 112,400.00 }
     ├── rmdSubject: false
     ├── surrenderChargeSchedule: SC-10YR, **year 3 continuing**,
     │     originalIssueDate 2024-03-15 retained
     └── rider GLWB-2024: benefitBase 107,000.00 **carried forward**
6. Post FA-0443  CONTRIBUTION_ROTH_CONVERSION_IN    A-77330  +112,400.00
                 basis 112,400.00 (converted amounts are basis in the Roth)
                 linkedConversionId CONV-99341 | linkedActivity FA-0442
7. Re-create segments on A-77330 per allocation instruction
8. Update PARTY ROLES: Owner/Annuitant P-0001 on A-77330, effective 2026-08-22
```

Note the paired, cross-referenced activities. FA-0442 and FA-0443 must reference each other and the conversion ID. This pairing is what makes the transaction auditable, reconcilable, and reportable as one economic event.

STAGE 8 — Financial Activity & Tax Qualification Change

Entity	Before	After
Product type	FIXED_INDEXED_ANNUITY	FIXED_INDEXED_ANNUITY *(unchanged — §5.1)*
Tax qualification	TRADITIONAL_IRA	ROTH_IRA
Basis	$0 (pre-tax IRA)	**$112,400** (converted amount becomes Roth basis)
RMD subject	true	**false**
5-year clocks	n/a	firstContributionYear 2026; conversion clock 2026
GLWB benefit base	$107,000	$107,000 *(carried)*
Surrender charge year	3 of 10	3 of 10 *(continued, not restarted)*

The conceptual payoff. Every one of Chapter 5's dimensions moved independently: the product didn't change, the contract form didn't change, the tax qualification changed, and a transaction effected it. A model that fused these into one plan code cannot represent what just happened.

STAGE 9 — Withholding

Amount withheld: $0.00.

This is the correct outcome **only because the owner explicitly elected it**. The architecture must:

- permit a 0% election on a conversion (a system that enforces a 10% default actively harms the customer — the withheld amount would itself become a taxable, potentially penalized distribution that never reaches the Roth);
- record the election, its source (W-4R), and its date;
- still generate the withholding *record* with a zero amount, so the reporting projection is complete and the absence of withholding is affirmatively documented rather than merely missing.

STAGE 10 — Accounting

Dr	Cr	Amount	Notes
Annuity Reserve — Traditional IRA (A-77120)	Annuity Reserve — Roth IRA (A-77330)	112,400.00	Internal reclassification
(no cash entry)		0.00	**No money leaves the enterprise**

The lesson for the accounting interface. A taxable, fully reportable transaction of $112,400 occurs with **zero cash movement**. Any accounting or reconciliation process keyed on cash will miss it entirely; any tax process keyed on payment issuance will fail to report it. This is exactly the "taxable-with-no-cash" category flagged in §7.3.

STAGE 11 — Tax Reporting

Form	Contract	Content
1099-R	A-77120 (source)	Gross distribution $112,400.00; taxable amount $112,400.00; federal withheld $0.00; **distribution code identifying a conversion**; IRA/SEP/SIMPLE indicator; payer = carrier; recipient = P-0001
5498	A-77330 (destination)	**Conversion amount** $112,400.00 (reported in the conversion box, *not* as a regular contribution); year-end FMV; RMD indicator = not applicable

[Specific box numbers and distribution codes must be verified against current-year IRS instructions — they are effective-dated reference data, not constants (§4.28).]

Two reporting controls worth building explicitly:

1. **Pairing check** — every conversion 1099-R must have a matching 5498 conversion entry for the same owner, same year, same amount. An unmatched pair is a defect, detectable automatically, and it will otherwise surface as an IRS notice to the customer.
2. **Reproducibility** — the 1099-R must be regenerable from the append-only ledger years later (§4.29). The FMV valuation reference (`VAL-2026-08-22-0117`) must be retained, not just the resulting number.

STAGE 12 — Correspondence

Document	Trigger	Contents
Conversion confirmation	FA-0443 posted	Both contract numbers, FMV converted, taxable amount, withholding, effective date
New contract package	A-77330 created	Contract, endorsements, rider schedule, **surrender-charge schedule showing the carried-over start date**
Tax notice	Conversion complete	Plain-language explanation: taxable in 2026, 5-year clock, no RMDs, recharacterization unavailable
Source-contract termination notice	A-77120 status change	Confirms termination by conversion, not surrender
Year-end forms	January 2027	1099-R and 5498

STAGE 13 — Downstream Systems

System	What it receives	Why it matters
Data warehouse	Both activities, both contracts, qualification change	Retention analytics must classify this as **conversion, not lapse** — otherwise persistency reporting is wrong
Commission	Conversion event	Typically **no new commission** and no chargeback; misclassification as new business creates a real financial error
Distributor / DTCC	Position close + position open	Advisor's book must show continuity, not a lost account
Actuarial	Reserve reclassification, rider continuity	Rider liability persists on the new contract
Compliance archive	Full request, quote, acknowledgment, confirmation	The irreversibility acknowledgment is the key artifact if the conversion is later disputed
AI / semantic layer	Updated entity state	An assistant asked "why did my contract number change?" must be able to answer from the linked activities

---

15.14 What This Case Study Demonstrates

1. **Four dimensions moved independently** (§5.1) — product constant, qualification changed, transaction executed, contract form re-issued.
2. **Tax qualification is versioned, dated, and reason-coded** — not overwritten (§5.5).
3. **A material transaction with zero cash movement** breaks any design that couples tax reporting to payment issuance (§7.3).
4. **Two-phase quote/confirm is a compliance control** — the customer saw a $112,400 taxable amount before an irreversible act (§9.3).
5. **The FMV calculation is deterministic and documented**, never inferred and never AI-generated (§14.4).

6. **Paired, cross-referenced activities** make one economic event auditable across two contracts.
7. **Carried-over surrender schedule and benefit base** are deliberate configuration decisions with direct customer-harm implications if made carelessly.
8. **Downstream classification matters as much as processing** — the same event mislabelled becomes a false lapse, a wrong commission, and a misleading persistency number.

Chapter 16 — Knowledge Check

*150 questions. Answers and explanations follow in §16.5 — work through the questions first. These are original questions written for this guide. They are **not** reproductions or reconstructions of ACORD ACE examination content.*

16.1 Conceptual Questions (1–50)

1. What does ACORD stand for, and in what year was it founded?
2. Is ACORD an insurance company? What is its legal/organizational form?
3. What was ACORD's original scope, and how did it expand?
4. Name the two broad categories of ACORD output relevant to an L&A architect.
5. What problem does a standards body solve that bilateral integration cannot?
6. Explain the N×N integration problem and how a canonical model changes it.
7. What is the current ACORD Life & Annuity XML standard release, and what does its version cadence imply for trading-partner agreements?
8. Name the main components of an ACORD L&A Standards release.
9. What is the ACORD Reference Architecture, and how does it differ from ACORD Data Standards?
10. Name the six publicly identifiable facets of the Reference Architecture described in this guide.
11. What is the ACORD Business Glossary, and roughly how many terms does it define?
12. What is the ACORD Information Model, and why is it called a "semantic hub"?
13. How does the Data Model differ in purpose from the Information Model?
14. What does the Capability Model describe, and what does it deliberately *not* describe?
15. How many high-level capabilities does the Capability Model define, and how is it decomposed?
16. What is the Component Model used for in a merger or consolidation scenario?
17. Give three uses of the Process Model for a Business Architect.
18. Define "canonical data model" in one sentence.
19. Why is the Information Model, rather than a message schema, the right place for canonical meaning?
20. What is TXLife, and what is its relationship to TXLifeResponse?
21. What is OLifE, in general terms?
22. Why does this guide decline to reproduce ACORD element names and type codes?
23. What is the ACORD Testing and Certification Facility, and what are its two validation stages?
24. What is a "type code" (tc) in ACORD schema terms?
25. Why do ACORD schemas provide defined extension points?
26. What risks arise when a carrier extends a standard schema without discipline?
27. Contrast XML and JSON for L&A integration on four dimensions.
28. Why is exposing `POST /txlife` with a type code in the body considered an anti-pattern?
29. What is the DTCC's relevance to L&A data exchange?
30. What is an anti-corruption layer, and what two directions does it protect?
31. What is the strangler pattern, and when is "never finishing" it acceptable?
32. Name four adapter strategies for exposing an IBM i PAS.
33. Why is CDC off database journals described as high-leverage in a legacy estate?

34. What does “the PAS is the system of record; everything else is a projection” mean operationally?
35. Why should business rules be configuration rather than code?
36. What distinguishes a Business Analyst’s ACORD knowledge from a Business Architect’s?
37. Which role needs the deepest knowledge of message schemas and certification?
38. Which role owns the *vertical* traceability chain, and why does that matter?
39. What is the traceability chain from capability to PAS field?
40. Why is a shared vocabulary a cost-saving measure and not merely a nicety?
41. What is a “holding” and why is the abstraction useful?
42. Why is Party separated from Role? Give three independent reasons.
43. Distinguish authentication, role, and authority.
44. Why must beneficiary designations be versioned rather than overwritten?
45. What are the four orthogonal dimensions that must not be conflated in an L&A data model?
46. Why is `PLAN_CODE` overloading described as an anti-pattern?
47. What does “append-only financial ledger” mean, and why is it non-negotiable?
48. Name three activity types that change value with no cash movement.
49. Name two taxable events that involve no cash movement.
50. Why must tax reporting be a projection over the ledger rather than a query over current balances?

16.2 Scenario Questions (51–100)

51. A customer requests a $12,000 withdrawal from an FIA with a $10,000 free-withdrawal amount and a GLWB allowing 5% annually on a $107,000 benefit base. What must the quote disclose?
52. A distributor’s system times out during a withdrawal submission and retries. What single design feature prevents a duplicate withdrawal?
53. An owner requests a transfer between index strategies three months into a 12-month segment. What is the likely answer and why?
54. A trust owns a non-qualified deferred annuity. What tax question must the system be able to raise?
55. An IRA owner asks to add their spouse as joint owner. How should the system respond?
56. A contract has an irrevocable beneficiary. The owner requests a beneficiary change. What happens?
57. An address change is submitted, followed two days later by a full surrender request. What control should fire?
58. A 1035 exchange arrives with no cost basis from the ceding carrier. What should the receiving carrier do?
59. A customer wants to 1035 exchange an annuity into a life policy. Is this permitted?
60. A UL policy has zero account value but an intact no-lapse guarantee. Should it lapse?
61. A policy lapses with an outstanding loan and embedded gain. What is the tax consequence, and why do customers dispute it?
62. A SIMPLE IRA owner requests a Roth conversion 14 months after their first contribution. What rule applies?
63. An RMD-age IRA owner requests a full Roth conversion. What must happen first?
64. A customer takes an indirect rollover and asks the carrier to confirm they’re within the one-per-year limit. Can the carrier confirm this?
65. A beneficiary calls to request a withdrawal while the owner is alive. What is the outcome, and at which control point?
66. A POA holder requests a beneficiary change. What must be verified beyond identity?
67. An owner dies on a contract where the annuitant is a different living person. Which death benefit rules apply?
68. A primary beneficiary predeceased the owner and no contingent is named. Where do proceeds go?
69. A minor is named as beneficiary and the owner dies. What complication arises?
70. Two beneficiaries are named at 60% and 50%. What happens at intake?
71. An owner surrenders a contract with a $180,000 GLWB benefit base and a $120,000 account value. What do they receive, and what is the disclosure obligation?
72. A GLWB rider terminates automatically on an ownership change. What must the ownership-change quote show?
73. A customer requests “an allocation change.” What clarifying question is essential?

74. A contract's index strategy references an index that is discontinued mid-segment. What governs?
75. A segment matures on a market holiday. What must the system handle?
76. A RILA contract with a 10% buffer experiences a 25% index decline at segment maturity. What is credited?
77. An FIA experiences a 25% index decline at segment maturity. What is credited?
78. A customer receives a renewal rate below the contractual guaranteed minimum. What has gone wrong?
79. A MYGA reaches term end and the owner takes no action. What should happen, and what is the regulatory exposure?
80. A contract reaches maturity and a default annuitization option is applied automatically. What is the recurring complaint risk?
81. A free-look cancellation is requested on a RILA whose value has fallen 8%. What determines the refund amount?
82. A free-look cancellation follows a 1035 exchange that funded the contract. Why is this messy?
83. A customer's state of residence changes from Texas to California. What must be recalculated?
84. A foreign-address change is processed. What new requirement arises?
85. An owner elects 0% federal withholding on a Roth conversion. Should the system permit it, and why does it matter?
86. A distribution is made to a payee with no valid TIN. What follows?
87. An eligible rollover distribution is paid directly to the participant. What withholding applies?
88. A systematic withdrawal plan and an RMD election both exist on the same contract. What is the risk?
89. An RMD calculation requires prior-year 12/31 FMV on an FIA with a GLWB. Why is this harder than it looks?
90. A gratuitous transfer of a non-qualified annuity is made to an adult child. What is the tax consequence and the reporting oddity?
91. A new-business application is submitted by a producer licensed in Texas for an owner residing in Oklahoma. What fails?
92. A RILA application is submitted by a producer with an insurance license but no securities registration. What fails?
93. Additional premium is received on an FIA. What is created, and what is *not* necessarily increased?
94. A carrier onboards to a TPA platform with plan codes that collide with an existing carrier's. What design decision saves you?
95. An analytics team wants to write corrected values back into the warehouse and have the PAS pick them up. What is wrong with this?
96. A statement shows a different account value than the API. What has likely gone wrong architecturally?
97. A quote is issued at 14:00 and confirmed at 22:00 the same day. What control matters?
98. A conversion 1099-R has no matching 5498. What does this indicate?
99. A customer disputes a 2023 index credit. What must the system be able to do?
100. Withholding rates change on January 1. What determines whether your system handles it gracefully?

16.3 Architecture Questions (101–125)

101. Draw or describe the six-layer flow from Business Capability to Insurance Application.
102. What are the ten layers of the PAS reference architecture in this guide?
103. State the three boundary rules that prevent architectural decay.
104. What must the canonical/ACORD integration layer never do?
105. What must the correspondence layer never do, and why?
106. Why must analytics never be a write path?
107. In a TPA/BPO context, what becomes a partition key, and where does it apply?
108. Contrast shared-schema multi-tenancy with per-carrier instances for an L&A TPA.
109. Why does the canonical model produce ROI faster at a TPA than at a single carrier?
110. Describe the five-stage legacy modernization sequence in Chapter 11.
111. Why is the read façade built before the write path?
112. What is the single rule that prevents an ACL from becoming a second PAS?
113. Why should core ledger functionality *not* be decomposed into microservices?
114. What is a parallel run, and when is it mandatory?

115. Why are legacy RPG/COBOL developers described as the most valuable people in a modernization programme?
116. Design the resource model for a withdrawal. Why is it a resource rather than a verb on the policy?
117. Why must `GET /policies/{id}/values` return multiple values?
118. What does `?asOf=` support on the participants endpoint enable?
119. What are the five layers of API validation, in order?
120. What belongs in an event payload, and what does not?
121. What is the correct behaviour when a repeated idempotency key is received?
122. How should distributor API tokens be scoped, and what additional claim is needed at a TPA?
123. Describe the AI reference architecture from sources to assistant.
124. State the deterministic boundary rule and the four reasons behind it.
125. What governance controls must an insurance AI assistant have?

16.4 Data-Model Questions (126–150)

126. Model the relationship between Party, Party Role, and Holding, with cardinalities.
127. Which party types may play the Annuitant role, and why is the restriction absolute?
128. What role-specific payload does a Beneficiary role carry that an Owner role does not?
129. How do you represent one person who is both Owner and Annuitant on the same contract?
130. What attributes make a Party Role reconstructable as of a past date?
131. List the six simultaneously valid "values" on a deferred annuity.
132. Why is a GLWB benefit base not stored as the contract's account value?
133. On what value is a GLWB charge computed, and from what value is it deducted?
134. What is a segment, and what attributes must it carry?
135. Why can a contract-level "index rate" field not represent an FIA correctly?
136. Model the five-level transaction hierarchy with an example of fan-out at each level.
137. Distinguish the three dates every financial activity must store.
138. How should a correction to a posted financial activity be represented?
139. What fields make cost basis defensible, and why is a single mutable field insufficient?
140. How should tax qualification be modelled so a Roth conversion is representable?
141. What must a qualification version record contain beyond the type?
142. How do you represent an inherited IRA's distribution regime?
143. Map `PLAN_CODE = 'FIA10Q'` to canonical attributes and identify what cannot be derived.
144. `QUALIFICATION_CODE = 'Q'` is ambiguous. What is the correct handling and what must *not* be done?
145. A legacy record has one `RIDER_CODE` field. What does this reveal about the book of business?
146. A legacy `STATE` field is ambiguous. What two things could it mean and why does the difference matter?
147. Why must `statusEffectiveDate` be sourced even though `STATUS` is present?
148. What does a missing rider benefit base in the legacy master indicate, and how serious is it?
149. Design the canonical representation of a 1035 exchange spanning two carriers.
150. What is the minimum entity set needed to represent a death claim with three beneficiaries electing different payout options?

16.5 Answers and Explanations

Conceptual (1–50)

1. **Association for Cooperative Operations Research and Development**, founded **1970**.

2. No. It is a **non-profit, industry-owned standards-setting body**. It does not underwrite, process claims, or provide coverage.
3. Originally formed by US P&C carriers and agents to standardize paper forms for new business and claims; expanded in the late 1970s into electronic data standards, then into additional lines (including Life & Annuity) and a cross-line enterprise architecture framework.
4. **(a) ACORD Data Standards** — line-of-business message standards (LA XML, EDI); **(b) ACORD Reference Architecture** — the cross-line enterprise architecture framework.
5. It converts every bilateral negotiation about *meaning* into a one-time mapping to a shared definition — and it does so with an authority no single carrier could impose on its trading partners.
6. N systems require ~N×(N–1) point-to-point mappings; each new system multiplies the burden. A canonical model makes it N mappings (each system maps once to the hub), turning multiplicative growth into additive.
7. **v2.49.00 Final (Dec 2025)**, following v2.48.00 (Dec 2024) and v2.47.00 (Dec 2023). The roughly annual cadence means trading partners must **agree on and certify against a specific version**, and version negotiation is part of partner onboarding.
8. A **Help File** (the primary reference document), an **Object Reference Model** (the high-level object hierarchy), and **schema files** for XML validation.
9. Data Standards are line-specific wire formats for exchanging data between organizations. The Reference Architecture is a technology-neutral, cross-line **enterprise architecture framework** of interrelated models used to design and organize an insurer's own architecture. One is for the wire; the other is for the enterprise.
10. Business Glossary, Information Model, Data Model, Capability Model, Process Model, Component Model. *(A seventh exists; this guide flags it as requiring verification rather than guessing.)*
11. A set of standardized, non-technical insurance definitions with synonyms and line-specific usage — **6,000+ terms**.
12. A formal, technology-neutral conceptual model of insurance data across all lines. It is the "semantic hub" because both storage models and message formats map back to it, giving any-to-any integration through a single source of meaning.
13. The Information Model captures **meaning** (conceptual, technology-neutral). The Data Model captures **structure for storage** (logical/physical), and is derived from it — used for canonical stores, ODS, and warehouses.
14. It describes **what** an organization does — abilities and activities across the value chain. It deliberately does not describe **how** it's done, nor the org structure, processes, or people.
15. **Ten** high-level capabilities, decomposed across **three additional levels**, grouped into tiers supporting strategy, planning, execution, and analysis.
16. It decomposes business functions into modular, technology-neutral components with standardized interfaces, so architects can decide which capabilities from each merging entity to retain, refine, or retire — preserving valuable legacy capability while removing redundancy.
17. Process engineering/redesign; aligning transaction messages to industry norms so exchanges interoperate; and identifying where automation or AI fits within a defined process.
18. A single system-agnostic representation of a business object that all systems and messages map to, instead of mapping directly to one another.
19. Because messages change format (XML→JSON→whatever's next) and vary by trading partner, while meaning must be stable. Anchoring canonical meaning to a wire format couples your semantics to a serialization decision.
20. `TXLife` is the request envelope carrying authentication/routing plus one or more transactions; `TXLifeResponse` is the corresponding response envelope carrying result/status and any returned data.
21. The Life & Annuity business object model beneath the transaction — the hierarchy containing Party, Holding, Policy, and related objects. *(Current structure requires verification against licensed documentation.)*
22. Because they are member-licensed content, and because reproducing them from memory or third-party mirrors risks propagating outdated or wrong definitions into an implementation — a worse outcome than an honest gap.
23. ACORD's validation portal. **Stage 1:** schema/DTD syntax and structure. **Stage 2:** manual review for business-rule and data-quality compliance. Passing both yields a certification report.
24. An enumerated code value from a controlled ACORD code list, used to constrain a field to a defined vocabulary (relationship types, transaction types, etc.).
25. So carriers can carry proprietary data without breaking base-schema validation or forking the standard — preserving interoperability while accommodating real product variation.
26. Extensions become de-facto private standards; trading partners can't consume them; future standard versions may conflict; and the extension often encodes a concept that *should* have been raised to the standards body. Undisciplined extension is how a standard stops being one.
27. **Schema validation:** XML mature and certifiable; JSON has OpenAPI/JSON Schema but no industry certification. **Adoption in L&A:** XML dominant on industry rails; JSON growing internally. **Verbosity:** XML high, JSON low. **Legacy tooling:** extensive for XML, requires investment for JSON.
28. Because it is RPC wearing REST's clothes — one opaque endpoint with a type discriminator. You lose resource identity, cacheability, meaningful HTTP semantics, and the ability to model a transaction's lifecycle (quote → pending → confirmed), while inheriting all the message model's coupling.

29. DTCC provides industry clearing/communication rails for annuity and insurance transactions between carriers and distributors — position feeds, applications, commissions, values. ACORD L&A EDI/XML specifications support these exchanges.
30. A deliberate translation boundary between a legacy model and a modern canonical model. It prevents **legacy concepts leaking outward** into the enterprise model and **modern concepts leaking inward** into the legacy system.
31. Incrementally replacing a legacy system capability-by-capability behind a stable façade until it can be retired. Never finishing is acceptable when the remaining legacy scope is stable, small, and cheaper to run than to replace — **provided it is named as a decision rather than treated as a failure**.
32. (a) Expose existing RPG logic as callable services; (b) read DB2 for i directly for query paths; (c) consume batch extracts where real-time isn't required; (d) CDC off journals to generate events.
33. Because it makes a batch system emit near-real-time events **without changing the batch system**. It decouples every downstream consumer from the nightly cycle at low risk and without touching proven code.
34. Only one system may authoritatively hold contract state and values. Every other store (warehouse, cache, statement system, distributor feed) is a derived copy that must be reconcilable back to it and must never be independently corrected.
35. Because rates, schedules, tables, and state variations change on legislative and product cycles, not release cycles — and they vary across 50+ jurisdictions. Effective-dated configuration can be changed and audited without code deployment; hard-coded rules guarantee an annual defect.
36. The BA needs working knowledge of the Glossary and of the entities and rules within their process scope. The Business Architect needs breadth across all facets and ownership of the traceability chain — capability through to field — plus the ability to make the business case.
37. The **Integration Architect** — schemas, transformations, error handling, versioning, extensions, and TCF certification.
38. The **Business Architect**. Analysts own process slices; solution and data architects own layers. Only the Business Architect is accountable for the vertical connection actually being complete — which is what makes impact analysis possible.
39. Capability → Process → Information Model entity → Data Model attribute → Message/API field → PAS field, with rules and controls attached at the relevant levels.
40. Because definitional disagreement consumes real project time in every requirements session, every test cycle, and every defect triage — and produces systems that disagree with each other. A cited external definition ends the argument rather than re-litigating it per project.
41. The generic container for "the thing a party holds" — a life policy, annuity contract, or account. It's useful because it lets one structure carry heterogeneous holdings, enabling portfolio-level queries across product lines without unioning unrelated structures.
42. (a) One person holds many roles across many contracts — duplicating them multiplies maintenance and error; (b) roles are time-bounded and must be reconstructable as of a past date while the person is not; (c) roles carry role-specific data and differing cardinality that cannot be expressed as fixed columns on a policy record.
43. **Authentication** = are you who you say you are. **Role** = what capacity do you hold on this contract. **Authority** = does that role, as scoped and documented, permit this specific action now. All three must pass; passing one or two is the classic fraud gap.
44. Because a death claim must be adjudicated against the designation **in force at the date of death**, which may be years before the claim. Overwriting destroys the evidence and is a direct litigation exposure.
45. **Product type, contract type, tax qualification, transaction type.**
46. Because it encodes multiple orthogonal dimensions in one string. It becomes wrong when qualification changes, prevents cross-qualification product reporting, causes combinatorial code explosion, and loses distinctions permanently (e.g. "Q" can't separate Traditional IRA from 403(b)).
47. Financial activities are never updated in place; corrections are new reversing plus correcting entries. It's non-negotiable because tax reporting, statement reproduction, audit, and dispute resolution all require reconstructing values exactly as they stood at any past date.
48. Interest crediting, index crediting, rider charges, surrender charges, MVA, GLWB rollup/step-up — any of these. (Benefit-base rollup is the purest example: contractual value changes, no cash, no account-value movement.)
49. Gratuitous transfer of a non-qualified annuity to a non-spouse (§72(e)(4)(C)); lapse of a policy with an outstanding loan and embedded gain. A Roth conversion within one carrier is a third.
50. Because balances are mutable and reflect *today*; a 1099-R must reflect a specific past year and be reproducible years later, including after subsequent transactions and corrections. Deriving reports from live balances makes prior-year reproduction impossible.

Scenario (51–100)

51. Gross amount required, surrender charge on the $2,000 excess, any MVA, federal/state withholding, taxable amount, net proceeds, **and — most importantly — that $12,000 exceeds the $5,350 GLWB annual allowance, triggering a pro-rata benefit-base reduction and a permanent decrease in guaranteed lifetime income**, with before/after figures.
52. **Idempotency keys** on the write endpoint, with the key stored against the resulting resource so a repeat returns the original result rather than creating a second transaction.
53. Almost certainly declined. FIA/RILA index strategies generally permit transfers **only at segment maturity**, because the segment's economics (option cost, cap) are set for its full term. This is the most common customer misunderstanding in indexed products.

54. Whether **§72(u)** applies — a non-natural owner on a non-qualified deferred annuity generally loses tax deferral unless the trust holds as agent for a natural person. The model must be able to express "owner is an Organization" for the rule to fire at all.

55. Decline. **IRAs are individual by statute** — joint ownership is not permitted. The correct response is to explain and redirect, likely to a beneficiary designation instead.

56. The change **cannot be completed by the owner alone** — the irrevocable beneficiary's consent is required. The system should surface this as a blocking constraint at quote time, not discover it at processing.

57. **Enhanced verification / potential account-takeover control.** Address change followed closely by a distribution request is a classic fraud pattern; it should trigger out-of-band verification against the *prior* contact details.

58. **Not guess.** Pursue the basis from the ceding carrier as a formal requirement, retain their statement, and treat unresolved basis as a blocking data issue — because a wrong basis produces a wrong 1099-R and a customer tax problem the receiving carrier will own.

59. No. **Annuity → life is not a valid §1035 exchange.** Life → annuity is valid. Direction matters and must be validated.

60. **No.** The secondary/no-lapse guarantee must be tested before any lapse. A zero account value with an intact guarantee is a fully in-force policy; lapsing it is a serious error with severe consequences.

61. The lapse triggers **taxable income to the extent of gain**, with the loan treated as received — so the owner gets a 1099-R with no cash. Customers dispute it because they received nothing at lapse; the taxable event occurred years earlier in economic substance.

62. The **SIMPLE IRA 2-year rule** — within 2 years of the first contribution, distributions/rollovers to non-SIMPLE plans are penalized (25%) and conversions are restricted. At 14 months, the rule applies.

63. The **RMD for the year must be distributed first**. RMDs are not eligible for conversion; converting an RMD is an excess contribution to the Roth.

64. **No.** The one-per-12-month IRA rollover limit applies across *all* the owner's IRAs at all institutions; a single carrier cannot see the others. The carrier must obtain an **owner attestation** and document it.

65. **Denied at the authority check** (§6.5) — the beneficiary may authenticate successfully and hold a genuine role, but has no authority to transact while the owner lives. This is precisely why role ≠ authority.

66. That the POA document **grants beneficiary-change authority specifically** — many POA forms exclude it, and many carriers require explicit language or their own form. Also: document validity, scope, non-revocation, and any state-specific requirements.

67. **Owner-driven** death rules under **§72(s)** apply — the owner's death triggers required distribution rules even though the annuitant is alive. Contracts and systems that assume annuitant-driven death benefits get this wrong, and the error surfaces at claim time.

68. To the **contingent** beneficiary if one exists; here none is named, so to the **estate** by default — which loses designated-beneficiary status and worsens the tax outcome. This is why encouraging contingent designations is a genuine customer-service function.

69. A minor cannot receive proceeds directly. A **guardian, custodian (UTMA), or trust** must be established — often requiring court involvement, causing significant delay. The system must flag minor beneficiaries at designation time so the owner can plan.

70. **Rejected at intake.** Percentages must total 100% **per class**. The requirement should be raised immediately with the specific defect identified, not discovered at claim time.

71. **$120,000 less applicable charges.** The $180,000 benefit base is notional — payable only as guaranteed income, never as cash. The disclosure obligation is substantial: the quote must make clear that surrendering forfeits $60,000 of notional value and the entire income guarantee.

72. That the ownership change **destroys the GLWB benefit base and the income guarantee**, with the specific dollar figures. A routine-seeming administrative change silently eliminating a six-figure notional value is exactly the harm the quote/confirm pattern exists to prevent.

73. **"Do you want to change where *future* premium goes, or move *existing* value?"** These are two different transactions (allocation change vs. transfer/rebalance) with different rules and — on an FIA — radically different timing. Customers describe both the same way.

74. The contract's **index substitution language**, which every FIA/RILA form contains. The carrier substitutes a comparable index per the contract terms. This has been invoked in practice and must be a supported system function, not a manual exception.

75. The contract's stated convention for non-business days — typically the **next (or prior) available index publication date** per the contract language. It must be configured and applied consistently, and it must be reproducible for later dispute.

76. With a 10% buffer, the carrier absorbs the first 10%; the owner bears the remainder: **–15%**. The RILA account value declines.

77. **0%.** FIAs have a 0% floor — the owner is credited nothing, but loses no principal to index performance.

78. A serious defect. The renewal rate **cannot fall below the contractual guaranteed minimum**. This indicates either a rate-setting error or a missing floor validation, and it is a regulatory issue as well as a customer one.

79. The contract's default renewal provision applies — typically renewal at the current rate. The **regulatory exposure is the advance notice**: state rules require notice within a specified window before term end, and whether a **new surrender-charge period attaches** is a significant consumer-protection issue that must match the filed form.

80. That the owner **didn't understand or didn't receive the notice**, and a default option (often life-only annuitization) permanently and irreversibly changed their contract. Advance notice, clear election options, and a documented default are essential.

81. **State law and contract language.** Most states require **return of premium**; some require **return of account value**. On a RILA that has fallen 8%, this distinction is the difference between the owner receiving their full premium and receiving 92% of it.
82. Because the **ceding contract may already be terminated**. Unwinding requires coordination with the ceding carrier, potential reinstatement, and careful handling so the customer isn't left with a taxable distribution from a 1035 that no longer has a valid receiving contract.
83. **State withholding defaults** (California has its own rules and mandatory-withholding interactions), **premium tax jurisdiction**, correspondence, and any state-specific contract provisions. Residence state and issue state now differ, and both matter for different rules.
84. **W-8BEN** to establish foreign status and any treaty rate; absent it, **30% NRA withholding** applies. Also 1042-S reporting rather than 1099-R, and enhanced sanctions screening.
85. **Yes, it must permit it — and it matters enormously.** Withheld amounts don't reach the Roth; they are themselves a taxable distribution and, if the owner is under 59½, potentially penalized. A system that force-defaults 10% on conversions actively harms customers.
86. **Backup withholding** at the statutory rate, plus a B-notice workflow to obtain a valid TIN. The reporting must proceed with the deficiency documented.
87. **Mandatory 20% federal withholding** on eligible rollover distributions paid to the participant — which is precisely why direct trustee-to-trustee transfers are preferable and why the direct/indirect distinction must be captured explicitly.
88. **Double distribution** — the RMD paid separately *and* satisfied again through the SWP, distributing more than intended and creating an unnecessary taxable event. The model must express "this SWP satisfies the RMD obligation" as an explicit relationship.
89. Because FMV for RMD purposes must reflect the **actuarial value of additional contract features**, not just the account value. It requires a deterministic valuation service with a documented methodology, retained inputs, and consistent application — not a per-case estimate.
90. Generally **recognition of gain under §72(e)(4)(C)** — a taxable event. The reporting oddity: a **1099-R is issued with no cash movement whatsoever**, which breaks any process that keys tax reporting to payment issuance.
91. **Producer licensing/appointment in the owner's state.** The producer must be licensed and appointed in **Oklahoma** (the owner's state), not merely Texas. This is one of the most common NIGO causes.
92. **Securities registration.** A RILA is an SEC-registered security sold by prospectus; the producer needs both insurance licensure and securities registration, with an affiliated selling broker-dealer.
93. A **new segment with the current rate set** is created (potentially with its own surrender-charge schedule on rolling-charge products). What is **not** necessarily increased is the **GLWB benefit base** — many riders credit additional premium at a lower rate or not at all.
94. **Carrier as a partition key** — plan codes are unique only *within* a carrier. Any design treating plan code as globally unique fails at carrier #2 and requires retrofitting under time pressure.
95. Analytics is a **read-only projection**, never a write path (§10.2). Corrections must be made in the system of record and flow outward; a write-back path creates two sources of truth, unreconcilable divergence, and an uncontrolled change route into financial data.
96. Correspondence is **recalculating values instead of rendering what it was given** (§10.2), or is reading from a stale projection. Either way, two layers now independently compute value — the exact condition that guarantees permanent manual reconciliation.
97. **Quote expiry.** Values move; an 8-hour-old quote on an indexed contract may no longer be accurate. The quote must carry an expiry and the confirm must revalidate — either re-quoting or rejecting if materially changed.
98. A **pairing defect** — either the destination contribution wasn't reported, the amounts disagree, or the transaction was misclassified. It will surface as an IRS notice to the customer. This is why an automated pairing check belongs in the tax-reporting controls.
99. Reproduce the exact calculation: the segment's start and end dates, start and end index values, the crediting method, and the cap/participation/spread **as of segment start** — plus retain the source of the index values used. Without segment-level, dated storage, this is indefensible.
100. Whether rates are **effective-dated reference data** (graceful — update the table) or **hard-coded in program logic** (a release, a regression test cycle, and a January defect window). This single design choice recurs annually and is the cleanest test of whether §10.2's rule was followed.

Architecture (101–125)

101. Business Capability → Business Process → Information Model → Data Model → Message/API → Insurance Application. A capability is realized by processes; processes consume/produce Information Model concepts; those are persisted per the Data Model and exchanged via Messages/APIs; the application executes.
102. Channels; API Gateway; ACORD/canonical integration; Business services; PAS; Rules engine; Accounting; Tax reporting; Correspondence; Data & analytics.
103. (a) The PAS is the system of record — everything else is a projection; (b) rules are configuration, not code, and live in one place; (c) the canonical layer translates, the business layer decides.
104. **Make business decisions.** It translates representations and validates structure; the moment it decides eligibility or calculates a charge, it has become an undocumented second PAS.
105. **Recalculate values.** It renders what it is given. Otherwise statements and the PAS will disagree, and the customer holds the document that says something different from the system.

106. Because a write path creates a second source of truth. Analytics stores are optimized for reading, often lossy or aggregated, and outside the control environment governing financial data.

107. **Carrier.** It applies to data, configuration, rules, correspondence templates, tax-reporting entity, and API authorization scope. Product/plan codes are unique only within a carrier.

108. Shared-schema gives operational leverage and cross-carrier analytics but demands rigorous partition enforcement and complicates carrier-specific regulatory requests. Per-carrier instances are safer and simpler to isolate but multiply upgrade, support, and infrastructure cost. Most L&A TPAs run a hybrid — shared services, partitioned data.

109. Because onboarding each additional carrier's legacy book becomes a **one-adapter problem** rather than a re-platform. The N-to-1 economics compound with every carrier added, which a single carrier never experiences.

110. (1) Read façade; (2) events via CDC; (3) write path for low-risk non-financial transactions; (4) write path for financial transactions with parallel-run reconciliation; (5) selective replacement of new products behind the same façade.

111. Because it is **low risk and high visible value** — nothing in the PAS changes, the business immediately gets real-time data and cheap channel addition, and it proves the canonical model before anything financial depends on it.

112. **The ACL translates; it never decides.** Enforced by explicit review of the ACL for business logic on a regular cadence.

113. Because **financial consistency across a ledger is exactly what a transactional monolith does well**. Distributing a ledger across services introduces eventual consistency, distributed-transaction complexity, and reconciliation burden into the one place where correctness is absolute.

114. Running the legacy and new calculation paths simultaneously against real production data and reconciling outputs daily. **Mandatory whenever any financial calculation is re-implemented rather than reused.**

115. Because they hold the business rules. Decades of undocumented product behaviour, edge cases, and regulatory accommodations exist only in the code and in their heads. They are the primary source for rule extraction, not obstacles to modernization.

116. `POST /withdrawals` creating a resource with a lifecycle (`quoted` → `pending_requirements` → `confirmed` → `settled`), with `/confirm` , `/cancel` , and `/requirements` sub-resources. As a verb on the policy, it can only return success or error — it **cannot represent NIGO, cannot be polled, cannot be cancelled, and cannot carry a quote**.

117. Because an annuity has at least six simultaneously valid values (account, cash surrender, guaranteed minimum, death benefit, free-withdrawal remaining, rider benefit base) plus per-segment detail. A single `value` field forces every consumer to guess which one it is.

118. Reconstruction of the participant set **as it stood on a past date** — which is what makes death-claim adjudication against the designation in force at date of death possible and defensible.

119. Schema → semantic → authority → business rule → regulatory. Return the most specific failure, and return all failures at once where possible.

120. **In:** entity IDs, event type, timestamp, correlation ID, and enough context to decide whether to act. **Not in:** full payloads — consumers re-read the API for detail, which keeps the event contract stable and avoids leaking a changing schema to every subscriber.

121. Return the **original result** of the first request — not an error, and not a new transaction. The key should be retained with the resulting resource for at least 24 hours.

122. Scoped to the distributor's **own book of business** — a broker-dealer must not be able to read another firm's policies. At a TPA, add a **carrier partition claim** so cross-carrier access is impossible by construction.

123. Sources (PAS, contract forms, product specs, procedures, regulatory material, legacy code) → ACORD-aligned semantic layer (canonical entities, Business Glossary vocabulary, entity extraction/linking) → storage (vector DB with entity metadata + knowledge graph) → RAG/orchestration (entity resolution, graph traversal, vector retrieval, live API calls) → assistant surfaces.

124. **Anything legally prescribed, financially consequential, or requiring identical reproducibility must be computed deterministically; the LLM may explain the result but must never produce it.** Reasons: legal correctness, reproducibility, auditability, and consequence asymmetry (a wrong explanation is corrected in the next message; a wrong withholding amount is an IRS penalty and a harmed customer).

125. Provenance on every answer; explicit abstention when retrieval misses; PII handling that respects the same entitlements as the API; human-in-the-loop tiered by risk; a regression suite of verified historical cases; and change control over the semantic layer as part of the control environment.

Data Model (126–150)

126. PARTY (1) → (0..*) PARTY ROLE (0..*) → (1) HOLDING. A party may hold zero or many roles; a holding has one or many roles; the role is the associative entity carrying the relationship, its dates, and its role-specific payload.

127. **Person only.** The annuitant is the measuring life; an organization has no life to measure. The same restriction applies to the Insured role. Both are absolute.

128. Beneficiary class (primary/contingent), percentage, per-stirpes/per-capita flag, and irrevocability. An Owner role carries ownership type instead (sole/joint/JTWROS/trust/custodial).

129. **Two role records referencing the same `partyId`** on the same contract — `roleType: OWNER` and `roleType: ANNUITANT` , each with its own effective date. They must **not** be deduplicated into one record.

130. `effectiveDate`, `terminationDate`, `status`, and never physically deleting or overwriting superseded records — plus `asOf` query support over that history.

131. Account value; cash surrender value; guaranteed minimum value; death benefit value; free-withdrawal remaining; rider benefit base. (Cash value as distinct from surrender value on a loan-bearing life contract is a seventh.)

132. Because it follows entirely different rules: its own rollup accumulation, its own step-up events, its own decrement on withdrawal, and it is **never payable as cash**. Storing it as the account value would make it appear withdrawable, which it is not.

133. Computed on the **benefit base**; deducted from the **account value**. Three independently configured facts: charge basis, charge source, and charge frequency.

134. A dated tranche of money in an index strategy. It must carry: segment ID, strategy reference, start date, start index value, term length, maturity date, principal, and the **rate set as of start** (cap, participation rate, spread, buffer/floor).

135. Because one contract holds multiple simultaneous segments with **different start dates, different maturity dates, and different rate sets** — created by additional premiums and staggered renewals. A single contract-level rate cannot represent even two segments correctly.

136. Business Transaction (1) → Policy Transaction (1) → Financial Activities (1..*) → *Accounting Transactions (1..)* → API/Message Events (1..*). Example: one $10,000 withdrawal request → one policy transaction → seven financial activities (three segment redemptions, surrender charge, withholding, benefit-base adjustment, parent) → four GL entries → several events (withdrawal.completed, policy.value.changed, rider.benefitBase.changed, tax.reportable.created).

137. **Effective date** (when it counts contractually), **processed date** (when the system executed it), and **receipt date** (when the request or money arrived). Collapsing them is the root of most backdating and reconciliation defects.

138. A **reversing activity** referencing the original (`reversalOf: <activityId>`), plus a new correcting activity. The original remains visible in history — because a statement showing it was already mailed.

139. `basisAmount`, `basisAsOfDate`, `basisSource` (premium / 1035-carried / after-tax contribution / conversion), and a **basis movement ledger** parallel to the activity ledger. A single mutable field cannot support prior-year 1099-R reproduction, cannot evidence a 1035 carryover, and cannot defend a customer dispute.

140. As a **versioned, dated entity** with effective/termination dates, a change reason, and a link to the financial activity that effected the change — not as a static column on the policy.

141. Type, effective and termination dates, cost basis and basis-as-of date, RMD applicability, plan/employer reference where relevant, Roth clocks (first contribution year and per-conversion clocks), change reason, and the linked activity ID.

142. Decedent identity, date of death, whether the decedent had begun RMDs, beneficiary classification (eligible designated beneficiary / designated beneficiary / non-designated), and the resulting applicable regime (life expectancy / 10-year / 5-year), with a pre-vs-post-SECURE indicator.

143. `productType: FIXED_INDEXED_ANNUITY`, `planCode: FIA10Q` (retained as source), `surrenderChargeScheduleId: SC-10YR`, and an *implied* qualified status. **What cannot be derived:** which qualified type it is (Traditional IRA vs. SEP vs. 403(b)), the rider benefit base, and any effective dates.

144. Map to an explicit `QUALIFIED_UNSPECIFIED` holding value with `remediationRequired: true` and the source code retained, then drive a data-archaeology project against source documents and 5498 history. **What must not be done:** default it to Traditional IRA. A plausible guess becomes indistinguishable from a verified fact the moment it's written.

145. That the legacy model is **strictly less expressive than reality** — meaning contracts with multiple riders are either misrepresented today, held in a shadow file, or the second rider isn't administered at all. It's a discovery finding requiring investigation, not a mapping problem.

146. **Issue state** (governs the filed form, product availability, and contract provisions) or **owner's current residence** (governs state withholding, premium tax, and correspondence). They diverge whenever an owner moves, and each drives different rules — so getting it wrong produces either the wrong contract terms or the wrong withholding.

147. Because `STATUS` records **what is**, not **since when**. Without an effective date you cannot answer "was this contract in force on the date of death?" — which is the question every claim, audit, and reinstatement turns on.

148. That a **contractual value the carrier owes may not be reliably stored** — the benefit base is the basis for guaranteed income. It is the most serious category of finding in a mapping exercise: not a translation problem but a potential financial-records gap requiring immediate investigation and, possibly, reconstruction from transaction history.

149. Two contracts at two carriers; two linked FINANCIAL ACTIVITY records (`EXCHANGE_1035_OUT` and `EXCHANGE_1035_IN`) cross-referencing an exchange ID; a **basis transfer** with the ceding carrier's documentation retained as a REQUIREMENT artifact; a replacement-disclosure reference; and a 180-day partial-exchange watch window on both sides.

150. PARTY (decedent + three beneficiaries); PARTY ROLE (beneficiary roles with class and percentage, effective-dated); POLICY (status, date of death); **three derived beneficiary interests** (each with its own elected payout option and its own TAX QUALIFICATION under the applicable inherited regime); FINANCIAL ACTIVITY per interest; PAYMENT per interest; TAX REPORTING per beneficiary; REQUIREMENT per beneficiary. **The key insight: one contract splits into three independently administered interests — not three payments off one record.**

Chapter 17 — Final Cheat Sheets

Cheat Sheet 1 — ACORD Architecture

ACORD	Association for Cooperative Operations Research and Development · founded 1970 · non-profit, industry-owned · global insurance standards body · 36,000+ participating orgs, 100+ countries · NOT an insurer
Two output tracks	**Data Standards** (line-specific wire formats) · **Reference Architecture** (cross-line enterprise framework)
L&A Data Standard	ACORD LA XML — current **v2.49.00 Final (Dec 2025)** · annual cadence · components: Help File, Object Reference Model, Schema Files · EDI track also exists (DTCC)
Reference Architecture	7 interrelated facets. Publicly identifiable: Business Glossary (6,000+ terms) · Information Model · Data Model · Capability Model (10 high-level, 3 further tiers) · Process Model · Component Model
Message pattern	TXLife (request) / TXLifeResponse (response) · OLifE (object model beneath) · *verify structure against licensed docs*
Certification	Testing & Certification Facility (TCF) — Stage 1 schema/syntax, Stage 2 business-rule & data-quality review
The flow	Capability → Process → Information Model → Data Model → Message/API → Application
Access	Schemas, Help Files, TCF require membership/participation program. Reference Architecture subscribable by members and non-members

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Cheat Sheet 2 — Life & Annuity Entities

Cluster	Entities
Party	Party (abstract) · Person (natural, can be Insured/Annuitant) · Organization (can't be Insured/Annuitant) · Producer · Carrier · Distributor
Contract	Holding (container) · Policy/Contract · Life Policy · Annuity Contract · Coverage · Rider · Benefit
Roles	Owner · Annuitant · Insured · Beneficiary · Payor · Producer · Trustee · POA · Assignee · Plan Sponsor
Financial	Premium · Payment · Financial Activity · Transaction · Loan · Withdrawal · Distribution
Investment	Fund · Subaccount · Index Strategy · **Segment** · Allocation
Values	Account Value · Cash Value · Surrender Value · Guaranteed Minimum Value · Death Benefit Value · **Rider Benefit Base**
Tax	Tax Qualification (versioned) · Basis · Tax Reporting Record
Supporting	Address (typed, dated) · Communication (consent) · Requirement (NIGO) · Underwriting Information

Three rules that carry the model: Party ≠ Role · Benefit Base ≠ Account Value · Financial Activity is append-only.

Cheat Sheet 3 — Party / Role Relationships

```
PARTY (1) ──< PARTY ROLE >── (1) HOLDING
              effectiveDate · terminationDate · roleType
              + role-specific payload
```

Constraint	Rule
Organization as Annuitant/Insured	**Prohibited**
Organization as Owner	Allowed → **§72(u)** check on NQ deferred
Owner = Annuitant	Normal
Owner ≠ Annuitant	Allowed → **§72(s)** owner-driven death rules
Joint owners on an IRA	**Prohibited**
IRA ownership change	**Prohibited** except divorce/death
Irrevocable beneficiary	Owner's rights constrained — consent required

Collateral assignee	Owner's transaction rights constrained
Beneficiary percentages	Must total **100% per class**

Authority = f(roleType, roleScope, documentValidity, contractState, requestedAction) — evaluated per request, logged with inputs. Authentication ≠ Role ≠ Authority.

Cheat Sheet 4 — Policy / Contract Model

```
POLICY ─┬─ COVERAGE ── RIDER ── BENEFIT
        ├─ INVESTMENT STRATEGY ── FUND/SUBACCOUNT/INDEX STRATEGY ── SEGMENT
        ├─ PARTY ROLE (1..*)
        ├─ TAX QUALIFICATION (versioned)
        └─ FINANCIAL ACTIVITY (append-only)
```

Product	Coverage branch	Investment branch
Term Life	Heavy	Empty
Whole Life	Heavy	Cash value table + dividends
UL / IUL / VUL	Heavy	Declared / index segments / subaccounts
MYGA / Fixed	Nominal	Single declared rate
FIA / RILA	None	**Index strategies + segments**
VA	None	Subaccounts (units)
SPIA / DIA	None	None — the contract *is* a payment schedule

Product-modelling checklist: measured life? · can value fall? · crediting type? · segments? · benefit-base rider? · surrender charge/MVA? · phase model? · which qualifications?

Cheat Sheet 5 — Financial Activities

The five levels: Business Transaction → Policy Transaction → **Financial Activity** → Accounting Transaction → API/Message Event.

Category	Types
Money in	PREMIUM_INITIAL · PREMIUM_ADDITIONAL · ROLLOVER_IN · EXCHANGE_1035_IN · TRANSFER_IN · LOAN_REPAYMENT
Money out	WITHDRAWAL · SYSTEMATIC_WITHDRAWAL · RMD · FULL_SURRENDER · LOAN_DISBURSEMENT · DEATH_BENEFIT · ANNUITY_PAYMENT · EXCHANGE_1035_OUT · ROLLOVER_OUT
Credits	INTEREST_CREDIT · INDEX_CREDIT (**can be negative — RILA**) · DIVIDEND · BONUS
Charges	SURRENDER_CHARGE · MVA (**±**) · RIDER_CHARGE · M&E · ADMIN_FEE · COI · LOAN_INTEREST
Internal	TRANSFER (paired) · REBALANCE · SEGMENT_MATURITY · SEGMENT_CREATION
Tax	FEDERAL_WITHHOLDING · STATE_WITHHOLDING · BACKUP_WITHHOLDING
Notional (no cash)	BENEFIT_BASE_ROLLUP · BENEFIT_BASE_STEPUP · BENEFIT_BASE_ADJUSTMENT · BASIS_ADJUSTMENT · TAXABLE_EVENT_NO_CASH
Corrections	REVERSAL · ADJUSTMENT · RESTATEMENT

Three dates always: effective · processed · receipt. **Corrections:** reverse + re-post, never update in place.

Cheat Sheet 6 — Tax Qualification

The four orthogonal dimensions: Product type (immutable) · Contract type (immutable) · **Tax qualification (changes!)** · Transaction type (per event). **Never conflate into a plan code.**

Qualification	Basis	Distribution	RMD (owner alive)

Non-Qualified	After-tax premium	**LIFO** — gain first	No
Traditional IRA	Usually $0	Ordinary income; pro-rata if basis	Yes
Roth IRA	Contributions	Tax-free if qualified (5-yr + trigger)	**No**
SEP IRA	Employer pre-tax	Ordinary income	Yes
SIMPLE IRA	Deferral + match	Ordinary income · **2-year rule**	Yes
401(k) / 403(b)	Per source	Per source	Yes
457(b)	Per source	Ordinary income · **gov't vs. non-gov't matters**	Yes
Inherited IRA	Per underlying	Per beneficiary regime	Yes, per regime

Qualification-changing events: Roth conversion (taxable, **irreversible**) · rollover · 1035 (carries basis) · death/inheritance · QDRO/divorce · NQ ownership change (§72(e)(4)(C), often taxable).

Model as: versioned, dated, reason-coded, linked to the effecting activity.

Cheat Sheet 7 — Annuity Transactions

Universal lifecycle: Intake → Identify → Validate → Require (NIGO) → Calculate (quote) → Commit → Settle → Notify.

Transaction	The thing people get wrong
New business	Producer licensing **in the owner's state**; RILA/VA needs securities registration
Additional premium	Creates a **new segment**; may not increase the benefit base
Withdrawal	GLWB **excess-withdrawal** test → permanent income reduction
Systematic withdrawal	Must stop on exhaustion; may double up with RMD
Full surrender	Benefit base is **forfeited**, not paid
RMD	Not on Roth (owner alive); FMV includes rider actuarial value
Roth conversion	**Irreversible**; must permit 0% withholding; RMD first if RMD-age
IRA rollover	Direct vs. indirect; one-per-12-month needs **attestation**
1035 exchange	Annuity→life **invalid**; basis must come from ceding carrier
Transfer	FIA/RILA: **only at segment maturity**
Annuitization	Irreversible; phase change; exclusion ratio
Death claim	Owner-driven vs. annuitant-driven; SECURE beneficiary regime
Beneficiary change	**Version, never overwrite**; irrevocable needs consent
Ownership change	May be taxable with **no cash**; may kill the rider
Address change	**Account-takeover vector**; recalc state withholding
Rider add/remove	GLWB usually issue-only; removal usually irreversible
Allocation change	**Future premium vs. existing value** — always disambiguate
Segment maturity	Rate set is **as of segment start**, not current
Renewal	Rate ≥ guaranteed minimum; advance notice is regulated
Lapse	**Test the no-lapse guarantee first**; loan → phantom income
Free-look	Return of **premium vs. account value** varies by state
Withholding	Effective-dated tables, never hard-coded
Tax reporting	A projection over the ledger; must be reproducible years later

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Cheat Sheet 8 — Message Architecture

Six parts: Header · Intent · Party · Holding · Activity · Result.

Key patterns:

- Parties declared **once**, referenced by ID from each role — never duplicated
- **Errors block, warnings inform** — a GLWB reduction is a warning, not an error
- `PENDING_REQUIREMENTS` is a **first-class status** (that's NIGO)
- Response carries the **requirement objects** so the caller can resolve without another call
- Correlation IDs thread request → response → async completion

Error class	Retry?
Syntax/schema	No — fix and resend
Semantic	No
Authorization	No
Business rule	No
Requirement (NIGO)	**Yes, after satisfying**
State conflict	Sometimes
Transient/system	**Yes, with idempotency key**

XML vs JSON: XML = mature validation, industry rails, verbose, legacy tooling. JSON = light, developer-friendly, no industry certification. **Answer: JSON at the edge, canonical in the middle, XML where the rails require it.**

Cheat Sheet 9 — API Architecture

Resource	Purpose
`GET /policies/{id}`	Contract summary
`GET /policies/{id}/values`	**Six+ values, never one**
`GET /policies/{id}/participants?asOf=`	Roles, dated, reconstructable
`GET /policies/{id}/transactions`	The ledger, parent/child, reversals
`POST /withdrawals`	**Resource with a lifecycle**, not a verb
`POST /beneficiary-changes`	Full set, versioned
`POST /roth-conversions`	Explicit withholding, irreversibility ack
`POST /1035-exchanges`	Ceding basis with documentation
`POST /rmd-elections`	Aggregation attestation
`POST /allocation-changes`	Explicit `scope`

Cross-cutting: Idempotency-Key on every write (repeat returns the **original** result) · OAuth2 + per-request authority evaluation · distributor-scoped tokens + carrier partition claim at a TPA · five validation layers (schema → semantic → authority → business rule → regulatory) · URI-path versioning, additive fields, never repurpose a field · events carry IDs and context, not full payloads.

Two-phase quote/confirm is a compliance control, not a UX nicety.

Cheat Sheet 10 — Legacy-to-ACORD Mapping

Legacy field	Canonical	Watch for
`POLICY_NUMBER`	`policyNumber`	**Not globally unique** — prefix with carrier

`OWNER_ID`	Party + Role (OWNER)	Explode to a dated role; find hidden joint owners
`ANNUITANT_ID`	Party + Role (ANNUITANT)	If = OWNER_ID, **two roles, one party**
`PLAN_CODE`	productType + planCode + scheduleId	**Overloaded** — decompose via maintained lookup; unmapped = hard stop
`ISSUE_DATE`	`issueDate`	CYYMMDD → ISO 8601
`QUALIFICATION_CODE`	`taxQualification.type`	**Ambiguous — never guess.** Map to `QUALIFIED_UNSPECIFIED` + remediation flag
`ACCOUNT_VALUE`	`values.accountValue`	Carry the **as-of date**; must equal Σ segments
`SURRENDER_VALUE`	`values.cashSurrenderValue`	Recompute if the cycle date is stale
`RIDER_CODE`	`riders[]`	**One field → array.** Benefit base often missing entirely
`STATUS`	`status` + `statusEffectiveDate`	Effective date usually **not in the record**
`STATE`	`issueState` and/or residence	**Ambiguous** — different rules follow from each
`RMD_INDICATOR`	`taxQualification.rmdSubject`	Cross-check against qualification type

The six discovery findings to lead with: overloaded codes · lost distinctions · cardinality gaps · missing data · ambiguous semantics · absent temporality. **The gaps are the deliverable, not the appendix.**

Appendix A — Verification Register

Items in this guide explicitly requiring confirmation against licensed or current ACORD documentation:

§	Item	Where to verify
1.4	The seventh Reference Architecture facet	acord.org/standards-architecture/reference-architecture
1.6	OLifE current object hierarchy and terminology	Licensed ACORD LA XML v2.49.00 Help File & Object Reference Model
1.7	Scope and maturity of ACORD Next-Generation Digital Standards / any official REST standard for L&A	acord.org
1.9	Schema extension mechanism specifics	Licensed schema documentation
4.1	LA XML new-business transaction type codes	Licensed LA XML documentation
4.8	Current RMD applicable age and excise-tax rate	Current-year IRS guidance
4.28	Current federal/state withholding defaults and mandatory-withholding states	Current-year IRS + state guidance
4.29	1099-R/5498 box numbers and distribution codes	Current-year IRS instructions
5.2	All contribution limits, ages, and thresholds	Current-year IRS guidance
8.1	Exact TXLife element hierarchy, cardinality, wrapper authentication structure	Licensed LA XML documentation
15.5	FMV valuation methodology for annuities with living benefits	Current IRS guidance + carrier actuarial policy

All XML and JSON examples in this guide are labelled "Illustrative ACORD-aligned example — not an official ACORD schema" and must not be used for implementation.

Appendix B — Publicly Accessible Sources

- ACORD — About: `https://www.acord.org/ACORD-about/ACORD-About`
- ACORD — Data Standards: `https://www.acord.org/standards-architecture/acord-data-standards`
- ACORD — Life & Annuity Data Standards: `https://www.acord.org/standards-architecture/acord-data-standards/Life_Annuity_Data_Standards`
- ACORD — Reference Architecture: `https://www.acord.org/standards-architecture/reference-architecture`
- ACORD — Reference Architecture update (Capability Model & Business Glossary, May 2025): `https://www.acord.org/ACORD-about/acord-news/2025/05/21/acord-reference-architecture-updates-capability-model-business-glossary`

- ACORD — Reference Architecture update (Claims/Process Model, Jul 2024): `https://acord.org/ACORD-about/acord-news/2024/07/10/acord-reference-architecture-update-provides-claims-processing-enhancements`
- ACORD — Reference Architecture update (Capability Model, Nov 2023): `https://acord.org/ACORD-about/acord-news/2023/11/29/acord-releases-update-to-the-reference-architecture`
- ACORD — Reference Architecture update (data facets, Aug 2023): `https://www.acord.org/ACORD-about/acord-news/2023/08/01/acord-releases-update-to-the-reference-architecture`
- ACORD Solutions Group — About: `https://www.acordsolutions.com/about/about-acord`
- IRS — retirement plans, IRAs, and annuity taxation guidance (current-year publications and form instructions): `https://www.irs.gov`
- Insurance Innovation Reporter — applications of the ACORD Reference Architecture: `https://iireporter.com/7-real-world-applications-of-the-acord-reference-architecture/`

Secondary sources were used only to corroborate publicly stated organizational facts. No schema, field-level, or courseware content was drawn from them.

End of guide.

ACORD Standards for US Life & Annuity — A Practical Business & Technology Architecture Study Guide *17 chapters · original synthesis · verification register at Appendix A*